Nonlinear Optics and Solitons

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Abstract

Beams of light tend to broaden as they propagate in linear medium. However, in a nonlinear medium (in which the index of refraction is proportional to the intensity) self-focusing can compensate for diffraction. In this case, the beam can create an index change similar to the one of an optical fiber and to guide itself in this self-induced fiber. These beams keep their shape and dimensions invariant along propagation. We called this type of beams spatial solitons. The beam that creates the fiber can be composed of more than one mode (e.g. TEM_{00} + TEM_{01}+...), in this case we called this type of soliton: composite soliton.

Here, I report on a new kind of composite solitons that lives in a 3-dimensional bulk; one dimension for propagation and the two transverse dimensions in which the beam is self-trapped. More specifically, I present the first observation of a dipole soliton: a multimode soliton composed of a single-hump component and a double-hump (dipole-type) component that jointly self-trap in two transverse dimensions and propagate in the third dimension. This discovery opened the door for new directions in soliton science. The progress from planar systems (e.g. slab waveguides) to 3D media enabled us to demonstrate phenomena that have no counterpart in 2D, such as a new type of vector soliton for which the energy structure rotates while propagating. These rotating propeller solitons consist of a dipole that rotates throughout propagation and makes a double helix structure in space, jointly trapped with a rotating bell-shaped component. We have studied these propeller solitons experimentally, analytically, and numerically and set forth the experimental and the theoretical foundation of the first ever composite solitons carrying angular momentum. The combination of rotation and propagation in 3D enabled us to go forward and investigate new phenomena such as collisions in space between rotating vector solitons.

In the second part of this research, I report on spontaneous pattern formation in incoherent cavities. Organized structures of trains of solitons can sometimes emerge spontaneously from a homogeneous state with random noise. We call this phenomenon spontaneous pattern formation. This type of structures can arise

in systems with different degrees of coherency (from a fully coherent laser beam to white light from a regular incandescent bulb).

In our recent series of papers, we have demonstrated pattern formation in incoherent cavities. We first demonstrated pattern formation in temporally incoherent cavities and showed that the bandwidth of the spatial frequency spectrum of the pattern narrows as the feedback increases. This opened the way to our subsequent work on cavities in which the circulating radiation is fully incoherent (both temporally and spatially). I demonstrated experimentally that such fully incoherent cavities exhibit two distinct thresholds: the first modulation instability threshold when self-focusing overcomes incoherent diffraction and the second cavity threshold when gain overcome cavity losses. These two different thresholds are experimentally distinguishable through the fact that the first threshold (the modulation instability threshold) is independent of the feedback, whereas the second threshold (the cavity threshold) is feedback-dependent.

Finally, as suggestion for further research, I discuss bright vortex soliton. The bright vortex is a ring of light that rotates along propagation and keeps it center (the storm eye) dark. This suggestion is based on materials that change their refractive index as a function of the temperature and as a function of the mechanical stress.

List of symbols

I	Total intensity
k	Wave vector
L_D	Diffraction length
n_0	Refractive index (linear)
P	Linear momentum
x, y, z	Spatial coordinate
Δn_0	Nonlinear refractive index change
λ	Wavelength
τ	Material response time
ψ	Slowly varying amplitude of the electric field
k	Thermal conductivity
E	Modulus of elasticity (Young modulus)
v_g	Group velocity
t	time
T	temperature
p	momentum
σ	stress
ν	spatial frequency
k	wave number
g	gain
q	heat

1. Introduction



Figure 1: In the summer of 1995, a water soliton was created (in front of the boat). This soliton was demonstrated in the same aqueduct in which Scott Russell made the first observation of such solitons in 1834.

Beams of light, water waves and other waves phenomena in nature tend to diverge while propagating in space. This broadening is called "diffraction". The fundamental origin of diffraction is the reason why we are not used to see waves which stay localized in space while keeping their shape invariant as they propagate. It is hence rather surprising that even though the origin of diffraction is so fundamental, waves that keep their shape stationary along propagation do exist in nature. These waves are called solitons and they exist in fluids [1] (see Figure 1), plasma waves [2], sound waves in liquid He^3 [3], temporal soliton in mechanical spring and mass systems [4], electric signals in telegraph lines, condensed matter, particle physics, and even in astrophysics and in DNA chains [5]. In optics, solitons manifest themselves in numerous forms: temporal solitons in fibers [6,7] that maintain their temporal width for distances of hundreds of kilometers , spatial solitons [8,9] that propagate numerous diffraction lengths without broadening (see Figure 2), spatio-temporal solitons [10,11],

as well as more sophisticated forms such as incoherent solitons that are made from white light from a regular incandescent light-bulb [12,13], discrete solitons [14,15,16,17], gap (or Bragg) solitons [18,19] and cavity solitons [20,21]. Solitons propagate and interact with one another while displaying properties that are normally associated with real particles [1,2].

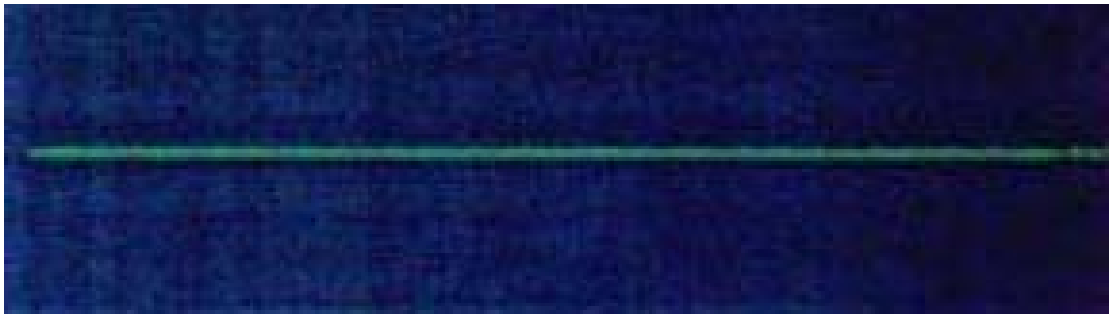


Figure 2: An optical spatial soliton[†]. Top view of a narrow beam launched (from the left hand side) into a nonlinear medium. The beam keeps it size and structure as it propagates.

Usually, propagating waves in nature tend to spread in space (time) due to diffraction (dispersion). In optics, a localized pulse in space propagating in a linear medium will broaden in space as it propagates. For pulses in space (beams), the broadening is caused by diffraction, while for pulses in time, the broadening is caused by dispersion. In general, the narrower the initial beam, the faster it diverges with propagation. For example, a Gaussian beam of 10 μm diameter at 0.5 μm wavelength diffracts to a 170 μm diameter after propagating 1 cm in glass; i.e. the beam diameter at the output face of the 1 cm long glass is 17 times wider than at the input face. An example is shown in
Figure 3 for beam with similar dimensions.

[†] Photograph by M. Shih [12]

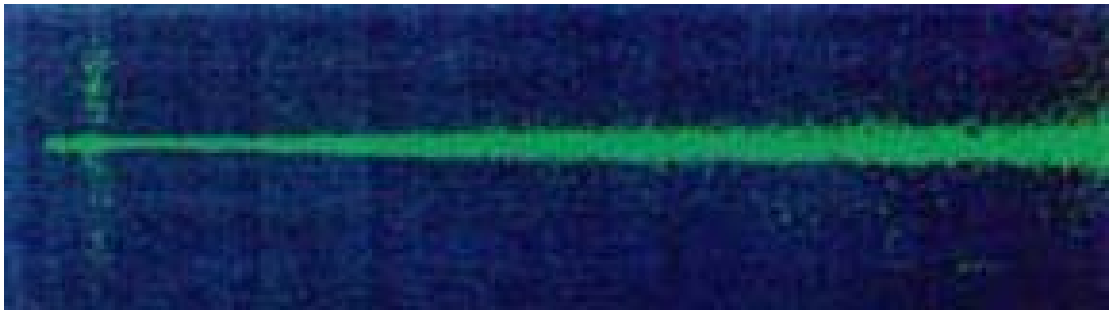


Figure 3: Top view of a narrow beam[‡] launched from the left hand side into a linear medium (the beam is visible because of stray scattering in the crystal). The beam gets broader as it propagates.

In most textbooks, linear diffraction in space is derived from the Maxwell equations (see for example [22]). This traditional derivation is well known and hence I had a motivation to shade light on diffraction from a wider perspective. I will try to treat different aspects of diffraction in the next section.

1.1. Diffraction

From the historical perspective, it was at one time suggested that interference and diffraction phenomena are caused by collisions between photons considered as small particles [23], or at least to some kind of complex interaction between photons. A simple experiment performed by a British physicist, Geoffrey Ingram Taylor, in 1908 excludes this possibility. He photographed the diffraction pattern of a needle and reduced the illumination until long exposures were needed to obtain an image. When the chance of two or more energy quanta passing through the apparatus simultaneously was made extremely small, the diffraction pattern was exactly the same as that obtained with a strong source of light.

This experiment is supported by many later experiments, showing that interference and diffraction are to be associated with single photons. When ordinary (incoherent) sources of light are used each photon can interfere only with itself and not with any other photon. An interference pattern can, of course, be photographed only

[‡] Photograph by M. Shih [12]

by recording the effects of many photons because one photon can activate only one grain on the photographic plate, but it does not matter whether the photons all arrive over a time span of a microsecond or of several weeks.

The photograph of an interference pattern is both a wave phenomenon because it shows the characteristic spatial periodicity and a quantum phenomenon because the whole energy of a photon can be used to activate a single grain. It is not possible to trace the path of one photon (regarded as a particle) through the apparatus and, at the same time, obtain the interference pattern. In Young's experiment there is no way of finding out through which slit a given photon passes—except by covering one slit and thus losing the interference fringes.

The energy and momentum of the photon are determined by the frequency of the waves. Fourier analysis of wave groups shows that, when the wave group is short (so that the position of the photon is fairly well determined), the frequency of the waves and, hence, the energy and momentum of the photon are known only within a wide range. The product of the uncertainty in position and the uncertainty in momentum can never be less than a certain minimum value. It will now be shown how these relations appear in an experimental situation.

In an experimental system a parallel beam of light is incident on a slit of width *d*, producing a diffraction pattern on a screen. The y-coordinate, at the moment when a photon passes through the slit, is known within an uncertainty $\Delta y = \pm d$. The direction of propagation when the photons emerge from the slit is uncertain because of diffraction, but most of the photons are observed within an angular range $\pm\theta$, in which $\sin\theta = \lambda/d$. There is therefore an uncertainty Δp_y in the y-component of the momentum (p) of about $\pm(h \sin\theta)/\lambda$, so that the product of the two uncertainties, Δp and Δy, is

$$\Delta p \Delta y \sim dh \sin\theta / \lambda \sim h \qquad (1)$$

in which the sign ~ indicates is of the "order of magnitude of."

More detailed analysis shows that, if the uncertainties Δy and Δp_y are the root mean square deviations of statistical theory, then their product is equal to or larger than $h/4\pi$; so that $\Delta y \, \Delta p_y \leq h/4\pi$.

In the context of solitons, Much understanding can be attained by considering the beam as a superposition of plane waves. In order to construct a beam, one can use plane waves of different intensities and propagation directions. In a linear medium, each plane wave maintain it's propagation direction (and energy) independently of the other components. However, off axis components are phase delayed in comparison with the on axis plane wave. Consequently, the waves that interfere constructively in the beam center becomes fewer and as a result of this the intensity at the beam center fades as the beam propagates. On the other hand, far a way from the beam center, there are dark regions in which the different components interfere destructively. But, as the beam propagates and the plane waves acquire different phases, the interference is no more destructive interference. This is the reason why light starts to appear far away from the beam center as the beam propagates. To conclude, beams propagating in a linear medium tends to broad as their intensity becomes lower at the beam center and higher far away from the beam center.

1.2. Optical spatial solitons

How can we make beams that maintain their shape and size along propagation? As explained in the previous section, the phase between the different plane waves constructing the beam should be invariant along propagation. In a linear medium, the phase between the different plane waves depends solely on the propagation angle. It is clear then, that if all the plane wave components have the same angles, they will also have the same phase velocity and no phase difference will be acquire during propagation. If we represent the propagation direction of these plane waves by arrows, these arrows will creates a cone in space. The associated beam will propagate in this linear medium with no broadening while it maintain it's shape and dimensions. This type of beams are called "Bessel beams" since their field is described by Bessel function.

In a nonlinear medium, however, each plane wave is influenced by all the others. This is because the index change is a function of the total intensity. For some nonlinearities it is possible to find an ensemble of plane waves which will delay the phase velocity of the on axis components (in comparison with the of axis components) Consequently, the phase delay between the different propagating plane waves (explained in the previous section) will be compensated by the other waves via the medium nonlinearity.

If the nonlinearity is such that the index at the beam center is higher than in the dark regions, then the plane wave that propagate on axis will experience the highest index and will propagate slower from those off axis. This high index experienced by the on axis wave can compensate for it's linear tendency to propagate faster (because it is on axis). One example in which the index of refraction fully compensate for the shorter trajectory is a hyperbolic secant solution in a Kerr type mediem[§]. The plane waves that construct the hyperbolic secant index profile (induced by the beam via the nonlinearity) have all the same phase velocity along propagation, and hence their interference (the hyperbolic secant shape) also maintain its shape and size as it propagates.

One can also look on solitons from the mathematical perspective. One has to find a stationary wave solution of governing nonlinear wave equation. The mathematical aspects of solitons are out of the scope of this thesis. Hence, here I will give just the final solution. For the exact mathematical derivation, the reader may refer to [24].

We start from the Maxwell equations with the following assumptions:

- The electric field in the propagation direction is negligible in comparison with the transverse electric field (paraxial approximation).
- The elecro-magnetic wave is monochromatic and its frequency is ω.

[§] In which the index change is proportional to the intensity of light

- The transverse electric field is $E(x,y,z,t)=\psi(x,y,x)\exp[i(\omega t-kz)]$ where the slowly varying field $\psi(x,y,x)$ changes much slower than $\exp[i(\omega t-kz)]$ [**].
- The change in the refractive index is much smaller than one.

For beams in Kerr-type medium (for which the index of refraction is the following function of the intensity: $n(I)=n_0+n_2 I$), one can start from the Maxwell equations and derive the nonlinear Schrödinger equation:

$$\frac{\partial}{\partial z}\psi(x,y,z)=[\frac{i}{2k}(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2})+\frac{ik\ n_2}{n_0}|\psi(x,y,z)|^2]\psi(x,y,z) \qquad (2)$$

Here, z is the propagation direction, k is the wave vector, and ψ is the slowly-varying amplitude of the electric field. Among all solutions for Equation (2) there is one family of particular interest:

$$\psi=\sqrt{\psi_0}Sech[\frac{x}{W_0}]\exp[i(\frac{z}{4z_0})] \qquad (3)$$

Here $W_0=\sqrt{\frac{2n_0}{n_2}}\frac{1}{k\psi_0}$ and $Z_0=\frac{kW_0^{\ 2}}{2}$. We can see that the intensity of this sub-family ($I(x,y)=|\psi|^2$) is z independent. For any chosen peak power, ψ_0 , we can find a sech solution with appropriate width (the width, W_0, is a function of the peak intensity and is wider for lower peak intensities). All solutions of this sub-family (equation (3)) will keep their shape and size invariant along propagation. Yet, in order to observe such solitons in nature, it is not enough to have a steady state mathematical solution in hand: one should also check for the stability of this steady state solution to noise and to deviations from ideal initial condition. If the solution exemplifies a state of stationary propagation - only then it can be considered as a soliton.

[**] The presentation of the electric field here is the phasoric presentation. For more on the phasoric presentation, see for example [22]

1.3. The same equations has the same solutions

In this section, I would like to show that solitons are not esoteric phenomena, but in fact, universal to a large variety of almost any wave phenomenon in nature. Any medium is in fact nonlinear when the amplitude of the wave is high enough. It is very useful to write the nonlinear Schrödinger equation (and also other nonlinear equations) formally in the form:

$$\frac{\partial \psi}{\partial A} = (\hat{D} + \hat{N})\psi \qquad (4)$$

where $\hat{D}$ is a differential operator that accounts for linear diffraction (or dispersion if the pulse is propagating in time) and $\hat{N}$ is a nonlinear operator that governs the effects of medium nonlinearities on pulse propagation. For particular solutions (ψ) diffraction ($\hat{D}$) is fully compensated by nonlinearity ($\hat{N}$) and the right hand side of equation (4) becomes zero. This particular solution (ψ) is invariant along propagation in A and hence this solution describes a soliton that maintain its shape and energy along propagation.

In general, diffraction and nonlinearity act together during propagation. As an example, in equation (2), $\hat{D} = \frac{i}{2k}(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2})$, $\hat{N} = \frac{ik}{n_0}\Delta n_0 |\psi|^2$, and the beam propagates along A=z. For solitary wave solutions [like equation (3)], the diffraction $\hat{D}$ is fully compensated by the medium nonlinearity $\hat{N}$ in such a manner that the intensity ($\psi\psi^*$) is A independent. However, to get a soliton in the laboratory (or in nature) not only must the mathematical solution be stationary along propagation, but this solution should also be stable. Stability analysis of solitons is done in numerous texts (see for example [24, 26]) and will be discussed briefly below.

In view of the fact that similar equations have the same solutions [25], we must conclude that solitons are not just an odd solution of some exotic system. Solitons are universal in nature: in any place one finds a solitary wave – one should look for the governing nonlinear equation, and in any place he sees a nonlinear equation he should search for solitary solutions. In the next subchapters, I will give examples for various kinds of solitons in order to demonstrate soliton diversity.

1.3.1. Optical spatial solitons

For simplicity we will reduce equation (2) to only two dimensions (x,z):

$$\frac{\partial \psi}{\partial z} = (\frac{-i}{2k}\frac{\partial^2}{\partial x^2} - i\frac{k\Delta n_0}{n_0}|\psi|^2)\psi \qquad (5)$$

As shown in section 1.2, the solitary solution for (5) is:

$$\psi = \sqrt{\psi_0}\,Sech[\frac{x}{W_0}]\exp[i(\frac{z}{4z_0})] \qquad (6)$$

Where $W_0 = \sqrt{\frac{2n_0}{n_2}}\frac{1}{k\psi_0}$ is the beam width and $Z_0 = \frac{kW_0^{\,2}}{2}$, is the diffraction length. This solution describes a beam profile uniform in *y*, bell-shaped in *x*, and traveling in *z* with no change of form or width. The intensity structure in *x* is equal to that of the mode of a graded-index waveguide with refractive index $n(x) = n_0(1+\frac{\sec h^2(x/W_0)}{k^2 W_0^{\,2}})$, so that self-consistency is assured. The solution (6) is uniform in *y*. As it propagates, this beam will disintegrate in $\hat{y}$, and some spatial frequency will start to form in $\hat{y}$ due to modulation instability (discussion on the origins of modulation instability will follow in the next chapter). One can eliminate this modulation instability in $\hat{y}$ by launching this beam into a slab waveguide made of Kerr nonlinear material. Stability of this solution (6) in *x* will be shown in the next section.

Circular solitons, which are self-trapped in two transverse dimensions [e.g. $\sim Sech(\frac{x}{Wo})Sech(\frac{y}{Wo})$] are not stable in Kerr media[††] since the balance between diffraction length and nonlinear length will yield $\psi_0 = \lambda^2/(\pi\, n_0\, \Delta n_0)$, which means that the peak power of the beam must be a critical constant value. Any small perturbation that will make the beam power smaller than the critical value will make

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[††] However In contrast with the Kerr nonlinearity ($\Delta n \propto I$), The photorefractive nonlinearity ($\Delta n \propto 1/(I+1)$) is of a different character and do support circular spatial solitons.

the beam diffract, and any deviation which will make the beam power larger than the critical value will lead to catastrophic self-focusing of the beam.

1.3.2. Fiber solitons

The study of most nonlinear effects in optical fibers involves the use of short pulses with widths ranging from ~10 ns to ~10 fs. When such optical pulses propagate inside the fiber, both dispersive and nonlinear effects influence their shape and spectrum.

A pulse of light propagating in glass will usually becomes broader as it propagates. The reason for that is that every pulse of light consists of a Fourier spectrum of different color components. The same mechanism responsible for the chromatic dispersion in a glass prism is the one that also makes the red components of the pulse propagate slower than the blue components and therefore causes the pulse to broaden as it propagates (this chromatic dispersion is also known as group-velocity dispersion).Agrawal [26] starts from the electric field wave equation and gets the governing equation for light pulse propagating in a optical fiber

$$\frac{\partial \psi}{\partial z} = (\frac{-i\,\beta_2}{2}\frac{\partial^2}{\partial T^2} + i\,\gamma\,|\psi|^2)\psi \qquad (7)$$

were ψ is the slowly-varying amplitude of the electric field, β_2 is the second-order dispersion of the wave number, and γ is the nonlinearity coefficient (of the medium) and T is the temporal profile ass seen by a viewer with the same velocity as the pulse group velocity ($T = t - z/V_g$). As its twin equation from the previous section, equation (7) also has a family of solutions which keep their profile unchanged when propagating. Given a negative β_2, and pulse width T_0 , the solitary solution is:

$$\psi = \sqrt{\psi_0}\,Sech[\frac{T}{T_0}]\exp[i(\frac{z}{2L_D})] \qquad (8)$$

Where ψ_0 is the peak power, $L_D = T_0^{\,2}/|\beta_2|$ is the dispersion length of the pulse (the length needed for the pulse to become twice wider when propagating in linear medium). This solution describes a pulse that keeps its temporal width invariant as it

propagates and thus is called a *temporal soliton*. It is important to note that for any given pulse width, T_0, one can find the proper peak intensity ($\frac{|\beta_2|}{\gamma T_0^{\,2}}$) that will make this pulse a soliton. The temporal power profile for this soliton is of course

$$I(t,z) = |\psi|^2 = \psi_0 Sech^2[\frac{t - z/v_g}{T_0}] \qquad (9)$$

The following gives a typical scale for the dimensions of such a soliton: in a silica fiber at the 1.55 μm wavelength, a 50 mW peak power soliton will be $T_0 = 10\,ps$ long, which is a pulse of about 2 mm length (in *z*). From an experimental point of view, it is not enough to have a steady state solution in hand. This steady state solution must also be stable. To give an example of an unstable solution, we can consider a pencil that is standing on its tip. This pencil on its tip system is in equilibrium. Yet, it will not take long before this pencil will fall flat on the ground. Mathematically (and physically) speaking, this equilibrium is not a stable one. For the fiber soliton, a balance should be achieved between the dispersion length $L_D = \frac{T_0^{\,2}}{|\beta_2|}$ and the nonlinear length $L_{NL} = \frac{1}{\gamma\psi_0}$, which are the length scales over which dispersive or nonlinear effects make the beam become wider or narrower. For a soliton, there is balance between the two and hence $L_D = L_{NL}$,which leads to:

$$T_0^{\,2}\psi_0 = \frac{|\beta_2|}{\gamma} = Const \qquad (10)$$

The right hand side of this equation is a function of the fiber parameters and hence is a constant. Equation (10) implies that fluctuations in the power can be compensated by changes in the pulse width and vice versa. This property is responsible for the robustness of this type of solitons, i.e. they are highly stable and any optical excitation with parameters close to the soliton parameters usually evolves with propagation into a stable soliton. We may note here that the solitary spatial solution in the previous section (Equation (6)) is stable in *x* because of the same reason that make the solitary temporal solution which is presented above (Equation (8)) stable. This is because the governing equation here is identical to the governing equation in the previous section.

1.3.3. Surface waves in water.

Solitons are not limited to optics, For example, surface gravity[‡‡] waves in water, described by the (dimensionless) Korteweg-de Vries equation:

$$\frac{\partial \psi}{\partial t} = (-\psi \frac{\partial}{\partial x} - \frac{\partial^3}{\partial x^3})\psi \quad , \qquad (11)$$

have solitary solution of the form:

$$\psi = 3\eta \, Sech^2[\frac{1}{2}\sqrt{\eta}(x - \eta t)] \quad . \qquad (12)$$

Here, ψ stands for the dimensionless height of the water wave front. This soliton is the one that had been found in the canal by Scott-Russell (see Figure 1)

Equations (11) and (12) also describe ion acoustic waves in plasma.

1.3.4. Superfluids.

Diffraction phenomena are not limited to one or two dimensions. The macroscopic description of superfluid dynamics can be described by the (dimensionless) Gross-Pitaevskii equation [27,28]:

$$\frac{\partial \psi}{\partial t} = [i\nabla^2 - i(|\psi|^2 - 1)]\psi \qquad (13)$$

(Where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ stands for diffraction in *x,y* and *z*) This equation supports a vortex line of the form:

$$\psi = \psi_0(r)\exp(i\theta) \, . \qquad (14)$$

Here θ is the angle in cylindrical coordinates, and $\psi_0(r)$ must satisfy equation (13) . $|\psi|^2$ is the superfluids density and the superfluid flux is $2\,\mathrm{Im}(\psi^* \nabla \psi)$. As angular momentum is conserved, vortex lines will tend to go in closed loops and not just start

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[‡‡] Water gravity waves are water waves larger than a few cm in dimension. For these waves, the restoring force is gravity. For water waves with a scale smaller than a few mm, the dominant restoring force is surface tension.

or stop (exactly like lines of magnetic field). The natural shape of the closed loop is a ring and the vortex line of Equation (14) can close a loop to become a vortex ring. Rayfield was the first to observe such vortex rings in superfluid helium in 1964 [29]. This type of vortex ring was proved to be perpetually stable by Pismen and Nepomnyashchy [30]

Up to this point, we have discussed various kinds of solitons trapped either in the temporal domain, keeping their structure against dispersion (fiber solitons in chapter 1.3.1 and water solitons in chapter 1.3.3), or in the spatial domain keeping their structure against diffraction (as the optical spatial solitons in chapter 1.3.2). Solitons can also be spatial and temporal at the same time while propagating without neither dispersion nor diffraction. It is interesting to compare the different kinds of solitons and to see that, even though the equations are sometimes different, the similar character of all those equations do support balance between nonlinearity and dispersion which can lead to solitary solutions as demonstrated in Figure 4.

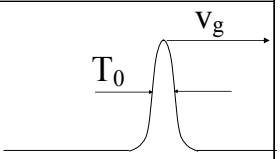
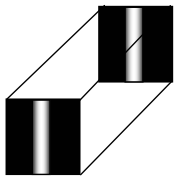
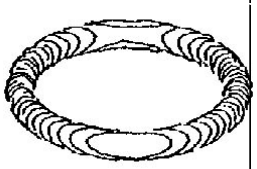
When the governing equation is $\frac{\partial \psi}{\partial A} = (\hat{D} + \hat{N})\psi$ where the					
	Propagation is in A= :	Dispersion $(\hat{D})$ is:	Nonlinearity $(\hat{N})$ is:	The solitary solution is $\psi =$	
Fiber	z	$\frac{-i\beta_2}{2}\frac{\partial^2}{\partial T^2}$	$i\gamma\lvert\psi\rvert^2$	$\sqrt{\psi_0}Sech[\frac{T}{T_0}]\exp[i(\frac{z}{2L_D})]$ $L_D = T_0^2/\lvert\beta_2\rvert$	
Optical spatial	z	$\frac{-i}{2k}\frac{\partial^2}{\partial x^2}$	$i\frac{k\Delta n_0}{n_0}\lvert\psi\rvert^2$	$\sqrt{\psi_0}Sech[\frac{x}{W_0}]\exp[i(\frac{z}{4z_0})]$ $Z_0 = \frac{kW_0^2}{2}$	
Water	t	$-\frac{\partial^3}{\partial x^3}$	$-\psi\frac{\partial}{\partial x}$	$3\eta\, Sech^2[\frac{1}{2}\sqrt{\eta}(x-\eta t)]$	
Superfluid	t	$i\nabla^2$	$-i(\lvert\psi\rvert^2-1)$	Vortex ring	

Figure 4: Comparison between different kinds of solitons. (the picture of the superfluid vortex ring was taken from a simulation by J. Koplik [31])

One can of course add many rows with different kinds of solitons to the table above. It is sufficient here to highlight just four examples of solitons from different fields in physics, in different nonlinearities and in different dimensions. Despite all these differences, one thing is common for all: the balance between diffraction and the stationary solutions along propagation.

1.4. Modulation instability, pattern formation and solitons

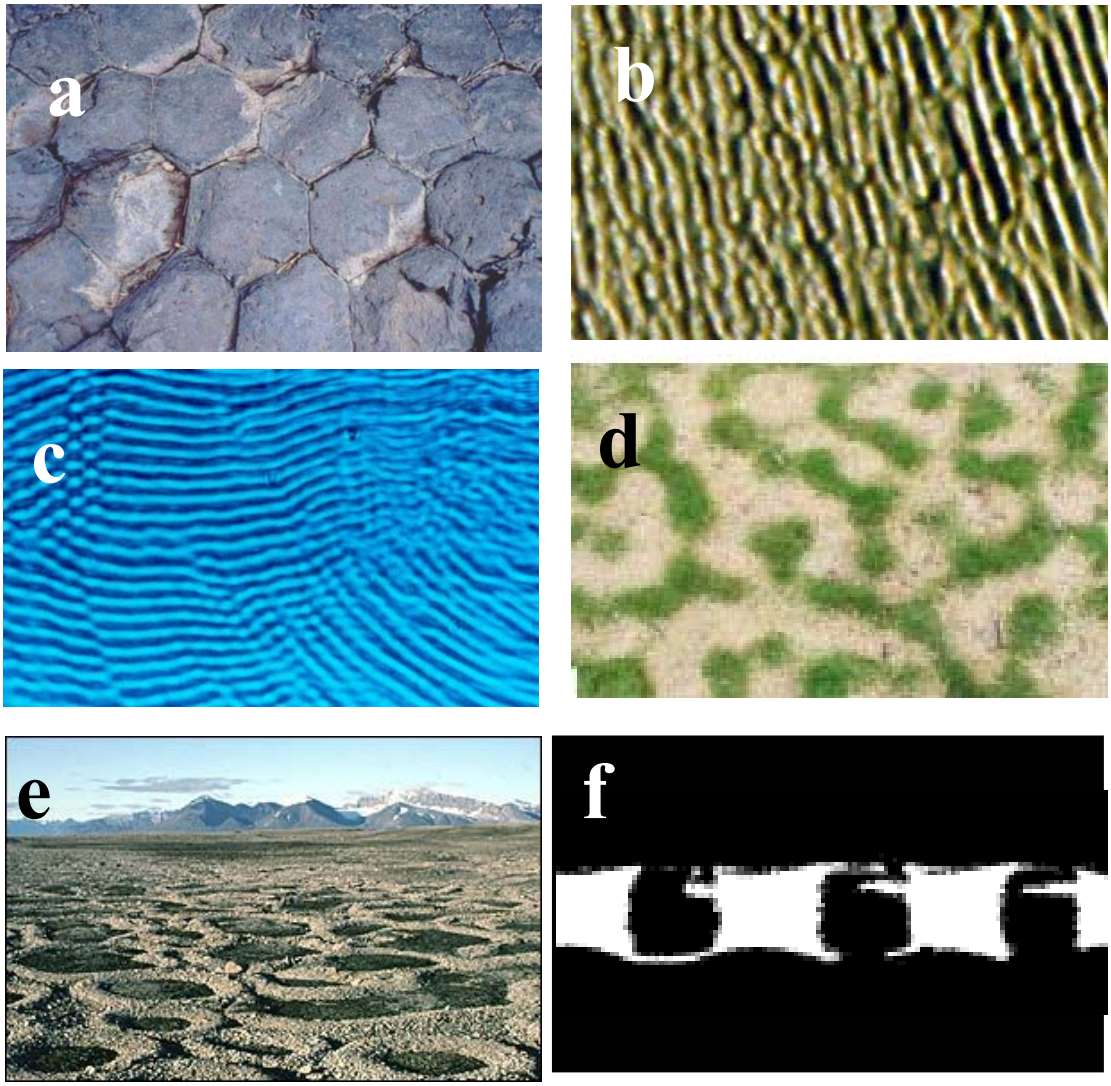


Figure 5: Pattern formation in various systems, a: Frozen Lava. b: Incoherent white light. c: Coherent light, d: Grass, e: Soil formation, f: Granular material. (source: a,c by the author, b by T. Schwartz in collaboration with the author [32], d by J. von Hardenberg [35], e by M. A. Kessler [34], f by O. Zik [36])

Alan Turing was the first to recognize that the formation of organized structures (evolving from a homogeneous state) can arise from the interplay between reaction and diffusion (morphogenesis) [33]. Such structures are universal and form in a variety of different systems. The common things for all the systems in which pattern

do form (from homogeneity) is the presence of at least two interacting fields (at least one of these fields is propagating); various example for such systems such as ground [34], vegetation [35], granular materials [36], incoherent light [37] and many other systems [38]. Some examples of pattern formation are shown in Figure 5.

In optics, almost all of the research on pattern formation was done with coherent light. Here, I will first explain pattern formation in a coherent optical system and then explain pattern formation in incoherent optical systems.

Let us start from the simple case of coherent pattern formation. When a plane wave of coherent light is propagating in a self-focusing medium, filamentation will develop as the beam propagates. A picture of such filamentation at the output face of a nonlinear ~5 *mm* long sample (taken with a regular still camera) is shown in Figure 5c. The distance here between filaments is about 40 μm.

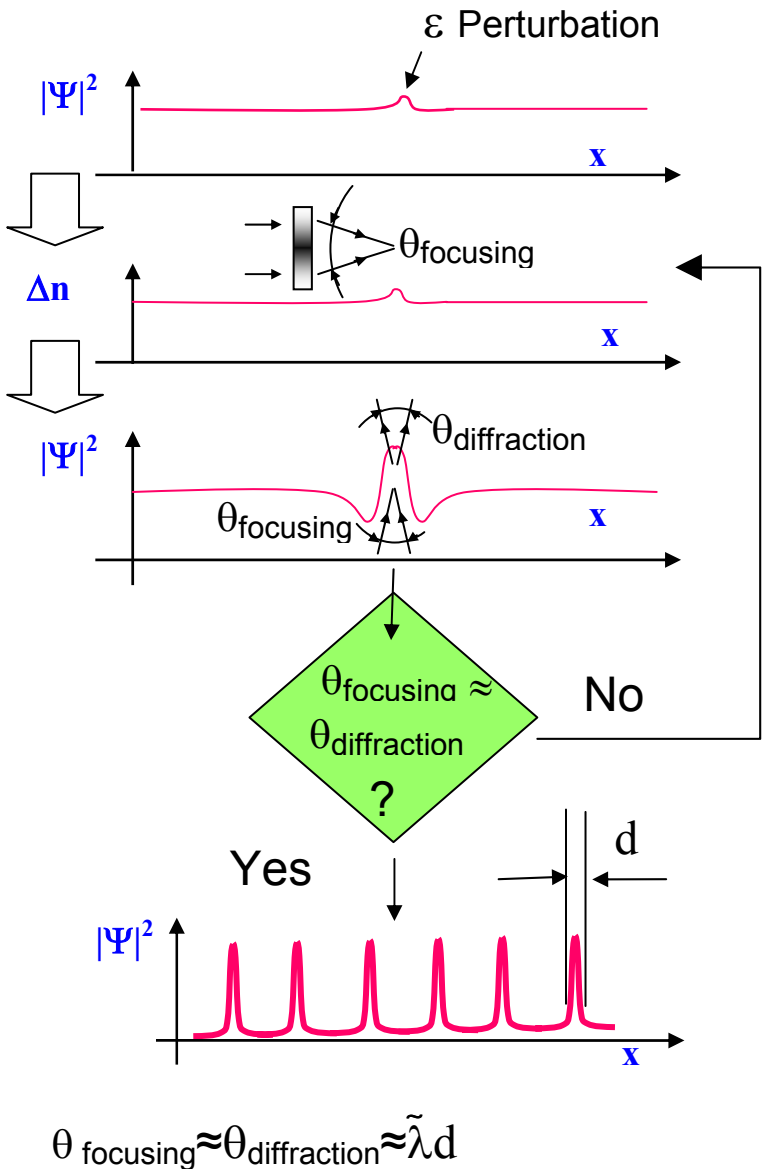


Figure 6: Coherent plane wave is transformed into a train of solitons via amplification of small perturbation.

This fragmentation occurs despite of the fact that a plane wave propagating in a homogeneous self-focusing medium is a solution of the nonlinear wave equation, and is invariant under propagation in z. This filamentation is the result of noise amplification. In nature, no matter how homogeneous the medium is, and how uniform the wave is, there will always be some noise present, such as a perturbation in the intensity or in the refraction index. The origin of this noise can come from many

sources such as imperfections in the medium or even quantum noise due to the discrete character of photons.

What is the physical reasoning behind this perturbation amplification mechanism? From the optical point of view, one can see in Figure 6 that a small intensity perturbation induces a local region of higher index of refraction. This higher index resembles a tiny graded index lens (GRIN lens). Consequently, light is focused from the neighboring areas into the perturbation center. The perturbation keeps growing along propagation for as long as the focusing angle of the induced lens is larger than the diffraction angle (which is proportional to the wavelength divided by the typical perturbation size). The perturbation grows into an isolated filament (pulse in space). We therefore call these pulses in space stripes. When a balance between focusing and diffraction is achieved, the stripe no longer evolves as it propagates.

After we have understood the physical origin of modulation instability, we can look for its mathematical formulation. Taking the (dimensionless) nonlinear Schrödinger equation for a self-focusing Kerr[§§] medium:

$$\frac{\partial \psi}{\partial z} = [\frac{i}{2}\frac{\partial^2 \psi}{\partial x^2} + i|\psi|^2]\psi \qquad \textbf{(15)}$$

Substitution of a perfect plane wave solution $\psi = A_0 \exp(i\varphi z)$ reveals that φ is equal to A_0^2. This means (as expected) that in a focusing medium that increases its index of refraction as intensity goes up, the only effect on a perfect plane wave will be its slowing down. The perfect plane wave solution is hence:

$$\psi = A_0 \exp(iA_0^2 z) \qquad \textbf{(16)}$$

Adding a small periodic perturbation with spatial frequency ν to this perfect plane wave reveals

[§§] Here we chose Kerr medium as a test case for several reasons: (1) There is a simple analytic solution to the problem. (2) A series expansion of (almost) any self-focusing nonlinearity in nature reveals that the first term in the Taylor series is Kerr-type and hence, to the lowest order (when the intensity perturbation is still small), every nonlinearity will behave like Kerr. (3) This nonlinearity exists in nature (in contrast with logarithmic nonlinearity for example).

$$\psi = \left\{A_0 + \varepsilon \exp(i\nu x + g\, z) + \varepsilon^* \exp(-i\nu x + g^* z)\right\} \exp(iA_0{}^2\, z) \qquad (17)$$

And substitution of this perturbed plane wave into the governing equation (15) reveals that the gain g as a function of the spatial frequency, ν, behaving as:

$$g = \sqrt{\nu^2 (A_0{}^2 - \nu^2 / 4)}\,. \qquad (18)$$

The dependence of this gain, g, on the spatial frequency of the periodic perturbation and on the intensity, $A_0{}^2$ is shown in Figure 7. In nonlinear medium, as the intensity goes up, the index change goes up; consequently, focusing is larger and can support the stronged diffraction of higher spatial frequency, so that the specific frequency that has maximum gain moves to higher values. Another interesting feature is the cutoff frequency. No matter what the intensity is, there is always a frequency high enough so that its diffraction is strong enough to overcome nonlinearity. Above this cutoff, the gain is negative, so perturbation at such high spatial frequencies are suppressed along propagation.

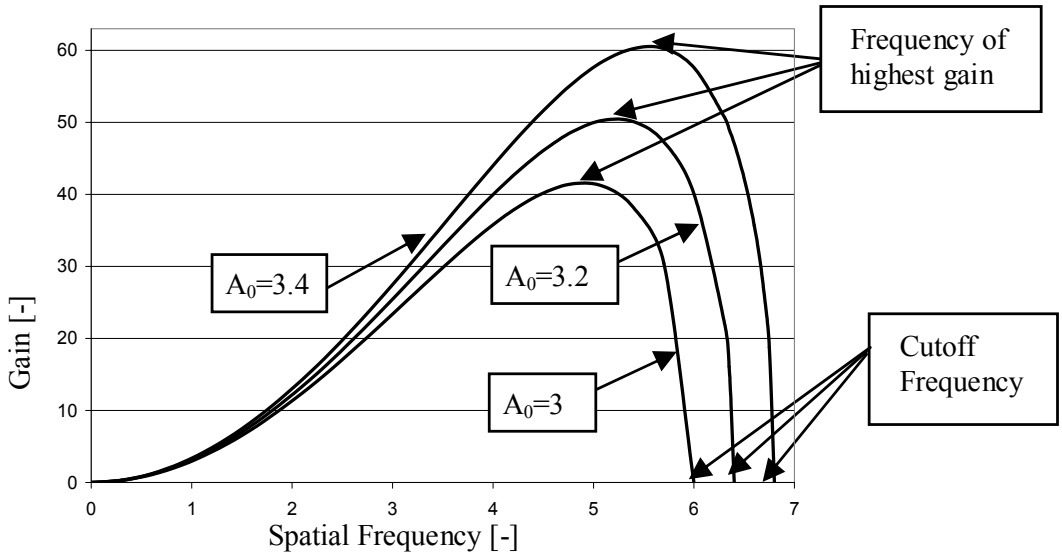


Figure 7: Gain as a function of spatial frequency for various values of intensity ($A_0{}^2$) in a Kerr self-focusing medium.

Until a few years ago, solitons and pattern formation were considered to be solely coherent phenomena. However, experimental observations of solitons made of partially spatially incoherent light [12] and of temporally and spatially incoherent (white) light [13] have opened the door to the observation of pattern formation with

partially spatially incoherent light [37] and of temporally and spatially incoherent (white) light [32]. To give a wider perspective, we may note that patterns in nature usually appear in systems in which there is no phase correlation between different places in space (see examples in Figure 5) . These types of systems are usually called classical systems.

Before moving on to pattern formation in incoherent systems, let us first discuss diffraction of both types of limits. Diffraction of coherent light is fundamentally different from diffraction of incoherent light [39]. The diffraction angle of incoherent light is proportional to the wavelength divided by the *correlation distance* and is independent of the size of the light beam (as long as correlation distance << beam width), while for coherent light the diffraction is, of course, equal to the wavelength divided by the *width of the beam*. This dissimilarity leads to an important difference: for coherent light, there is no threshold value for pattern formation and even the smallest nonlinearity is sufficient for a pattern to evolve, whereas for incoherent light, a pattern will form only if the nonlinearity is above some threshold value that depends on the degree of spatial correlation (coherence). This difference needs further explanation: (i) For a coherent periodic perturbation, there is no diffraction at all. One can understand this by viewing this periodic perturbation as an interference of one plane wave with another arranged at symmetric angles relative to the propagation direction. In this case, we will get a periodic perturbation which is invariant throughout propagation. In view of the fact that there is no diffraction at all for such a periodic perturbation, there is nothing to oppose self-focusing and even the slightest self-focusing nonlinearity is sufficient for patterns to form. For a small nonlinearity, the balancing diffraction will also be small and consequently the periodicity of the patterns will be large, but filamentation will always occur whenever a self-focusing nonlinearity exists, small as it may be (ii) For incoherent light, a periodic perturbation in the time-averaged intensity will diffract as the inverse of the *correlation distance*. Because of this, when we start from a linear medium and increase the nonlinearity, nothing will happen in the beginning since the nonlinearity is not strong enough to overcome incoherent diffraction. Small intensity perturbations will fade away. Increasing the nonlinearity further will take us to the threshold value, above which the nonlinearly-induced focusing overcomes incoherent diffraction and

the perturbation will grow into a pattern of high modulation. The sharp increase in modulation is a characteristic of the modulation instability threshold.

1.5. Summary

To summarize the introduction, I have presented solitons, diffraction, and spontaneous pattern formation. The reader may probably find in many textbooks statements claiming that the localized character of solitons (and pattern-formation) is in some manner surprising. But why is it so unexpected to see localized solitary waves? Here, I have tried to answer this question by examining the opposite question: Why is it so expected to see diffracting waves? As shown above, it would seem that wave spreading is inevitable and therefore irreducible, since it originates at a fundamental level from the uncertainty principle of quantum mechanics. However, surprising as it may be, diffraction can be fully compensated by nonlinearity, despite its fundamental origins.

Another important point is that solitons (and pattern formation) are universal in nature. Here, I tried to shed light on the versatile character of solitons by comparing their properties in the table of Figure 4 in such a manner that the reader can go along the column and compare various generic aspects of solitons.

The next 3 chapters are the core of this thesis as they contain my experimental observations on solitons composed of more than one component and on patterns in incoherent optical cavities.
In the next chapter, I will describe a new kind of solitons that consist of more than one component (usually referred to as composite solitons) and live in a 3-dimensional bulk. I will describe the first experimental observation of a dipole-type composite soliton: a multimode soliton composed of a single-hump component and a double-hump (dipole-type) component that are jointly self-trapped in two transverse dimensions, propagating as a single entity in the third dimension conserving its structure, power, linear momentum, and angular momentum. Then, I will describe a rotating propeller solitons consist of a dipole that rotates throughout propagation and

makes a double helix structure in space, jointly trapped with a rotating bell-shaped component. This propeller soliton is the first ever composite solitons carrying angular momentum.

In chapters 3 and 4, I will describe pattern formation in an incoherent cavity.

Chapter 3 is about patterns in a temporally incoherent (optical) cavity. In this chapter, I will show that patterns spontaneously form in temporally incoherent cavities above a specific threshold. I will also show that the bandwidth of the spatial-frequency spectrum of the pattern narrows as the feedback increases.

Chapter 4 is about patterns in a fully incoherent optical cavity. In this chapter, I will demonstrate experimentally that such fully incoherent cavities can also support spontaneous formation of pattern, yet they exhibit two distinct thresholds. The first modulation instability threshold when self-focusing overcomes incoherent diffraction, and the second cavity threshold when gain overcome cavity losses. These two different thresholds are experimentally distinguishable through the fact that the first threshold (the modulation instability threshold) is independent of the feedback, whereas the second threshold (the cavity threshold) is feedback-dependent.

The two last chapters are about suggestion for experimental observation of bright[***] vortex solitons, which were never observed experimentally before (chapter 5) and on other projects I have worked on, but did not play the lead role (chapter 6)

[***] Bright solitons are solitons for which the intensity is bell shaped in space, the intensity goes to zero far away from the bell center. On the contrary, dark solitons are dark holes immersed in a uniform distribution of light. For dark (bright) solitons, the dark hole (bell shaped intensity distribution) is invariant along propagation.

2. Composite solitons in 3D

An optical beam propagating in a self-focusing nonlinear medium can be self-trapped and forms a spatial soliton when it is guided in its own self-induced waveguide [40,41,42]. When the soliton consists of a single optical field (a scalar soliton), the induced waveguide can guide a single mode only (e.g. in Kerr media), or multiple modes (in a saturable nonlinearity such as photorefractives [43]). In general, however, a soliton can involve more than one optical field, in which case it is called a vector soliton. This happens when the field components jointly induce a waveguide and trap themselves in it by properly populating the guided (eigen-) modes [40] (see Figure 8). In the degenerate case, all the field components populate the fundamental mode and the total intensity structure of the soliton has a single-hump [44,45]. Vector solitons, however, can also form when the field components belong to different modes of their jointly induced waveguide, as predicted in the temporal [57,54 ,46] and spatial [47] domains, and observed in photorefractives [48]. Such solitons are called composite or multimode solitons and their intensity can exhibit a single-hump or a multi-hump structure [48]. Single-hump and double-hump multimode solitons which are trapped in one transverse dimension [(1+1)D] were found to be stable for a large range of parameters [48,49].

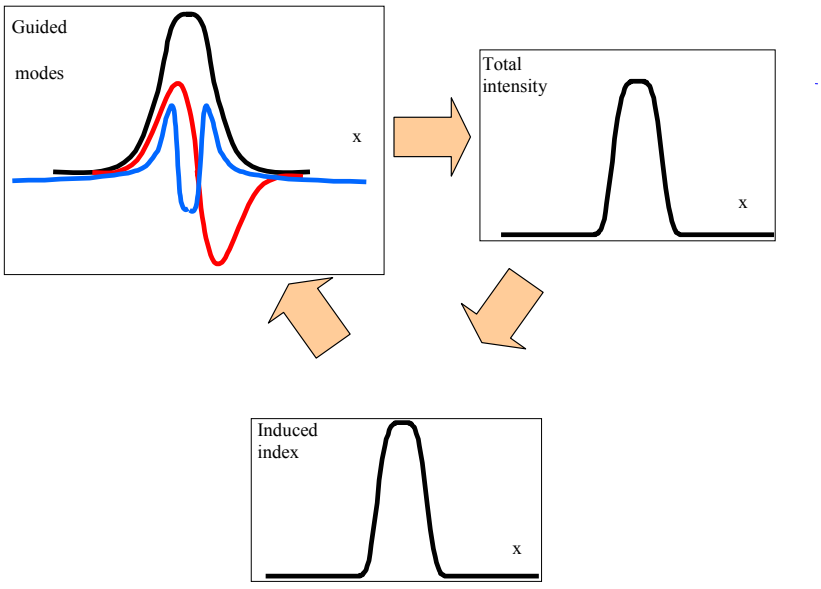


Figure 8: Self-consistency of vector soliton

This is the place to note that, in order to form vector or multimode solitons, the interference between the field components must not contribute to the nonlinear index change, Δn. Otherwise, the induced waveguide changes during propagation, and thus the field components self-trapped within it are not stationary. One method to satisfy this condition is to have the field components polarized orthogonal to one another [44,45]. Another method is to have them at two widely-spaced optical frequencies, so that the interference term is asynchronous with either of the components [50]. A third method relates to nonlinearities that have a non-instantaneous temporal response with a time constant τ. In this technique, the components are of the same wavelength and polarization but are mutually incoherent with respect to one another [51]. Thus, the phase of the interference term varies (randomly) in time much faster than the nonlinearity can respond, and therefore the contribution of the interference to Δn averages out. In my research, I have generated composite multi-mode solitons by employing the coherence based method [51] and the photorefractive screening nonlinearity [52, and appendix A].

Until the year 2000 all composite solitons studied, experimentally or theoretically, were trapped only in one transverse dimension (see Figure 9). This 2D configuration in which the soliton propagates along one dimension and traps in the other (transverse) dimension is, of course, limiting since the soliton must either propagate in a slab waveguide or be very large in the third dimension (in this case it will suffer modulation instability). Moreover, in (1+1) dimensional configuration, it is not possible to obtain vector solitons with angular momentum. It is also not possible to make a collision with angular momentum in a such a 2D geometry. These are only few of the spurs that prompted us to search for a 3D vector solitons that are trapped in two transverse dimensions while propagating in the third dimension [(2+1)D][†††].

[†††] For a discussion on the different kind of angular momentum that light can carry, see appendix B.

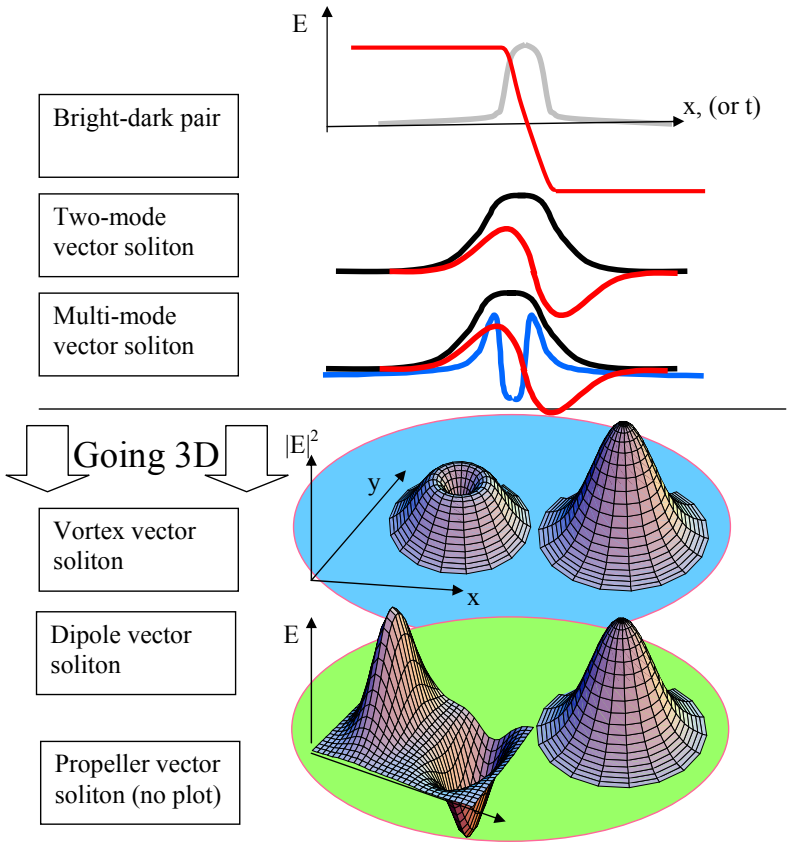


Figure 9: The history of composite solitons. A bright-dark pair predicted by Christodoulides [53] and Taratnik[54] in 1988. Experiment by Shalaby [55] in CS_2 in 1992 and by Chen [56] in Photorefractive in 1996. A soliton made of the first and the second modes. idea by Christodoulides [57] in 1988 and by Trillo [58] in 1988 (in the temporal domain). Experiment by Mitchell [59] in 1998 (with spatial solitons). Multi-mode solitons; experiment by Mitchell [60] 1998 (spatial). Vortex type composite solitons; idea by Musslimani [60] 2000. Dipole type composite soliton; Idea by Kivshar [61] in 2000. experiment by Carmon [68] in 2000 and Krolikowski [62] in 2000. Rotating Propeller Solitons: experiment by Carmon [63], 2001. Today: ideas for experiments with rotating quadropoles, hexapoles.

2.1. Going 3D, the way from (1+1)D to (2+1)D vector solitons

Following the progress with scalar solitons in saturable nonlinearities, including the demonstration of self-trapping in two transverse dimension [(2+1)D] [42], it was natural to seek (2+1)D composite solitons. Such multimode solitons were first predicted in cylindrically symmetric systems, in which at least one of the

components carries an orbital angular momentum [64,65] of a topological type. That is, if a round trip around our beam on the surface of equal phase gives an $l\lambda$ jump, (l must be an integer) we say that the beam has a topological charge of l. When the intensity distribution of the beam is axi-symmetric each photon carries an angular momentum equal to $\hbar l$.

At this point, it is important to note that the angular momentum carried by this soliton is an *orbital angular momentum* and has nothing to do with the *internal angular momentum* (associated with the spin that of each photon). A complete discussion on the differences between the two kinds of angular momentum is given in Appendix B.

Shortly after the vortex-type composite soliton was theoretically studied, a dipole-type composite soliton was proposed. For this latter case, one of the modes resembles a TEM_{01} Gaussian mode [66]. (An example of a dipole component is shown in the second row of Figure 10). However, until 2000, all experimentally observed multimode solitons were solely in a (1+1)D geometry [48,67].

In 2000 we [68], and afterwards others [69], demonstrated experimentally a composite (2+1)D soliton. The two-component soliton was made of a fundamental component (Figure 10, first row) co-propagating with a dipole component (Figure 10, second row), together forming a stationary dipole-type composite soliton.

In this chapter, I review our recent progress in generating (2+1)D composite dipole-type solitons. We start by describing a stationary dipole-type composite soliton, and continue by presenting our latest results on rotating propeller-type composite solitons. The fact that this composite soliton is trapped in two transverse dimensions facilitates new types of experiments that cannot be performed with solitons trapped in a single transverse dimension. In fact, the demonstration of rotating propeller solitons is an elegant example of experiments that can be made only in (2+1)D geometry. Furthermore, collision experiments between (2+1)D solitons, when the soliton trajectories do not lie in the same plane, are fundamentally a full 3D task. We therefore envision a whole range of experimental and theoretical work that will emerge from

these recent studies of composite (2+1)D solitons, especially those which carry angular momentum (e.g., the propeller solitons).

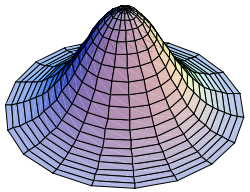
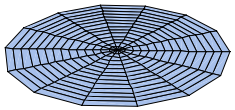
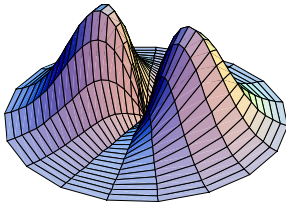
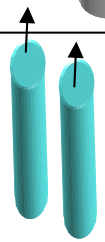
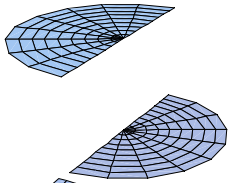
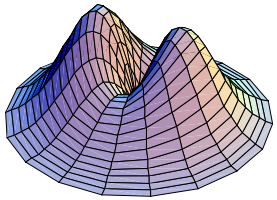
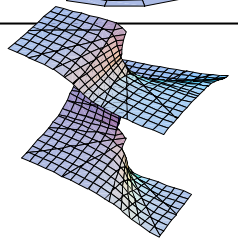
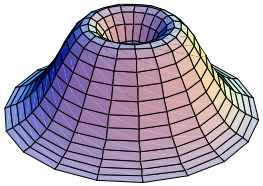
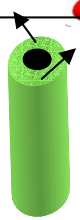
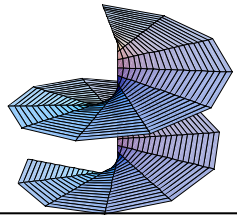
	Field Equation	Intensity Distribution	Intensity flow	Plane of equal phase
Fundamental	$A = u_1(r)\exp(i\mu_1 z)$			
Dipole	$A = u_2(r)\exp(i\mu_2 z)\cdot$ $\{\exp[i(\Omega z+\theta)]+\exp[i(\Omega z-\theta)]\}$			
Propeller	$A = u_3(r)\exp(i\mu_3 z)\cdot$ $\{R_+\exp[i(\Omega_+ z+\theta)]+R_-\exp[i(\Omega_- z-\theta)]\}$			
Vortex	$A = u_4(r)\exp(i\mu_4 z\pm\theta)$			

Figure 10: Field equation (slowly-varying field), intensity distribution, intensity flow, and equal-phase planes (in space) for various kinds of (2+1)D composite soliton components.

2.2. Dipole type composite soliton

In this section, I will present the first experimental observation of (2+1)D multimode (composite) solitons. A single-hump component and a double-hump (dipole-type) component are jointly self-trapped as a composite soliton in a biased photorefractive crystal.

In this bi-modal dipole-type composite soliton, we design the components such that the first is a fundamental (TEM_{00}-type) mode and the second is a dipole mode, exhibiting a TEM_{01}–like structure.

In our experimental system (Figure 11) we expand and collimate an Ar^+ laser beam at 488 nm, and split it into ordinary, -o-, and extraordinary, -e-, polarized beams using a polarizing beam splitter. The -o- and -e- beams are polarized perpendicular and parallel, respectively, to the optical or c-axis of the crystal. The -o- beam is used as a background illumination covering the crystal uniformly, as necessary for screening solitons [70,71]. The -e- beam is then split in two to create the first (fundamental) and the second (dipole) components. The dipole component is created by inserting a thin piece of glass in half of the beam. The glass is rotated until it induces a $(2m+1)\pi$ phase shift (m=integer) between the two halves of the beam, thus forming a dipole (TEM_{01} like) mode. The first component is made incoherent with the dipole by having the optical path difference between the beams (1 m) exceed the coherence length of the laser (10 cm). Since the phase between these modes is varying much faster than the response time of the crystal (τ » 1 sec), no stationary interference pattern forms on the time scale of the crystal's response [51,52]. The two modes are combined (with a beam splitter) and focused on the input face of a 13 mm long $Sr_{0.6}Ba_{0.4}Nb_2O_6$ crystal. The input and output faces of the crystal are imaged onto a CCD camera. To view the individual components, we block one component and sample the other in a time interval (~1 msec) much shorter than τ, so that Δn does not change within the observation window.

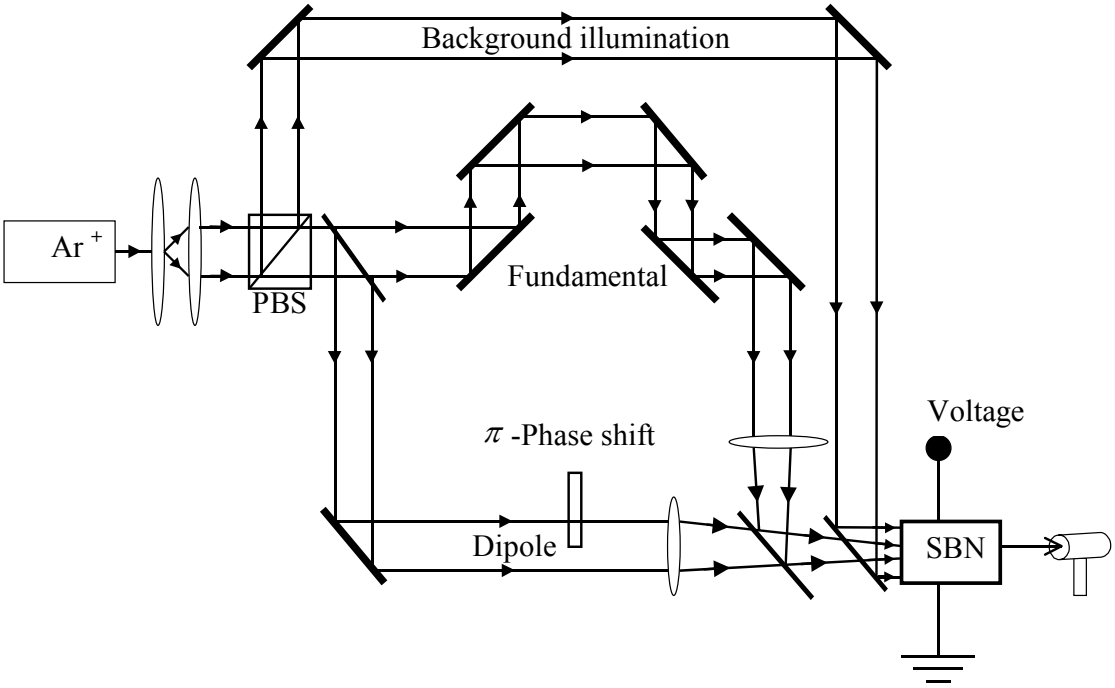


Figure 11: The experimental setup for generating a dipole-type composite soliton as demonstrated by the author [68].

In the first experiments (Figure 12), we launch a 19 μm FWHM fundamental component, u, and a 20 μm (peak-to-peak separation) dipole, d. The peak intensity of the fundamental component (normalized to the background intensity) is $|u_o|^2 = 30$. The peak dipole intensity is $|d_{peak}|^2=10$. The intensities of $|u|^2$ (fundamental), $|d|^2$ (dipole), and $|u|^2+|d|^2$ (total intensity) at the input face of the crystal are shown in Fig. 2(a), (b) and (c), respectively. The normally-diffracting outputs after 13 mm propagation (~ 3 diffraction lengths) of $|u|^2$, $|d|^2$, $|u|^2+|d|^2$ are shown in (d), (e) and (f). When 4060 Volts/cm are applied along the crystalline c-axis direction, a composite soliton forms. The intensities at the output face of the crystal, $|u|^2$, $|d|^2$, $|u|^2+|d|^2$ are shown in (g), (h) and (i). This composite soliton is single-humped and looks exactly like the input. The two modes self-trap when they are launched together, forming a composite soliton. However, the stand-alone components do not trap on their own at the value of the nonlinearity that supports the composite soliton. The reason is the parameter range that supports a composite soliton: the existence range. Scalar solitons are governed by the soliton existence curve, that is, a soliton of a given peak intensity and width-to-wavelength ratio forms at a particular value of the maximum index change [72, 73, 74]. A two-mode soliton, however, has an existence range that in addition depends on the relative strength of the first to the second mode [48]. From the theory of [(1+1)D]

multimode solitons [48], we know that for a given intensity of the first mode there is a range of intensities of the second mode for which self trapping occurs. Intuitively, if the second mode (dipole) constituent is too large, then the composite soliton turns into two separate solitons that repel each other (due to their relative phase difference of π).

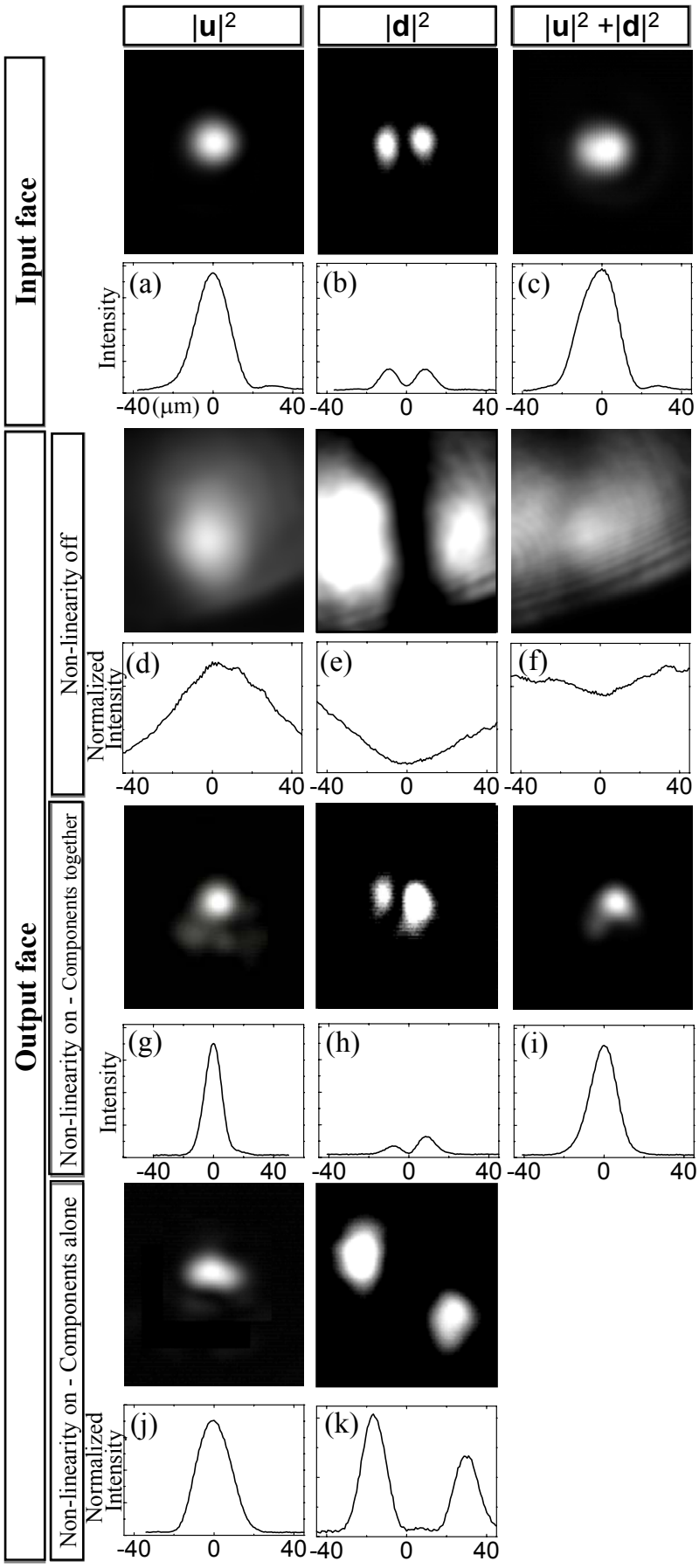


Figure 12: Multimode [(2+1)D] soliton (u is the fundamental component and d is the dipole) with $|u_o|^2$ =30, $|d_{peak}|^2$ = 10.

(a)-(c): Input intensities: (a) $|u|^2$, (b) $|d|^2$, (c) $|u|^2+|d|^2$.

(d)-(f) Diffracted output intensities with nonlinearity off: (d) $|u|^2$, (e) $|d|^2$, (f) $|u|^2+|d|^2$.

(g)-(i) The composite soliton output intensities at 4060 Volts/cm: (g) $|u|^2$, (h) $|d|^2$, (i) total intensity of the double-hump soliton, $|u|^2 + |d|^2$. The individual stand-alone components do not self-trap by themselves:

(j) $|u|^2$ by itself is distorted because the nonlinearity its too high for a scalar soliton, and (k) the stand-alone dipole forms two separate solitons which repel each other. The intensities are normalized to the peak intensity.

To see this experimentally, we launch the first and the second modes separately, and apply the same nonlinearity that gave rise to the composite soliton. For the stand-alone first mode, the nonlinearity is too high (because a composite soliton requires a higher nonlinearity than a scalar soliton of the same width [64]) and the output is distorted as shown in Figure 12(j). On the other hand, the stand-alone dipole forms two separate solitons that repel each other to an output separation of 75 μm (~4 times larger than as the input) as shown in Figure 12 (k). After showing a single hump dipole soliton, we move forward and change the ratio between the components so that the total intensity is double-humped and repeat the same procedure described above. The results are shown in Figure 13.

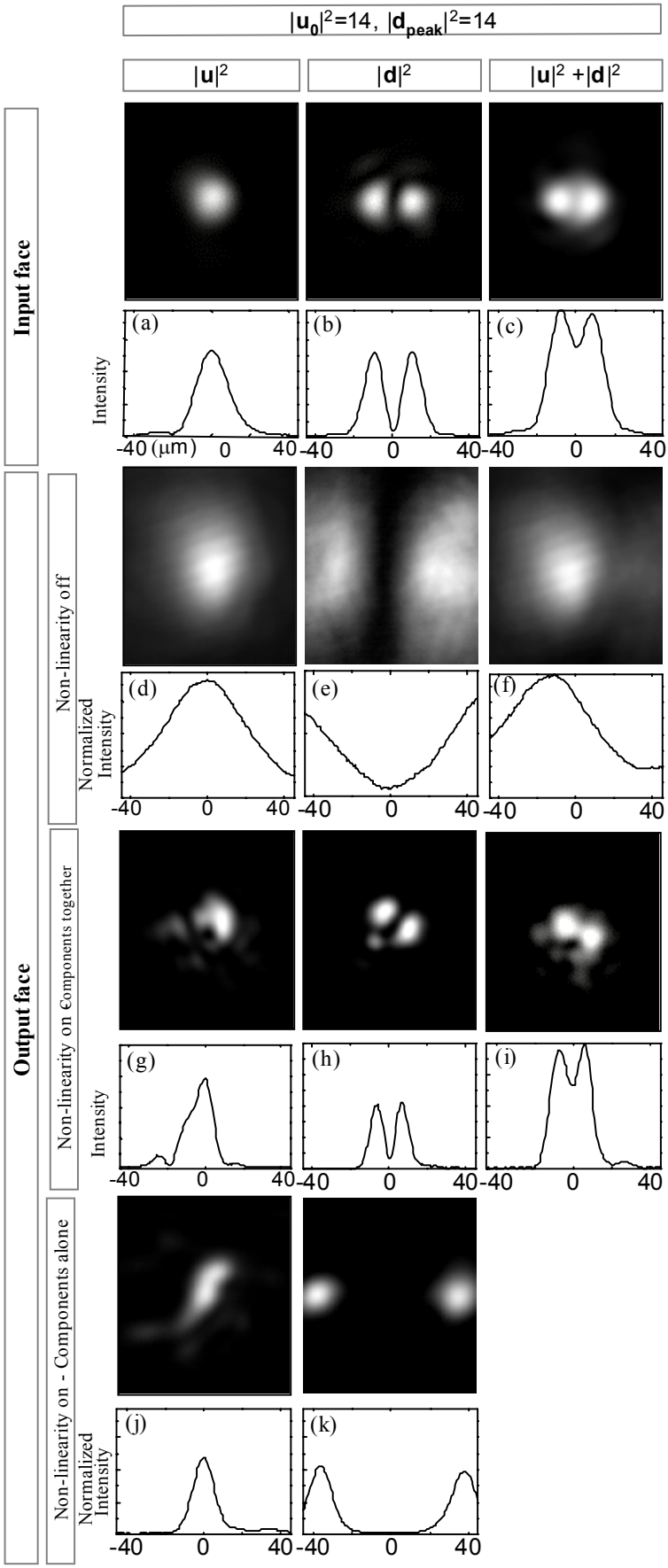


Figure 13: Experimental results for a dipole-type composite soliton. Double-hump multimode [(2+1)D] soliton with $|u_o|^2 = |d_{peak}|^2 = 14$.

(a)-(c): Input intensities: (a)$|u|^2$, (b) $|d|^2$, (c) $|u|^2+|d|^2$.

(d)-(f) Diffracted output intensities with nonlinearity off: (d) $|u|^2$, (e) $|d|^2$, (f) $|u|^2+|d|^2$.

(g)-(i) The composite soliton output intensities at 4230 Volts/cm: (g) $|u|^2$, (h) $|d|^2$, (i) total intensity of the double-hump soliton, $|u|^2+|d|^2$. The individual stand-alone components at 4230 Volts/cm do not self-trap by themselves:

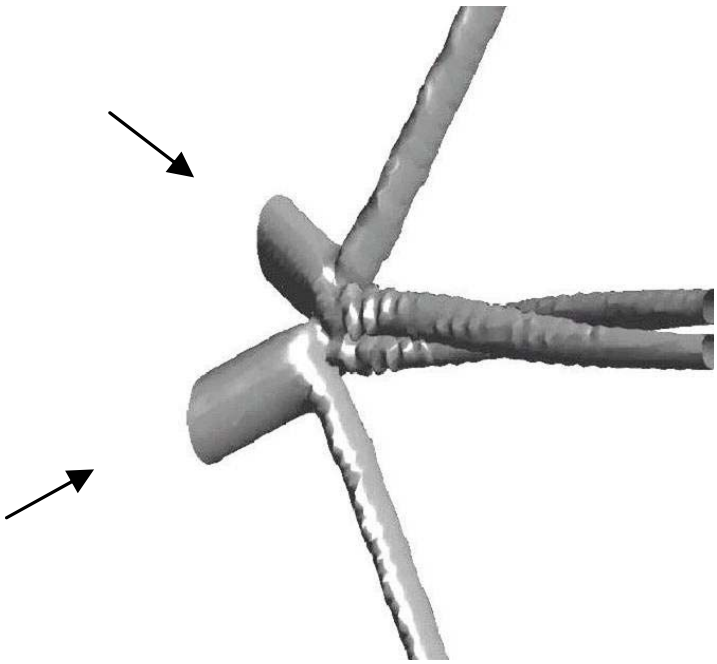
(j) $|u|^2$ by itself is distorted because the nonlinearity its too high for a scalar soliton, and (k) the stand-alone dipole forms two separate solitons which repel each other. The intensities in each figure are normalized to the peak intensity

To conclude this section, we have experimentally demonstrated the first composite (2+1)D soliton. The self-trapping of this multimode soliton opened up the way to create such dipole type composite solitons that rotate along propagation. Below are few hints that made us look for composite solitons for which one of the components is a rotating dipole:

(a) We saw in our experiments that vortex-type composite solitons disintegrate in the screening nonlinearity due to the anisotropy of this nonlinearity[†††]. The vortex mode disintegrated into two poles.

(b) While simulating collisions of two vortex-type composite solitons, we saw that the collision product resembles a rotating dipole (Figure 14). It was reasonable to conclude that, if this rotating dipole survives such an extreme event as a collision, it must be very stable.

Figure 14: Collision between two vortex-type composite solitons [64]. The collision product is a rotating dipole.



The combination of stability and rotation (of both the energy-structure and phase) made this rotating dipole soliton very attractive. We named this soliton propeller because its equal-phase plane resembled propeller blades and because of its rotation. The propeller solitons are presented in the next section.

[†††] See chapter 5.

2.3. Rotating propeller composite soliton

For a rotating propeller composite soliton, the intensity structure of both components rotates throughout propagation. In particular, the dipole mode has a double helix form, and its equal-phase planes are tilted like propeller blades. Hence, we name it a propeller soliton. *This is the first soliton whose intensity structure (not only its phase) rotates during propagation*.

This propeller soliton consists of two components: a bell-shaped component and dipole component. Both components rotate in unison as they propagate. The dimensionless equations for the slowly-varying beam-envelopes, The dimensionless equations for the slowly-varying beam-envelopes, Ψ_1 and Ψ_2 are

$$i(\partial\Psi_{1,2}/\partial\zeta)+\nabla_{\perp}^{2}\Psi_{1,2}+\Delta n(I)\Psi_{1,2}=0 \quad \textbf{(19)}$$

where ζ is the propagation distance[‡‡‡], $\nabla_{\perp}^{2}=\partial^{2}/\partial\xi^{2}+\partial^{2}/\partial\eta^{2}$, $\Delta n(I)=-1/(1+I)$ the refractive-index change, and $I=|\Psi_1|^2+|\Psi_2|^2$ the total intensity. Interference terms (e.g., $\Psi_1\Psi_2^*$) do not contribute to Δn, so the induced-waveguide does not vary during propagation. As discussed in previous sections, there are several methods of eliminating the contribution of such terms to Δn [9,68,69]. We employ the method in which Ψ_1 and Ψ_2 are mutually-incoherent, so the phase of $\Psi_1\Psi_2^*$ terms varies

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[‡‡‡] The transformation of the dimensionless equation into dimensional units is through $\Delta n(I)=-\Delta n_0\frac{1}{1+I}$, ξ and η coordinates are scaled by $\frac{\lambda}{2\pi}\sqrt{\frac{n}{2\Delta n_0}}$ and ζ is scaled by $\frac{\lambda}{2\pi}\frac{n}{\Delta n_0}$. Typically, as in our experiments, $\Delta n_0/n$ is of the order of 2×10^{-4} for $10\,\mu m$ FWHM soliton at $0.5\,\mu m$.

randomly in time much faster than the nonlinearity can respond, and their contribution to Δn averages out [51]. We seek self-trapped solutions, including those rotating about ζ. Such solutions are of the form

$$\Psi_{1,2}(\rho,\theta,\zeta)=\psi_{1,2}(\rho,\theta,\zeta)\exp(i\mu_{1,2}\zeta) \quad \textbf{(20)}$$

where $\mu_{1,2}$ are the (real) propagation constants. The simplest solution is the radially-symmetric, scalar, stationary in ζ, soliton, for which $\psi_1=u_1(\rho)$ and $\psi_2=0$. A second case is the vortex-type composite-soliton [64,65] for which $\psi_1=u_1(\rho)$ and $\psi_2=u_2(\rho)\exp(im\theta)$, where m (integer) is the orbital charge. A third case is the dipole-type composite soliton $\psi_1=u_1(\rho,\theta)$ and $\psi_2=u_2(\rho,\theta)$ [68]. For these three cases, the intensity structure is stationary in ζ. Here we are interested in propeller–type composite-solitons: dipole solitons that rotate about ζ. Thus, neither ψ_1 nor ψ_2 have radial symmetry, and their intensity structure rotates while propagating. Such solitons can be found numerically, yet much insight can be obtained by looking into the limit $|\psi_1|^2>>|\psi_2|^2$, for which ψ_1,ψ_2 can be expanded as $\psi_1(\rho,\theta,Z)=\psi_1^{(0)}+\varepsilon^2\psi_1^{(2)}+\ldots$ and $\psi_2(\rho,\theta,Z)=\varepsilon\psi_2^{(1)}+\varepsilon^3\psi_2^{(3)}+\ldots$ where ε is the smallness parameter and $Z=\varepsilon^2\zeta$. In this case, $\psi_1^{(0)}=u_1(\rho)$, coinciding with the scalar case, and $\psi_2^{(1)}(\rho,\theta,Z)$ satisfies the eigenvalue equation

$$\nabla_\perp^2\psi_2^{(1)}-\mu_2\psi_2^{(1)}+\Delta n(|u_1(\rho)|^2)\psi_2^{(1)}=0 \quad \textbf{(21)}$$

The solution is $\psi_2^{(1)}=u_2(\rho)[C_+(Z)\exp(i\theta)+C_-(Z)\exp(-i\theta)]$, where $C_\pm(Z)$ are found through the second- and third-order equations in ε and satisfy $idC_\pm/dZ=-C_\pm(a|C_\pm|^2+b|C_\mp|^2)$, where a and b are constants. Defining $C_\pm(Z)=R_\pm\exp(i\phi_\pm)$ yields $\partial\phi_\pm/\partial Z=(aR_\pm^2+bR_\mp^2)\equiv\Omega_\pm$, where $R_\pm$ and $\Omega_\pm$ are real constants. Hence, $C_\pm(Z)=R_\pm\exp(i\Omega_\pm Z)$ and

$$\psi_2^{(1)} = u_2(\rho)[R_+ \exp i(\Omega_+ Z + \theta) + R_- \exp i(\Omega_- Z - \theta)] \quad (22)$$

This family of solutions includes the $m = 1$ vortex-type composite soliton [64,65], recovered for $R_- = 0$, $\psi_2^{(1)} = u_2(\rho)[R_+ \exp i(\Omega_+ Z + \theta)]$, whose phase rotates, but whose intensity, $|\psi_2^{(1)}|^2$, has radial symmetry and displays no rotation. A different member of this family is the non-rotating dipole-type composite soliton [75] $\psi_2^{(1)} = u_2(\rho)[R \exp i(\Omega Z)\cos(\theta)]$, recovered when $R_\pm = R/2$, $\Omega_\pm = \Omega$. This radially-asymmetric dipole does not rotate, not even in its phase. Except for these particular cases, all other solutions described by Eq.(22)are radially-asymmetric and both their intensity structure and their phase rotate during propagation. These are the propeller-type solutions. We emphasize that the rotation rates of the intensity and the phase are different: the intensity,

$$|\psi_2^{(1)}|^2 = u_2{}^2(\rho)\ \{R_+{}^2 + R_-{}^2 + 2R_+R_-\cos[(\Omega_+ - \Omega_-)Z + 2\theta]\}, \quad (23)$$

is rotating with an angular velocity $(\Omega_- - \Omega_+)/2$, whereas the phase of $\psi_2^{(1)}$ is rotating with a **different** angular velocity and in an opposite direction. Because the rotation rates of the intensity and the phase are different, there are no stationary solutions in **any** rotating frame. The rotation direction is determined by which is larger R_+ or R_-. The Poynting vector forms a helix during propagation, because the energy flow is in a direction perpendicular to the equal-phase planes.

To find a propeller-type composite soliton solution, one can choose the parameters determining the rotation rates rather freely (as long as $|\Omega_\pm| << |\mu_2|$) and use a 1D relaxation code to find $u_1(\rho)$, $u_2(\rho)$, μ_1 and μ_2 from by replacing $\Psi_{1,2}$ by $u_{1,2}(\rho)\exp(i\mu_{1,2}\zeta)$. This procedure holds as long as $|\psi_1|^2 >> |\psi_2|^2$. When the modal intensities are comparable, 2D relaxation codes are used to find $\Psi_{1,2}(\rho,\theta,\zeta)$. To illustrate the propeller-type soliton, we show [Figure 16: Calculated dynamics of the fundamental and the rotating dipole modes of a propeller soliton. Here $|\psi_{1(\rho=0)}|^2 = 4$ and $|\psi_2|^2_{peak} = 0.45$ in units of the saturation intensity. $R_+ = 0.9$, and $R_- = 0.15$. The slightly elliptic fundamental mode (top) is rotating in unison with the dipole (bottom).]

the propagation of a solution with $|\Psi_1(\rho=0)|^2$ 9 times the peak intensity of the dipole mode $|\Psi_2|^2_{peak}$. In this example, $R_+ = 0.9$, and $R_- = 0.15$, displaying an 8μm FWHM almost-circular bell-shaped Ψ_1, and a rotating dipole-type Ψ_2 with a 12μm peak-to-peak separation. The two modes rotate together, in a self-trapped fashion, at a rate of 9° per mm.

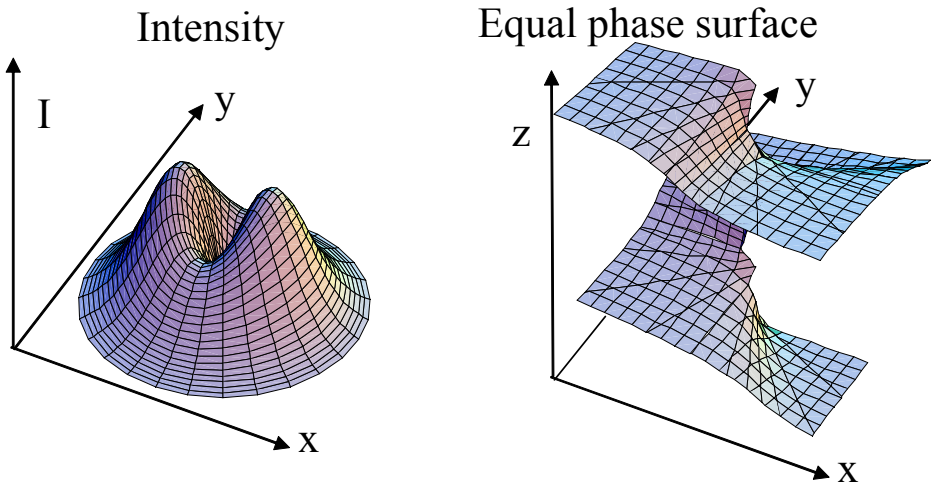


Figure 15: The rotating dipole (Ψ_2) - a typical intensity distribution (left) and the equal-phase surface as the soliton propagates (right).

We find $u_{1,2}(r)$ by a relaxation method and calculate the intensity flow by launching $\psi_{1,2}(z=0)$ into a beam propagation code. This propeller solution displays propagation (Figure 16) of an almost circular bell-shaped ψ_1, and a rotating dipole-type ψ_2. These two modes rotate together, in a self-trapped fashion, with a rate of roughly 9° per mm for a distance exceeding 30 propagation lengths (here, $|\psi_{1(\rho=0)}|^2 = 4$, and $|\psi_2|^2_{peak} = 0.45$ in units of the saturation intensity. $R_+ = 0.9$, and $R_- = 0.15$.).

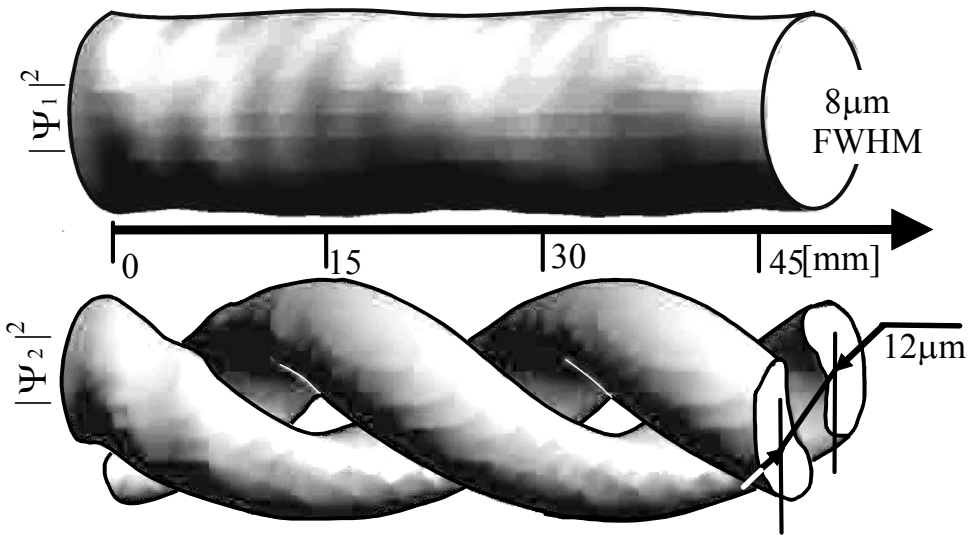


Figure 16: Calculated dynamics of the fundamental and the rotating dipole modes of a propeller soliton. Here $|\psi_{1(\rho=0)}|^2 = 4$ and $|\psi_2|^2_{peak} = 0.45$ in units of the saturation intensity. $R_+ = 0.9$, and $R_- = 0.15$. The slightly elliptic fundamental mode (top) is rotating in unison with the dipole (bottom).

As shown in equation (23) the rotation rate is exactly $\Omega_+ - \Omega_-$, Even so, usually we do not get the propagation constants (Ω_+, Ω_-) while solving equations (19) by using relaxation methods. In our first simulations, the rotation rate was very small. At that point, there was a need for a simple method for calculating the rotation rate. My own choice was to use a geometrical method to find the rotation rate by tracking the path of the Poynting vector in space. The Poynting vector is easy to calculate geometrically if one knows λ, R_+, R_-, (and the distance between the poles of the rotating dipole mode), since the Poynting vector is perpendicular to the equal-phase plane.

The Poynting vector at the intensity peak is a good indication of the direction in which light flows and hence can give an indication of the rotation. For large rotation rates, we need the Poynting vector to be tilted as much as possible relative to the z-axis. If we find the equal phase-plane, we will be able to draw the Poynting vector perpendicular to this equal phase-plane. Below, we find the equation for the equal-

phase plane, but first we should note that the electric field, Ψ, is a function of the slowly varying amplitude, ψ :

$$\Psi_2 = \psi_2 \exp[i(\omega t - kz)] \quad \textbf{(24)}$$

The places of equal phase, ϕ, in space obey:

$$\tan(\phi) = \frac{\mathrm{Im}[\Psi_2]}{\mathrm{Re}[\Psi_2]} = const \quad \textbf{(25)}$$

Substituting ψ_2 from equation (24) in equation (25) and assuming $\Omega << k$ reveals that the plane of constant phase $\phi = 0$ obeys:

$$R_+ \sin(\theta - kz) = R_- \sin(\theta + kz) \quad \textbf{(26)}$$

at t=0. The height (z) of this equal-phase plane as a function of the azimuthal angle (θ) is shown in Figure 17 (for various values of R_+ and R_-)

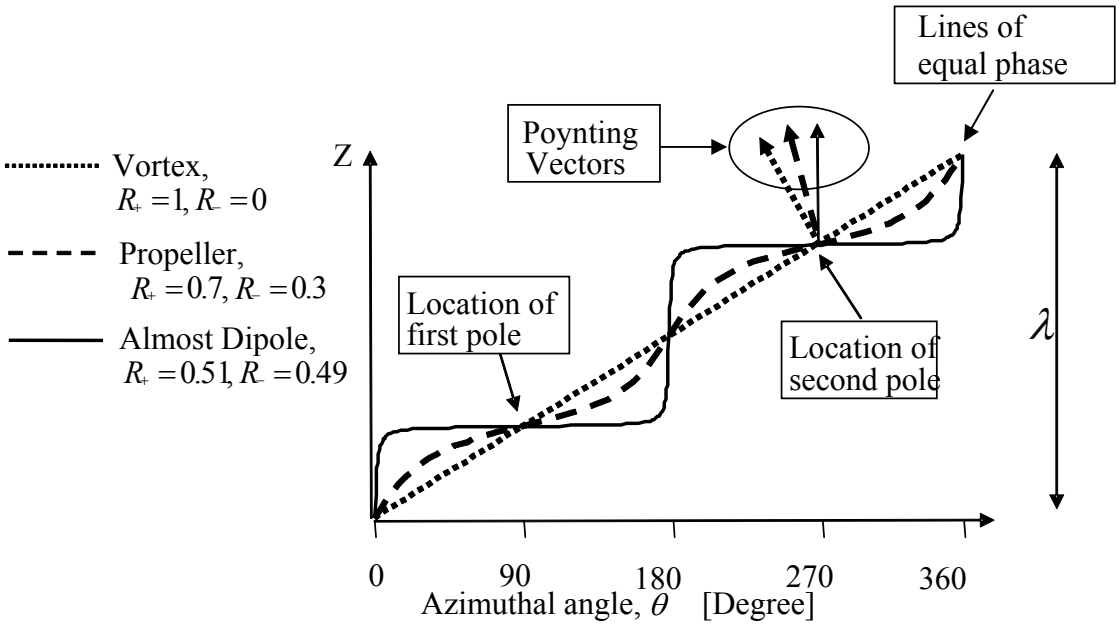


Figure 17: The height of the equal-phase plane (in z) as a function of the azimuthal angle. For the *vortex,* the slope is uniform and the Poynting vector is tilted relative to the z axis, so that the rotation is maximal. For a stationary *dipole*, the Poynting vector is parallel to z axis and hence it does not rotate. For the *propeller* (the rotating dipole), R_+ and R_+ should be different enough from each other so that the Poynting vector will be tilted relative to the z axis.

It is clear from this plot that if one chose R_+ to be almost equal to R_- (Figure 17, solid curve), the Poynting vector at the intensity peaks is almost perpendicular to the z-axis and in this case the propeller will exhibit almost no rotation. Changing R_+ to be different from R_- (Figure 17, dashed line), increased the rotation rate. By knowing the direction of the Poynting vector in space and the radius of the dipole (the distance between the poles divided by two), it is only a simple geometrical problem to calculate how much distance the propeller should pass in order to accomplish one rotation[§§§]. This geometrical calculation stands in good agreement with our simulations.

Experimentally, (see Figure 18) we generate the propeller solitons in photorefractive media, employing the screening nonlinearity [52] and the mutual incoherence method to generate multi-component solitons [51]. We expand and collimate a 488 nm laser beam and split it into Ψ_1 and Ψ_2. The Ψ_2 beam is first passed through a helical phase-mask (vortex mask), which multiplies the field by $\exp(i\theta)$, providing the angular momentum necessary for rotation. Then, to shape it as a dipole, the plane of the mask is reproduced (with a 4f system) and a metal wire is introduced so that it crosses the beam from side to side through its center. The components are combined (with a beam splitter) and imaged onto the input face of the crystal, so that the input face is a de-magnified image plane of the vortex crossed by the wire. This structure propagates perpendicular to the c-axis of a $Sr_{0.6}Ba_{0.4}Nb_2O_6$ (SBN) crystal, in the standard configuration [70,71]. The crystal is also illuminated uniformly with a background beam (to tune the degree of saturation of the nonlinearity), which is polarized orthogonally to the Ψ_1, Ψ_2 beams. The intensity distributions at the input and output faces are imaged onto a CCD camera. The modes can be inspected individually using the method described in [48,68].

[§§§] One should remember here that the wavelength in the medium is, of course, smaller than in air.

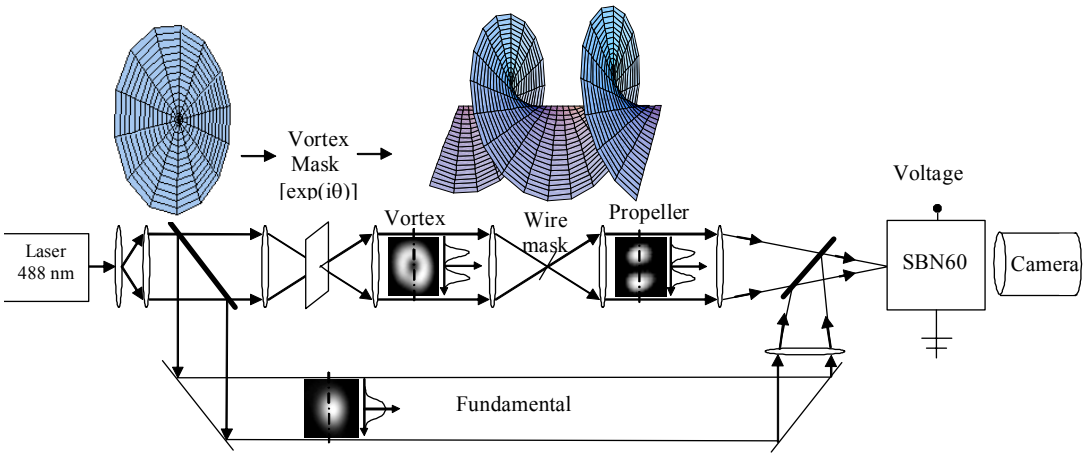


Figure 18: Propeller soliton – experimental system.

To monitor the rotation of the rotating dipole mode as a function of propagation distance, we cut a crystal in pieces of 1, 6 and 13 mm lengths and used them to sample the propagation dynamics at various distances. The results show rotation of about 19° per mm (Figure 19- a,b,c and d).

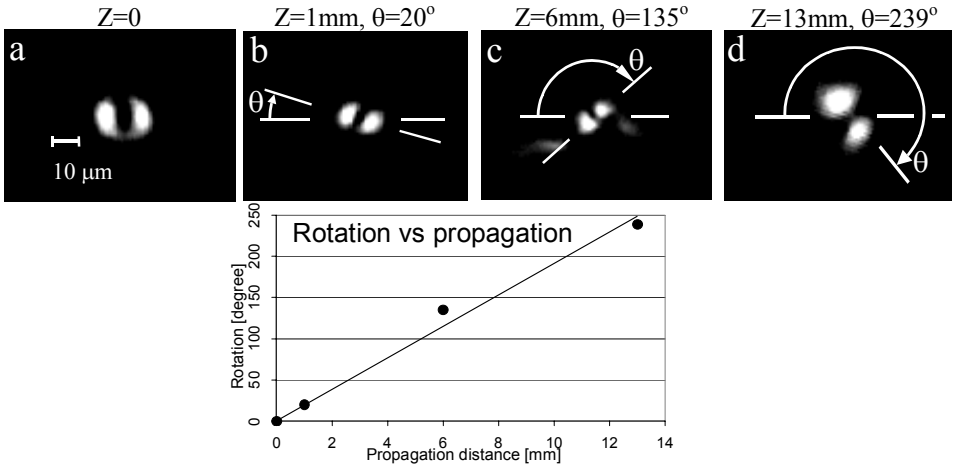


Figure 19: Photographs of the intensity distribution of the rotating-dipole component at the input face of a crystal (a), and after propagating 1, 6, and 13 mm (b,c, and d) in the nonlinear crystal while jointly trapped with a bell-shaped component (not shown here). The dipole rotation angle vs. propagation length is shown in the graph. Here the intensities are $|\Psi_1|^2 = 10$ and $|\Psi_2|^2 = 1$ in units of the saturation intensity.

To validate our observations, we performed a few tests. First, we changed the spin of the propeller from m= -1 to m=1. As expected, inverting the spin caused the

propeller to rotate in the opposite direction (Figure 20 b,c). In the second test, we turned off the fundamental component. Figure 20d shows that the dipole cannot self-trap all by itself: if the rotating-dipole component is launched alone, it splits into two solitons that repel and diverge away from one another. The rotation rate of this repulsive two-solitons system is reduced to about 1/6 of the rotation rate of the dipole component of the composite propeller soliton (compare Figure 20 d to c). This is because angular momentum is conserved, and thus the rotation rate decreases with the radius of rotation squared. In the third test, we turned off the nonlinearity in order to compare self-trapped propagation with linear diffraction. As expected, the dipole (and the fundamental) diffract (Figure 20 f) to a size much larger than the distance between the poles of the rotating-dipole component of the composite propeller-soliton (Figure 20 c).

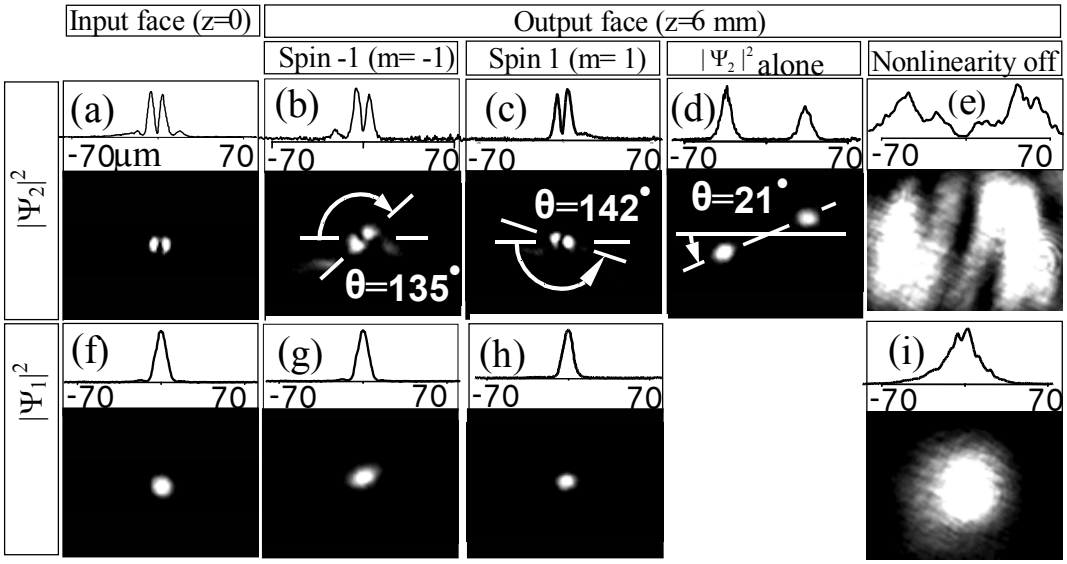


Figure 20: : Experimentally observed propagation of a propeller soliton with $|\Psi_{1\max}|^2=10, |\Psi_{2\max}|^2=1$ for 6 mm propagation (3 diffraction lengths). (a),(f): Input intensities of the modes. The separation between the dipole peaks in (a) is 11 μm , and the FWHM of the fundamental mode in (f) is 11 μm . (b),(g) Self-trapped output with 3300V/cm applied. The dipole mode rotates by 135^0 and its peak-to-peak separation is 13 μm . The fundamental mode is slightly elliptic. (c),(h): Same as (b),(g) with opposite spin $m=1$. As expected, inverting the topological charge leaves all dimensions and rotation angles similar to those of (b)-(g) but gives rotation in the opposite direction. (d): Output dipole mode at 3300V/cm when the fundamental mode is blocked. The dipole evolves into two solitons separated by a distance of 114 μm and overall rotation of 21^0.

In conclusion, we made the first stationary-dipole composite-soliton and then moved forward and demonstrated analytically, numerically, and experimentally a rotating propeller soliton: a composite soliton for which the intensity structure rotates throughout propagation. This was the first soliton for which the intensity structure rotates along propagation. And points out new directions for research, for example, following this work on propeller solitons, we investigated collisions between such solitons and found that two such rotating dipoles can form a new rotating dipole upon collision. In this process, each pole of the new dipoles comes from different source dipole [76] see Figure 21

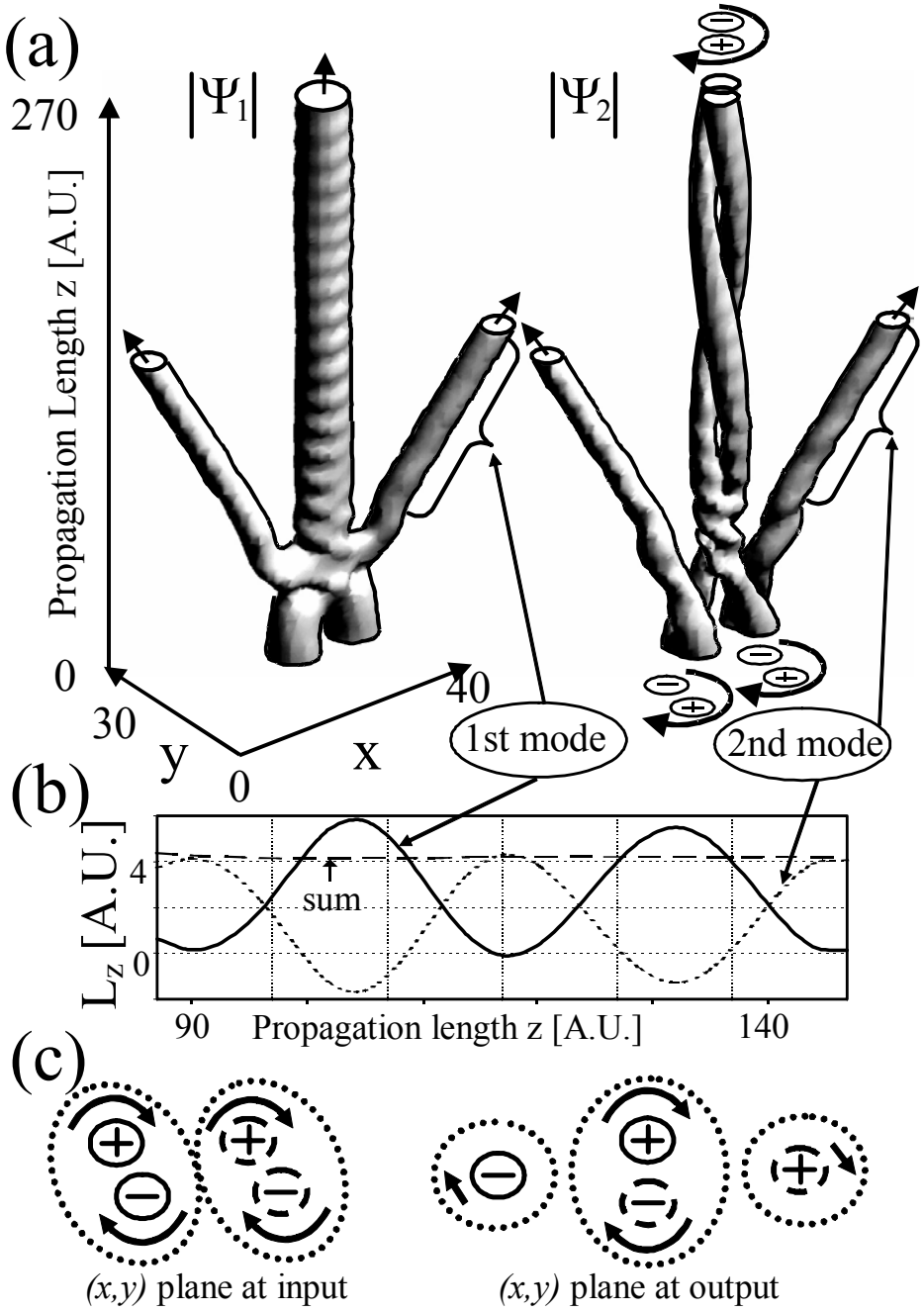


Figure 21: (a) Collision of two co-rotating, symmetrically arranged, propeller solitons. Propagation length of 100 units of z corresponds to 120 diffraction lengths. (b) Evolution of the angular momentum of the fundamental mode (solid), of the rotating dipole mode (dotted), and of the total intensity (dashed) in the right-most outgoing beam. Each component has an oscillating angular momentum, but their sum remains constant. The left-most outgoing beam exhibits identical behavior. (c) Illustration of the input and output configurations of the dipole mode.

Our work opened a new subfield in soliton science. Many follow-up papers have appeared in the past few years. Below is a partial list of the follow-up work by other groups:

Demonstrating mutual stabilization of a triple- and a double-humped transverse light structure, leading to the formation of a two-dimensional vector soliton without a stabilizing fundamental Gaussian shaped mode [77].

Transverse instability of vector solitons and generation of dipole arrays [78]

Investigation of dipole-mode vector solitons in anisotropic nonlocal self-focusing media [79]

Quadrupole vector solitons [80]

Internal oscillations and instability characteristics of (2+1)-dimensional composite solitons in a saturable nonlinear medium [81]

Symmetry-breaking instability of multimode vector solitons [82]

Stabilization and breakup of coupled dipole-mode beams in an anisotropic nonlinear medium [83]

Stability of spiraling solitary waves in Hamiltonian systems [84]

Spatial optical solitons and soliton clusters carrying angular momentum [85]

Rotating optical soliton clusters [86]

For future research, we envision the generation of rotating quadropoles, hexa-poles, etc.

3. Pattern formation in a temporally incoherent cavity

3.1. Introduction

First, we will define here a temporally incoherent cavity as a cavity for which the length of one cycle is longer than the temporal coherence length of the light, Consequently, interference effects between different cavity cycles (e.g. fringes) are averaged out. This chapter describes the formation of patterns in this type of cavity.

As explained in the introduction (chapter 1.4, page 18) *incoherent organized patterns* can spontaneously form from a homogeneous state. Organized patterns in an extended nonlinear medium can emerge from noise when an incoherent beam of uniform intensity is propagating in a self-focusing medium [37]. In the next two chapters we demonstrate what happens when the intensity distribution at the output face of a nonlinear medium is recycled back to the input face. In other words, we examined pattern formation when the nonlinear medium is suspended in a ring cavity. In this section, we examine pattern formation in ring cavity when the beam is spatially coherent but is temporally incoherent. That is to say, the field from different cycles circulating in the cavity are mutually incoherent with one another. From the experimental perspective, if we try to interfere beams from different cavity cycles, we see no stationary interference pattern as the fringes change faster than our measuring device (or our nonlinear medium) can track them (the typical time in which the fringes change is the velocity of light divided by the coherence length).

Pattern formation in optical cavities has been studied since lasers were discovered [87]. Spontaneous pattern formation occurs when particular spatial frequencies are selected from noise, because they experience higher gain, and, through the cavity feedback, stabilize to form a pattern. Examples include the formation of stripes, hexagons, spirals, vortices, shock waves, complex rings, and lattice-like features [for a review see Ref.[88]]. These phenomena were observed in a variety of materials: laser gain media, photorefractive crystals, thermal nonlinearities, quadratic

nonlinearities, and more [89]. What distinguish optical cavities from single-pass propagating systems is the existence of a typical cavity threshold for pattern formation and the existence of resonant frequencies. In incoherent cavities, pattern formation exhibits different features below and above threshold. In coherent cavity the patterns are also highly affected by the detuning between the frequency of the light beam and the nearest resonant frequency of the cavity. However, for all of the resonators studied previously, pattern formation has been a fully coherent process, that is, the coherence length of the laser light is much larger than the cavity length.

Here, we explore pattern formation in a different nonlinear cavity: a cavity in which the resonant frequencies are negligible**.** This is a passive cavity for which beams from different cycles are mutually incoherent with one another. In our cavity [for schematic sketch see Figure 22], beams from different cycles interact with one another through a nonlinear index change, Δn, that is a function of the sum of the intensities of the circulating beams but does not depend on interference terms between beams of different cycles. We find, theoretically and experimentally, that the patterns emerging in our incoherent cavity exhibit bandwidth narrowing in the spatial frequency domain as feedback is increased.

3.2. Theoretical background

We model our cavity when the feedback is small and the pattern is of a low modulation depth (visibility), so the entire dynamics (in temporal steady state) results from the interaction between the input beam, Ψ and the feedback beam Φ that went one cycle around the cavity. The index change Δn is a function of $I = <|\Psi + \Phi|^2> = <|\Psi|^2 + |\Phi|^2>$, where the time average $<>$ is taken over the response time of the medium, τ . Since in modulation instability studies, any saturable nonlinearity initially acts as a Kerr nonlinearity for the perturbation (with the nonlinear coefficient evaluated at the intensity of the input beam) [37] , we take $\Delta n \propto I$ (Kerr self-focusing). The dimensionless equations in one transverse dimension are

$$i\frac{\partial \Psi}{\partial z} + \frac{\partial^2 \Psi}{2\,\partial^2 x} + I\,\Psi = 0; \quad i\frac{\partial \Phi}{\partial z} + \frac{\partial^2 \Phi}{2\,\partial^2 x} + I\,\Phi = 0 \qquad . \tag{27}$$

We seek solutions of the form $\Psi(x,z,t)=\psi(x,z)\exp(i\mu z)$ and $\Phi(x,z,t)=\phi(x,z)\exp(i\nu z)\exp(i\alpha(t))$, where μ,ν are real propagation constants, $\alpha(t)$ is a stochastic phase that varies much faster than τ. Taking a fraction $\varepsilon(<<1)$ of $\Psi(x,z=L)$ from the crystal output and imaging it as feedback to the input $z=0$ implies $\varepsilon^2|\psi(x,z=L)|^2=|\phi(x,z=0)|^2$. Ψ starts as a plane wave ψ_0 and develops a periodic pattern due to MI. When the modulation depth of this pattern is low, ψ and ϕ each can be written as a sum of a plane wave and perturbation:

$$\Psi(x,z)=[\psi_0+\psi_1(x,z)]\exp(i\mu z) \qquad (28)$$
$$\Phi(x,z)=[\phi_0+\phi_1(x,z)]\exp(i\nu z)$$

where $|\psi_1|<<|\psi_0|, |\phi_1|<<|\phi_0|$. Substituting Ψ,Φ in Equation (28) yields $\mu=\nu={\psi_0}^2$. The boundary condition gives $\phi_0=\varepsilon\psi_0$. The linearized equation for ψ_1,ϕ_1 are:

$$i\frac{\partial\psi_1}{\partial z}+\frac{1}{2}\frac{\partial^2\psi_1}{\partial x^2}+{\psi_0}^2(\psi_1+{\psi_1}^*)+\phi_0(\phi_1+{\phi_1}^*)\psi_0=0 \qquad (29)$$

$$i\frac{\partial\phi_1}{\partial z}+\frac{1}{2}\frac{\partial^2\phi_1}{\partial x^2}=0 \qquad (30)$$

Here, $|\psi_0|<<|\phi_0|$, implies that ϕ propagates mostly linearly. Without loss of generality, we seek

$$\phi_1(x,z)=\tilde{\phi}_a\exp(if)\cos(qx)\exp(-iq^2z/2) \qquad (31)$$

$$\psi_1(x,z)=\tilde{\psi}_a\exp(if)\cos(qx)\exp(-iq^2z/2)+\tilde{\psi}_b\exp[i(qx+\Omega z)] \qquad (32)$$
$$+\tilde{\psi}_c\exp[-i(qx+\Omega^* z)],$$

where ~ denotes the real modal amplitude and f is real[****] . Substituting ψ_1 in equation (30) yields $\tilde{\psi}_a = \varepsilon\tilde{\phi}_a$ and $\Omega = \pm q\sqrt{(q^2/2) - (\psi_0{}^2)}$. For imaginary Ω, this pattern (with $2\pi/q$ periodicity) grows exponentially. We imposed $\varepsilon\Psi(x, z=l) = \Phi(x, z=0)\exp[i\alpha(t)]$. Because $\Delta n \propto < I >$, $\alpha(t)$ has no influence on the evolution of ψ . Using the boundary condition, the dependence of the spatial spectrum $\hat{\psi}_1$ on q is

$$\hat{\psi}_1(\nu, z=L) \propto \frac{\exp(|q|L\sqrt{\psi_0{}^2 - q^2/4})}{1-\varepsilon^2\exp(|q|L\sqrt{\psi_0{}^2 - q^2/4})}. \qquad (33)$$

For $\varepsilon = 0$, this relation converges to the known result of modulation instability in a system without feedback [90]. Equation (33) implies that increasing the feedback ε leads to bandwidth narrowing of the periodic pattern in the spatial frequency domain. As the feedback is increased and reaches a specific (threshold) value, there is a particular spatial frequency, $q_{\max}$, for which Equation (33) diverges. As the feedback approaches the threshold value, the bandwidth becomes very narrow as a particular q dominates. These features are apparent in Figure 23.a, which shows the calculated bandwidth of $|\psi_1|^2$ as a function of feedback.

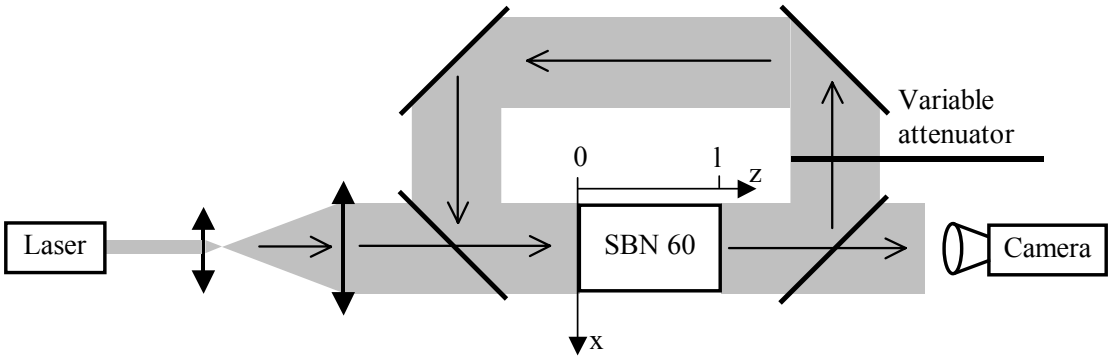


Figure 22: A schematic sketch of the temporally incoherent ring cavity.

[****] Going beyond lowest order analysis, the dominant modes are those of equal superposition of waves with q and -q, for which the visibility $|\psi_1|_{\max} / |\psi_0|$ is larger than that of nonequal superpositions.

3.3. Experimental results.

Experimentally, we use the photorefractive screening nonlinearity and a 488nm laser beam with a coherence length much smaller than the ring cavity length. The beam is launched into the 5.5 mm long crystal and covers its entire input face uniformly. Taking the beam exiting the crystal, we recycle a measured fraction of its intensity in a feedback loop of the ring cavity (see Figure 22). We use lenses in the feedback loop to image the output face of the crystal onto the input face (1:1 imaging). The lens diameter is large enough to pass all the spatial frequencies in the system. We monitor the patterns at the output face of the crystal (using a camera) while increasing the feedback (starting from zero feedback, and keeping all other parameters fixed, especially those which influence the nonlinearity). Figure 23 shows the bandwidth narrowing as feedback increases is very obvious in these pictures.

We compare the theoretical prediction and the experimental results by plotting the measured spatial bandwidth as a function of feedback intensity, as shown in Figure 23. The experimentally measured bandwidth is also shown in Figure 24. Comparing the prediction to the experimental results (Figure 23.a to Figure 23.b) shows that the theory provides an excellent description of the behavior of the experimental system all the way from zero feedback to the threshold limit. Very close to the threshold and above it, the assumptions of small perturbations made in the theory are no longer valid. Experimentally, however, we obtain the functional bandwidth-narrowing behavior above the threshold value as well, all the way to the narrowest bandwidth possible in our system.

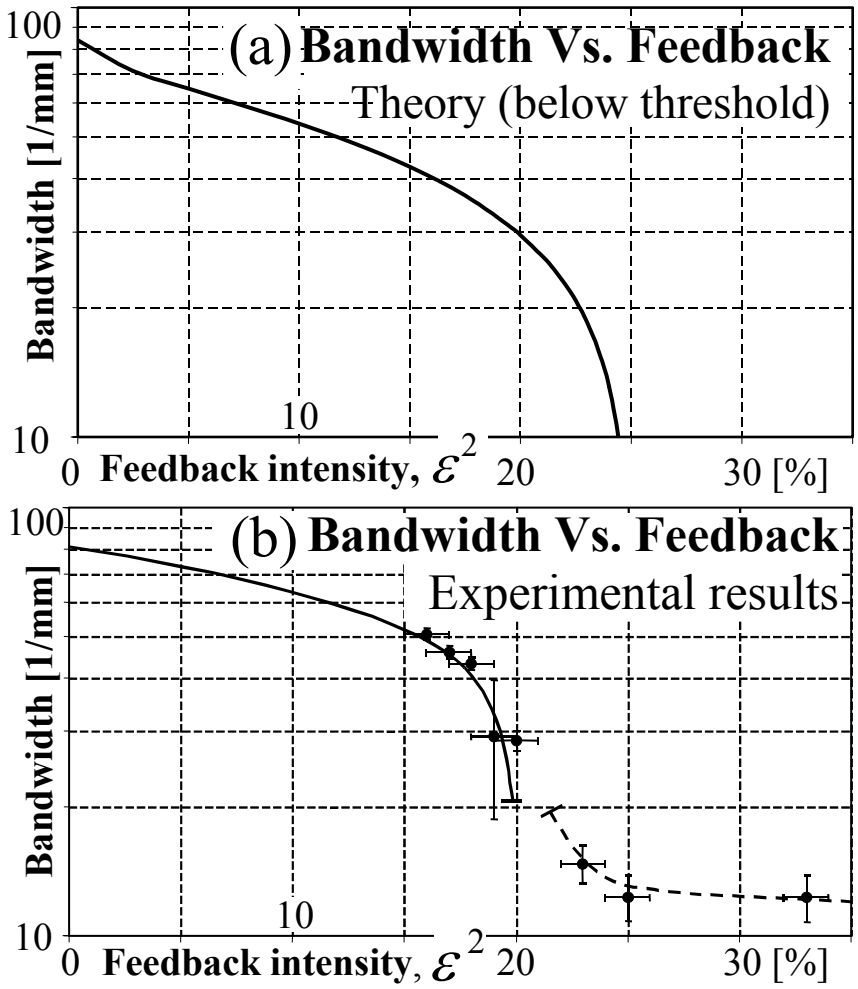


Figure 23: The bandwidth of the spatial frequency power spectrum (measured at FWHM) as a function of the feedback intensity. (a) Theoretical curve based on Eq. (33) and the parameters of our experiment. All of our experimental parameters have been measured independently in separate experiments, except for the absolute value of the nonlinear index change [the agreement between theory and experiments is excellent, below the threshold, where the assumptions made in the theory hold]. (b) Experimental results.

This work on the evolution of patterns in a nonlinear cavity in which the resonant frequencies are negligible [91] can bridge the gap between pattern formation in cavities and in single-mirror systems. The characteristic behavior resembles the linewidth narrowing in laser systems, and, in general, characterizes systems undergoing a phase transition. We expect that other phase transition phenomena

should exist in our cavity, and that they will have intriguing properties that are very different from those in coherent cavities.

The next step is to experimentally study cavities for which the light is not only temporally incoherent but also spatially incoherent. This cavity will be shown in the next chapter.

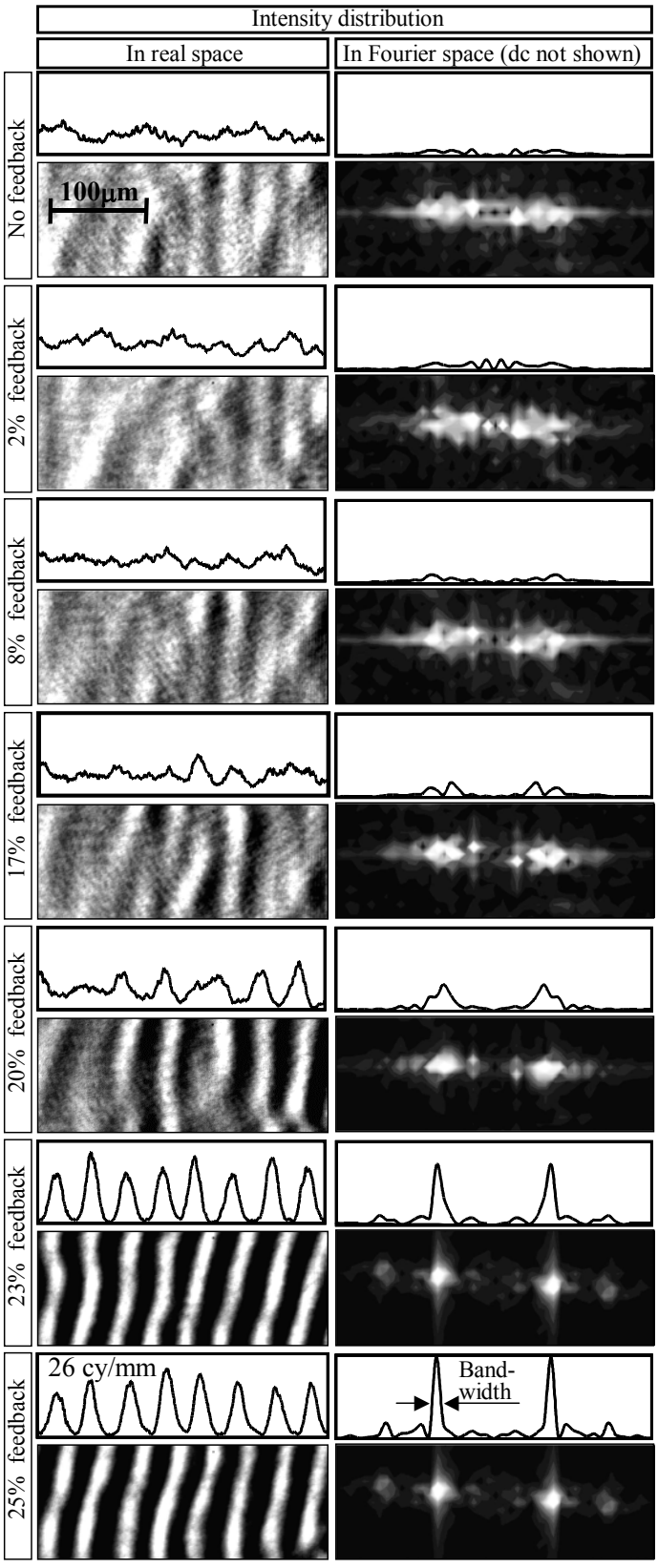


Figure 24: Experimental results, bandwidth narrowing with increasing feedback intensity. Shown are photographs of the intensity distribution in a specific region at the crystal output face (left), along with their calculated spatial (Fourier) power spectrum (right), at various feedback values. All measurements are taken without moving the crystal and without changing the nonlinearity. The bias field is 340 [V/cm], and the signal intensity is 2.1 times larger than the background intensity. The bandwidth narrowing is very obvious in these pictures, as the stripes become sharper and more regular with increasing feedback intensity, and the spatial power spectrum goes from multiple peaks (at no feedback) to a single isolated narrow peak at high feedback intensities.

4. Pattern formation in a temporally and spatially incoherent cavity

4.1. Introduction

First, we will define here a *spatially incoherent* cavity as a cavity for which the typical (transverse) correlation distance of the phase is smaller than the typical dimension of the patterns evolve in this cavity. As a result, different zones in the same stripe are not phase correlated. In this chapter, I will describe the spontaneous formation of patterns in a cavity in which the circulating light is both temporally (see 3.1) and spatially incoherent

In optics, the formation of patterns in nonlinear medium with feedback loop is a topic of extensive research [88]. The most common optical feedback loop is an optical-cavity designed to produce specific interference effects when excited by electromagnetic radiation. Consequently, the light circulating in such cavities is typically both spatially and temporally coherent. In the previous chapter, I described our work on the emergence of patterns in a temporally incoherent cavity. When feedback was increased we observed (i) formation of pattern, (ii) threshold with visibility jump right after (iii) bandwidth narrowing in the spatial frequency domain [91]. However, all nonlinear optical feedback systems studied thus far, including the one in the previous section, were fully spatially coherent. In this chapter, I described our work on pattern formation in a new system: a cavity in which the circulating light is both spatially and temporally incoherent.

4.2. Theoretical description of the incoherent cavity.

Our system is a passive ring cavity in which the cycle length is much larger than the coherence length (l_{coh}), so that there are no interference effects between the different cavity cycles. Furthermore, the transverse correlation distance (l_{cor}) of the light (in any given transverse plane) is only a few wavelengths. As a

result, there is no phase relation between neighboring stripes. A nonlinear medium of length L is suspended in the cavity. This nonlinear medium has a non-instantaneous response slower than the phase fluctuation. Hence, it responds only to the time-averaged intensity.

4.2.1. The instability threshold

From now on we assume that the correlation statistics is given by Lorenzian in Fourier space, and that the nonlinearity is

$$n_{(I)} = n_0 + \frac{\gamma I}{1 + I/I_s} \qquad \textbf{(34)}$$

where n_0 is the material index of refraction, γ is a parameter denotes the nonlinearity strength and I_s is a parameter denotes the degree of saturation. In the limit $I_s \to \infty$, the nonlinearity is of Kerr type and saturable otherwise. In our recent theoretical paper [see 92 and appendix c], we have shown that there exists a well-defined threshold at which the beam becomes modulationally unstable; if

$$\kappa I^{(0)}/(1-\varepsilon) > \theta^2 n_0 \qquad \textbf{(35)}$$

the beam is unstable. Here, $\kappa = \partial n(I)/\partial I\,|_{I=Itotal}$, $I^{(0)}$ is the intensity of the first cavity cycle at the input face of the nonlinear medium, ε denotes the cavity feedback parameter, and $\theta \propto \lambda / L_c$ is the degree of spatial incoherence.

We point out that the incoherent beam incident on the cavity is unstable if and only if the same beam with intensity enhanced by the factor $(1-\varepsilon)^{-1}$ is unstable in a single-pass system [39,37]. The factor $(1-\varepsilon)^{-1}$ simply reflects the increase of total intensity in the cavity due to feedback, $I_{total} = I^{(0)}/(1-\varepsilon)$. Thus, below and at the instability threshold, one cannot distinguish from the cavity system with incident intensity $I^{(0)}$, and the single-pass system with intensity $I^{(0)}/(1-\varepsilon)$. Physically, this equivalence is a consequence of the mutual incoherence between the fields of different cycles. The perturbations and the uniform parts of beams from different cycles do not interfere, but simply add up. This means that apart from the trivial enhancement of total intensity, the cavity boundary conditions do not affect the position of the MI

threshold. This is also confirmed by the fact that the position of the instability threshold, given by inequality [39,37], does not depend on the length of the sample L, which is embedded in boundary conditions. This result is in a sharp contrast to coherent cavities [93], where the instability process is influenced by the boundary conditions through the interference of fields from difference cycles.

4.2.2. The cavity threshold

In contrast to the instability threshold, the boundary conditions, through the interplay of nonlinear gain and cavity loss, induce a transition from the low to high modulation pattern. This second threshold occurs[††††] when the nonlinear gain overcomes loss in a single pass as

$$\varepsilon^{-1} < \exp(g(\alpha_{\max})L) . \qquad (36)$$

Here, L is the length of the nonlinear material, $\alpha_{\max}$ is the spatial frequency of the pattern and g is the nonlinear gain, $|g(\alpha)| = |\alpha| \left(\sqrt{\frac{\Delta n}{n_0}\left(1+\frac{I_0}{(1-\varepsilon)I_s}\right)^{-2}\frac{1}{(1-\varepsilon)}} - \theta \right)$.

We call this second threshold the cavity threshold since it is an inherent cavity feature. This cavity threshold is characterized by a sharp increase in the modulation depth.

In Figure 25, we present the calculated dependence of the modulation depth (of the pattern) by the nonlinearity for different values of feedback. Note that as the first threshold is feedback-independent, its location is also feedback-independent (see the circle). Further increasing the nonlinearity results in a sharp increase in the modulation depth of the patterns. The increase in the modulation depth characterizes the second cavity threshold (in which cavity gain overcomes losses). We see that the second cavity threshold is truly feedback dependent and appears in lower nonlinearities as feedback goes up.

[††††] See calculation in reference 92

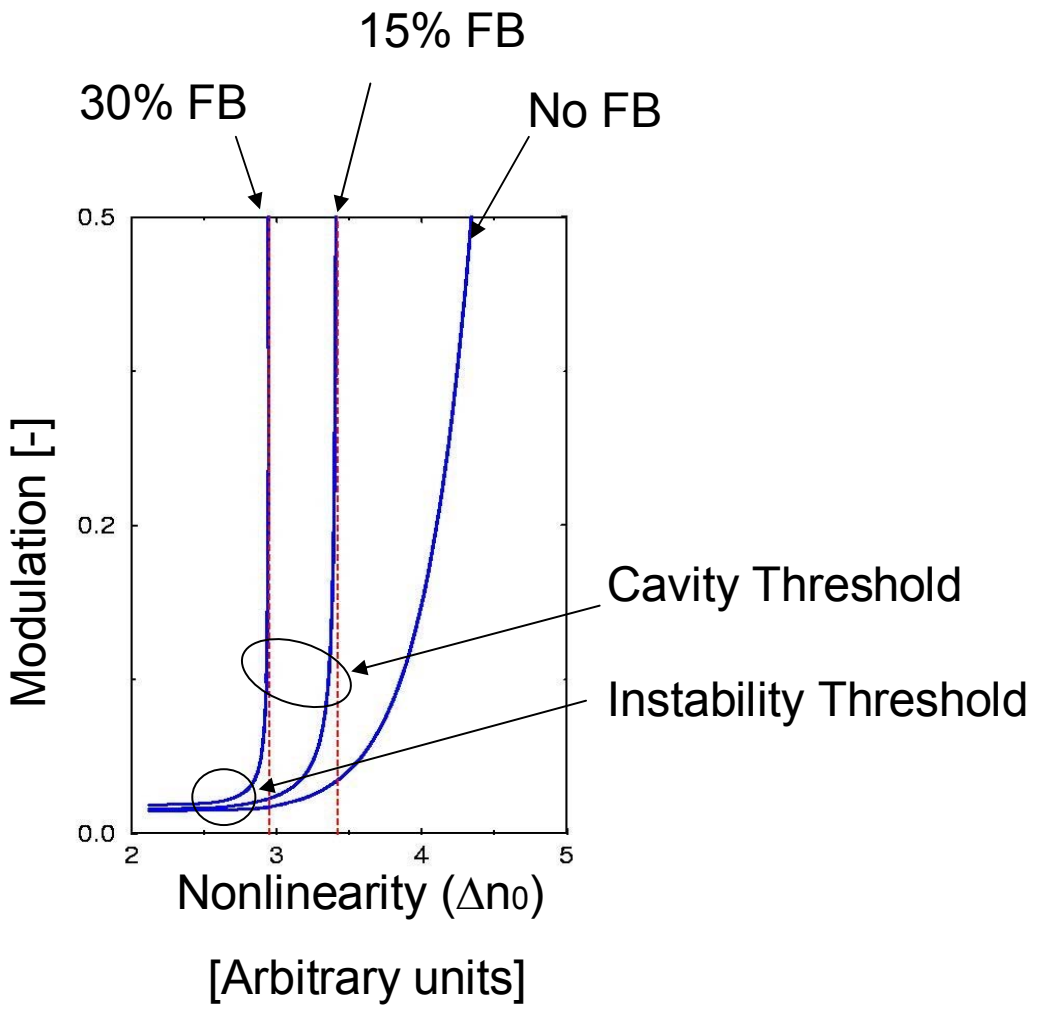


Figure 25: Plot of the calculated modulation (modulation depth of pattern) as function of the nonlinearity for 0%, 15% and 30% feedback. The modulation depth is shown from the instability threshold up to the cavity threshold (indicated by vertical dashed lines).

4.3. Experimental system and results

In our experiment [94,95] [Figure 26] we use a 488 nm Ar^+ laser beam with a coherence length much smaller than the cavity length so that there is no interference between different cycles in the cavity (as $L_{cavity} = 80cm$ and $L_{coh} \approx 8cm$). The beam is sent through a rotating diffuser in order to make it partially spatially incoherent. The beam then enters a cavity containing a nonlinear $Sr_{0.6}Ba_{0.4}Nb_2O_6$ (SBN) crystal, employing the screening nonlinearity [52]. We tune the strength of the nonlinearity by adjusting an external voltage which is applied to the crystal. We control the feedback

with a variable attenuator and monitor the intensity at the output face of the nonlinear medium with a CCD camera.

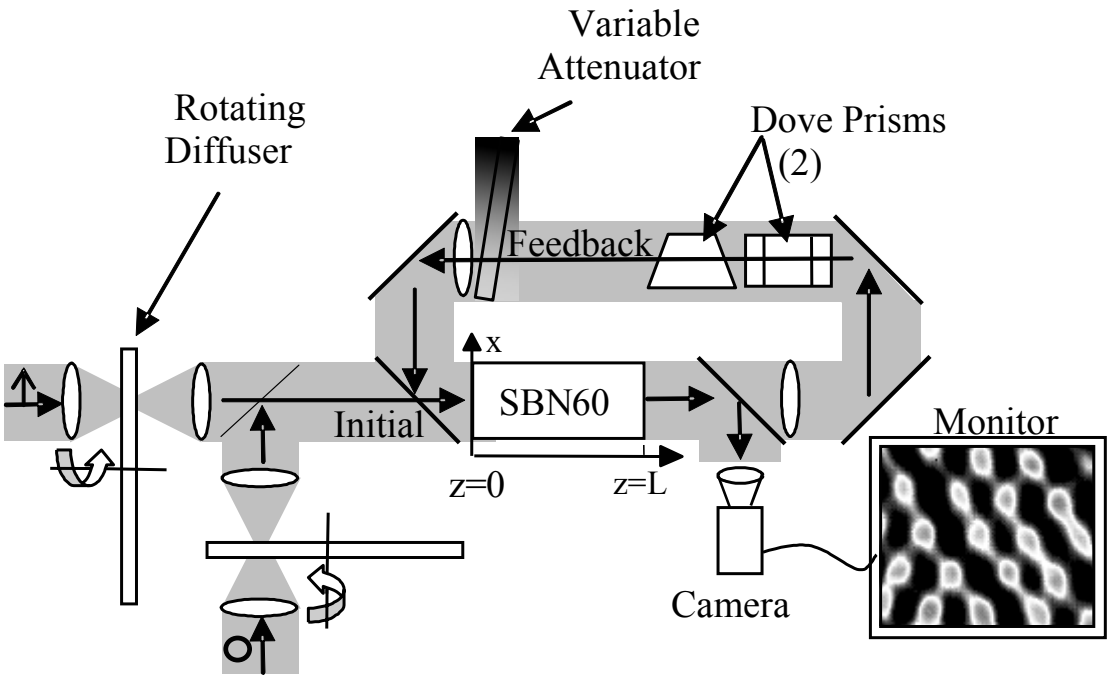


Figure 26: Experimental system.

We measure the modulation depth of the evolving pattern as a function of the nonlinearity.

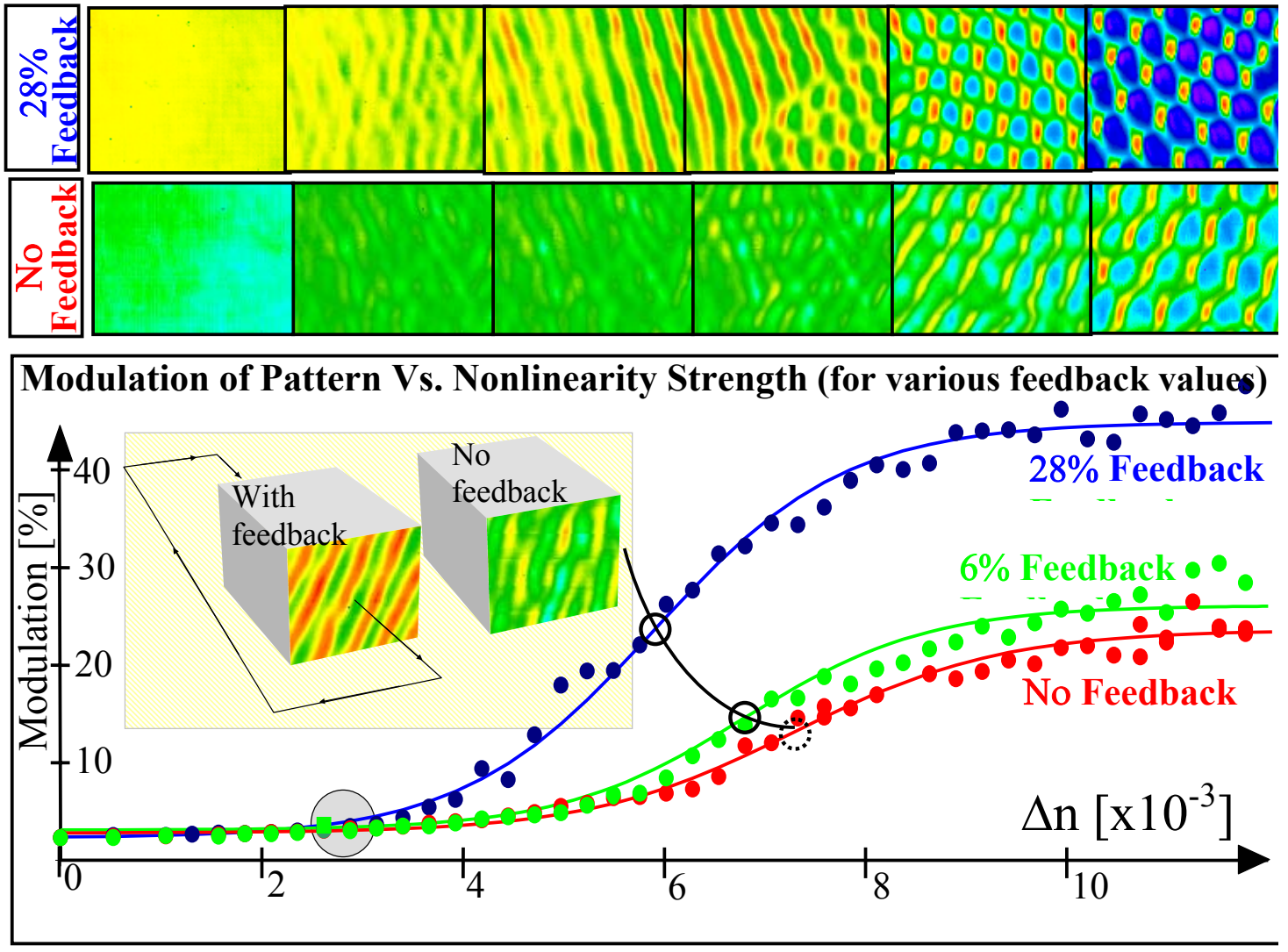


Figure 27: Experimental results. Above: Pictures of the intensity distribution at the output face of the nonlinear medium as nonlinearity increases when feedback is applied (top row) and with no feedback (bottom row). Below: plot of the modulation depth of pattern as a function of the nonlinearity for 0%, 6% and 28% feedback.

In the experiment, we monitor the modulation depth while changing the medium nonlinearity. We repeat the experiment for various values of feedback and see that as we start to increase nonlinearity (from zero), there is no immediate increase in the modulation depth. As we increase the nonlinearity further, bring us to the point in which nonlinearity compensates for incoherent diffraction; this is the point when the modulation depth start to increase as the nonlinearity becomes stronger. The increase in the modulation depth is much faster for large values of feedback. However, the modulation increase is not sharp as the one in the simulation (Figure 25) and as a result of that it is difficult to distinguish between the two threshold predicted by theory.

Theoretical analysis reveals that, when the noise levels are very low (and feedback is present) the increase in the modulation depth as a function of the nonlinear index change is very sharp. Hence, where the noise level is low, it is easy to distinguish the almost vertical modulation increase (when feedback is present) from the shallow increase of the no feedback case. For a large noise, however, the modulation increase does not approach a vertical line. In our experimental system, the noise is large and hence the modulation increase is not as sharp as in the low noise simulation. In this case, one can take the point of sharper modulation increase (the inflection point) as an indicator for the second threshold. This approach gives the correct cavity threshold point, but is nevertheless problematic on one issue; in the zero feedback case it also yield an inflection point where there should not be any cavity threshold present. Therefore, the best way to see the cavity threshold is to decrease the noise levels in the systems so that the sharp, almost-vertical, increase in the presence of a feedback will be clearly distinguished from the shallow increase when the feedback is off.

In conclusion for this chapter, we have made the first experimental observation of pattern formation in any incoherent optical cavity, that is, a feedback system in which the cycles are phase independent from each other and the beam itself is spatially incoherent. This work bridges the gap between pattern formation phenomena in correlated (coherent) systems and uncorrelated (incoherent) systems, in

analogy with the difference between the instabilities in phase-dependent (quantum) and phase-independent (classical) systems.

5. A suggestion for further research: an elasto-thermal rotating-ring soliton, proof of concept.

Bright optical vortices are structures taking the form of dark holes immersed in the center of a light ring; the phase around the vortex presents an integer number *l* of windings, which is usually called the topological charge of the vortex.

Bright vortices usually disintegrate in a medium of anisotropic nonlinearity of the refraction index. Many groups in the world focused their effort on analyzing the disintegration of composite solitons that one of their components is a bright vortex [79, 83].

Efforts were also made searching for a special nonlinearity that can support such a single component (scalar) bright vortex. In 1997, Teixeiro demonstrated theoretically that a bright vortex is stable in cubic quintic isotropic nonlinearity[‡‡‡‡][96] later on it was claimed that these bring vortex solitons display a symmetry-breaking azimuthal instability [97], which leads to breakup of the spinning soliton into a set of fragments, each being a stable nonspinning soliton. However, because this instability is very weak, such solitons may be observable in cubic quintic nonlinearity for a finite distance of propagation.

In any case, until now there was no experimental observation of such a bright vortex (neither in cubic quintic nonlinearity nor in any other type of nonlinearity).

This type of bright vortex soliton is important since there was never an experimental observation of:

[‡‡‡‡] In cubic quintic nonlinear term in the weve equation is $(a|\psi|^2+b|\psi|^4)\psi$ where ψ is the slowly varying electric field

- A bright soliton that consists of only the second anti-symmetric mode (such a soliton was not demonstrated neither in (2+1) dimensions nor in (1+1) dimensions).
- A single mode bright-soliton carries angular-momentum.
- A bright-soliton with a completely dark point at its center.

The bright vortex soliton that will be presented below answers all the properties listed above (1-3).

When looking for an isotropic nonlinear medium, I had noticed that optical glass is the best choice for almost perfectly homogeneous and isotropic media (glass is one of the basic building blocks in optics). It is also of much help that the exact optical parameters of glass are easily available. Looking in the data sheets of an optical glass, reveals that there are two properties that can help the ring build a high index of refraction in its dark center and create a waveguide that will guide the ring: the thermal induced index change, and the strain induced index change (photo-elastic effect).

Thermal-induced change of the index of refraction has been used in the past for self-guiding a bell-shaped beam [99], while the use of photoelastic effects for guiding a beam is not know to the author. The bell-shaped beam, guided by Dabby [99], consist of the first mode and does not carry angular momentum. The author of [99] also does not consider the possibility of stress-induced birefringence as the birefringence induced by thermal expansion of the glass (at beam center) reassemble the birefringence caused by sticking a tack in a transparent plastic. As I will explain in this section, my suggestion is to use the photoelastic effect in such a way that will increase the index of refraction at the beam center (and help us guide the ring) and at the same time will not cause birefringence of any kind.

The principle here is very simple: for every beam passing in glass, part of the energy will be absorbed by the glass. Let us assume that the intensity loss upon propagation through the entire bulk is only a few percents. Then, the heat generation in the glass is (approximately) constant along propagation. Because of the axial symmetry of the system, the direction of the heat flow is always radial and the heat can flow only outside. Because heat flow only from hot places to cold places, the

center of the beam must be hotter than any other place in the bulk, and if the temperature coefficient of the refractive index is positive (as in a few types of glass) then the index of refraction at the beam center will be higher than in its margins. In this way a waveguide is built.

The heat diffusion equation in cylindrical coordinates is:

$$\frac{1}{r}\frac{\partial}{\partial r}(kr\frac{\partial T}{\partial r})+\frac{1}{r}\frac{\partial}{\partial \phi}(k\frac{\partial T}{\partial \phi})+\frac{\partial}{\partial z}(k\frac{\partial T}{\partial z})+\dot{q}=\rho\, c_p\frac{\partial T}{\partial t} \quad . \qquad (37)$$

where T is the temperature, k is the heat conductivity, $\dot{q}$ is the heat generation, and $\rho\, c_p$ is the medium density multiplied by its heat capacity. If (i) $\dot{q}$ is almost uniform in z, (ii) the initial conditions are axisymetric, and (iii) we wait long enough for a steady state, equation (37) will reduce to:

$$\frac{1}{r}\frac{\partial}{\partial r}(kr\frac{\partial T}{\partial r})+\dot{q}=0 \quad . \qquad (38)$$

Having a beam with an intensity $I(r)$ and knowing the medium absorption, it is straightforward to calculate $\dot{q}(r)$. Substitution of $\dot{q}(r)$ in equation (38) gives $T(r)$ by integration, and from here we can calculate the thermal induced index change by using:

$$\Delta n_{th} \Delta T(r) \Delta \qquad (39)$$

.

where α is the temperature coefficient of the refractive index.

To achieve a balance between diffraction and self-focusing, we can assume (approximately) that the change in the refractive index, Δn_0, that is needed to guide a Gaussian beam (in a waveguide of quadratic index):

$$\Delta n_0=\left(\frac{\lambda}{2\pi\omega_0}\right)^2\frac{1}{n_0} \quad . \qquad (40)$$

Here, ω_0 is the radius if the Gaussian beam. Δn_0 in this equation (40) is the needed refractive index.

The index change we have in hand due to thermal effect is [98]:

$$\Delta n_0 = \frac{0.96\, b\, p\, \alpha}{k\pi} \quad . \qquad (41)$$

where b is the absorption coefficient and p is the beam power. It is not a mistake here that the index change is not a function of the beam radius. To emphasize this point, if we take a beam of 1 Watt, then the index change is independent of the beam radius and will be the same index change for a 10 μm width beam as well as for a 1 cm beam. The index change at the center of the beam is also independent of the intensity profile (as long as the beam is radially symmetric) and hence two beams of the same power will have the same index change at their centers, even of one of them is a Gaussian beam and the other is a Gauss Lagurre 01 beam (with zero intensity at its center).

In order to get a spatial soliton, the index change needed for guiding a beam (40) should be comparable to the one induced by the thermal effect (41). Compering the indexes of equations (40) and (41) yield

$$\omega_0 \approx 0.23\lambda \sqrt{\frac{k}{b\, n_0\, p\, \alpha}} \, . \qquad (42)$$

This equation implies that fluctuations in the power can be compensated by changes in the pulse width and vice versa (same as in section 1.3.2 here and as in reference [26]). Solitons which obeys this type of dependence between power and width are usually robust as any optical excitation with parameters close to the soliton parameters usually evolves with propagation into a stable soliton. However, I should note here that this analysis is usually considered as a rule of a thumb and it does not stand instead of a mathematical proof of stability.

We can also learn from Equation (42) that for a given laser (given power and wavelength) the glass which will enable us to make a thinner soliton is the one with the maximum value of $\frac{\alpha\, n_0\, b}{k}$.

In the past, this thermal nonlinearity was used to guide a 32 micron FWHM bell-shaped beam to a distance of 35 cm in lead glass [99]. As of yet, no one has made a bright vortex soliton, dealt with bi-refringence (that is caused by the photo

elastic effect and will be discussed below), or found the mathematical solution for thermal solitons.

For a bright vortex (of topological charge one), two opposite points on the ring circumference are π out of phase. In self-focusing nonlinearities of local character, light components of opposite phase repel each other [9]. This is the reason why a TEM_{10} mode in self-focusing medium tends to disintegrate into two solitons which repel each other [68]. If we take two points on the circumference of local vortex, and if these two points are azimuthally 180 degrees distant from each other, then the direction of the electric field of these two loci is opposite. Because of the opposite phase, these two points will also tend to repel each other in a local self-focusing medium. This is the reason why (in a nonlinearity with a local character) one has to prevent this repulsion-induced broadening of the finite vortex by using another bell-shaped component in order to glue the vortex together [63].

As shown above, with the thermal nonlinearity, the situation is completely different as the finite vortex induces a bell-shaped index and hence, in contrast with the local nonlinearity presented in [9], there is no repulsion here between the vortex components of opposite phase. The index change in the center of the finite vortex is maximal, despite the fact that the intensity in the center of the finite vortex is minimal and equal to zero. The vortex does not need another gluing component [see Figure 28]. The thermal nonlinearity is hence a natural choice for supporting a vortex. Never before has there been a bright soliton consisting of the second (asymmetric) mode only [not even in (1+1)D]. This thermal vortex soliton will be the first to be made of the second mode only in (2+1)D configuration (this mode also carries orbital angular momentum)

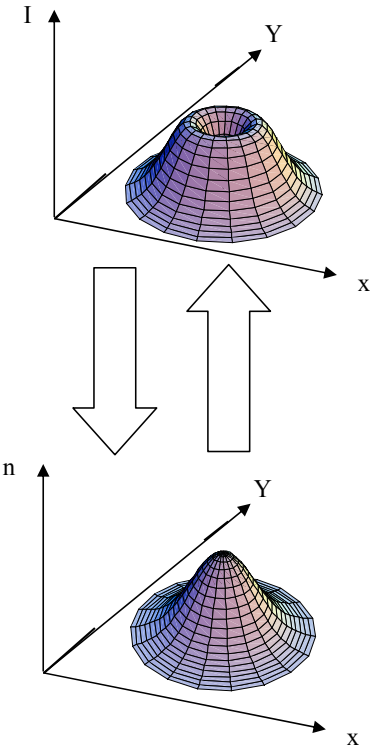


Figure 28: Illustration of self-consistency for the solitary mode of a thermal bright vortex of finite dimensions. Above: the intensity distribution, $I(x,y)$, of a finite vortex. Below: the corresponding index change $\Delta n(x,y)$.

If the second mode of the bell shaped waveguide is close enough to the bright vortex (above), self consistency is assured and the bright vortex will be guided by its self induced waveguide.

However, as heated materials usually expand, stress is induced in the medium and consequently, the photoelastic effect also contributes to the index change. Using the photoelastic effect, we earn twice: first, heat directly increases the index, and second, heat causes stress, which also causes the index to rise (see equations below). Knowing the intensity distribution, we can integrate Equation (38) to get the temperature distribution, $T(r)$. The radial stress σ_r induced by this distribution is given by [100,101]

$$\sigma_r(r_1) = \alpha E\left(\frac{1}{R^2}\int_0^R T(r)rdr - \frac{1}{r_1^2}\int_0^{r_1} T(r)rdr\right). \qquad (43)$$

Here, E stands for the modulus of elasticity for the medium (Young's modulus) and R stands for the radius of our bulk material (where $R >> \omega_0$, one can take R as ∞). The temperature (T) is relative to the temperature at the bulk margins. In the same manner, the tangential stress is

$$\sigma_t(r_1) = \alpha E\left(-T(r_1) + \frac{1}{R^2}\int_0^R T(r)rdr + \frac{1}{r_1^2}\int_0^{r_1} T(r)rdr\right). \qquad (44)$$

Here, negative stress stands for compression. Taking the stress optical coefficients ($K_\perp, K_\parallel$) from the glass catalog, it is easy to find the index change by using:

$$n_r(r) = K_\perp \sigma_t(r) + K_\parallel \sigma_r(r) \qquad (45)$$
$$n_t(r) = K_\perp \sigma_r(r) + K_\parallel \sigma_t(r)$$

where n_r stands for the index of refraction of the radial polarization and n_t stands for the index of refraction of the tangential polarization

While the stress helps us increase the index at the beam center, and therefore helps us to guide the beam, stress also causes bi-refringence problems (as $n_r(r) \neq n_t(r)$). Yet, it is easy to solve this bi-refringence problem by choosing the polarization to be perpendicular to a principal axis. In our case, to prevent bi-refringence, the polarization should be either radial or tangential (see Figure 29)

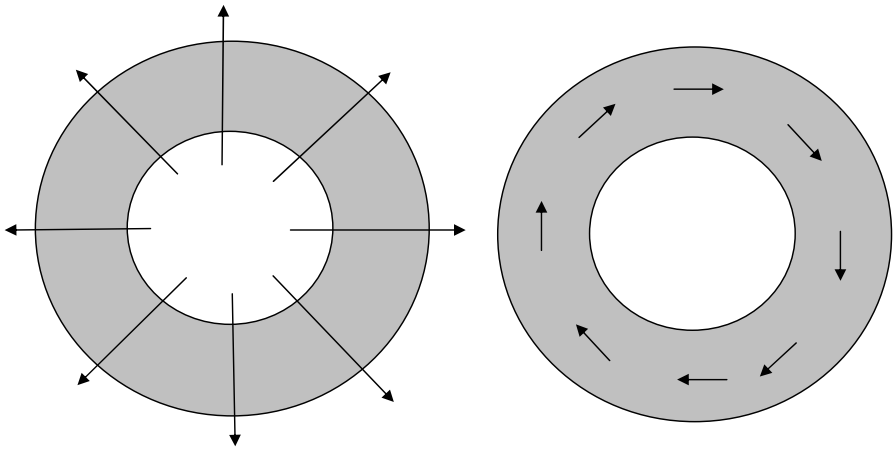


Figure 29: A finite vortex beam of radial polarization (left), and finite vortex of azimuthal polarization (right). In these configuration the polarization is perpendicular to a principal axis and therefore bi-refringence is eliminated.

Usually, such a polarization is not possible since, for example, a Gaussian beam cannot be polarized radially (or azimuthally) because polarization is then undefined at the beam center. With vortices, though, the intensity at the beam center is zero. there

are no photons there, and therefore the problem does not come up[§§§§]. Such radially and azimuthally polarized beams can be formed using a metal mask [102].

It is important to note here that when going smaller in dimensions, heating, in general, (and consequently the index change) becomes faster. The reason for this fast response is fundamental and results from the dimensionality of the problem. The delaying factors are (i) the dissipation of heat, which is proportional to the surface area ($\propto \text{Size}^2$) of the heated zone, and (ii) the heat capacity of this heated zone, which is proportional to the volume of the heated zone ($\propto \text{Size}^3$). To make this point clear, when going 10 times smaller in dimension, the delaying factors are reduced by factors of 100 and 1000 accordingly. To get an even faster index change in a specific zone, one can just increase the power of the heating beam (the time from turning the heating beam on until the heating process starts is extremely fast and is limited only by the velocity of light). From a technological point of view, it is possible to go smaller in dimension by taking the wavelength to be smaller (UV). High power lasers are also available.

To conclude, optical glasses and silica are a mature technology: the mediom is mass-produced, and its quality has high degrees of isotropy and homogeneity. In addition, dimensionality gives here a chance to make the thermal nonlinearity faster. These properties make this type of solitons important from the application standpoint and not only from the fundamental point of view. Its qualities make this type of thermal nonlinearity attractive for doing vortex solitons of finite dimension and other types of single-mode (scalar) solitons, such as rotating dipole, rotating quadropoles, hexapoles etc. These solitons are impossible to make in photorefractive nonlinearity as zones of opposite phase tend to repel each other [9,68]. Glass may also be the only medium which will enable us to watch experimentally the collisions investigated theoretically and numerically (for composite solitons) in [64,65,76]; as the homogeneity of the glass enable us to find the beam track by monitoring stray scattering in the glass.

[§§§§] Note that in the same manner, the phase is also not defined at the vortex center and this also does not cause any problems.

6. Miscellaneous.

This chapter includes several projects I have worked on, but did not play the lead role. In most cases my contribution was to instruct a new student in his experiment , raising ideas for a new project or designing the experimental system. Below I describe these projects briefly and list the relevant publication.

6.1. Discrete *solitons in optically induced periodic system*

In this project, I collaborated with Dr. Jason Fleischer on the first observation of solitons in optically induced periodic system.
the resultant publications are:

1) J. W. Fleischer, **T. Carmon**, M. Segev, N. K. Efremidis, and D. N. Christodoulides, *Observation of discrete solitons in optically-induced real time waveguide arrays*, Physical Review Letters, 90, 023902 (2003).
2) J. W. Fleischer, N. K. Efremidis, **T. Carmon**, D. N. Christodoulides and M. Segev, *Solitons in optically-induced nonlinear photonic lattices,* **Optics and Photonics News**, ***Special Issue: Optics in 2002.*** **13**,49 (2002)

Periodic nonlinear systems are ubiquitous in nature and are known to exhibit behavior that differs fundamentally from that of their homogeneous counterparts. In these periodic nonlinear lattices, the underlying dynamics are dominated by the interplay between linear coupling effects among adjacent potential wells and nonlinearity. Indeed, a balance between these two processes can result in a self-localized state, which can be either a time-periodic breather or a steady-state lattice soliton. Very often, such systems can be represented as a discrete array of coupled nonlinear sites, in which case the associated self-localized states are better known as discrete solitons (DS) [103,104,105,106]. Examples occur in abundance, in all branches of science, such as biology [103], nonlinear optics [104], solid-state physics [105], and Bose-Einstein condensates [106]. In the past few years, much progress has been made in attempting to observe discrete solitons in various physical systems. In

biological α-helices, time-dependent breather states have been inferred from pump-probe experiments [107]. In (linear) arrays of current-driven Josephson junctions, localized time-periodic rotobreathers (quantum vortex pairs exhibiting counter rotating phases) have been observed [108]. In antiferromagnets, spectral data from resonance experiments have indicated DS states for spin waves [109]. In transition metals, evidence of localization for vibrational states has been reported [110]. However, in all of these systems, direct observations of steady-state discrete solitons (not breathers) remain elusive. In fact, direct observations of DS have been reported only in a 1D array of nonlinear waveguides [111,112], but even that only for self-focusing nonlinearities that restrict the allowed class of soliton solutions. In all of the above, the experiments require specialized materials with fixed geometries that limit their potential application. In this project we have made the first experimental observation of discrete solitons in an array of optically induced waveguides. The waveguide lattice is induced in real time by illuminating a photorefractive crystal with a pair of interfering plane waves. We demonstrated two types of bright discrete solitons: in-phase self localized states and the staggered (π out-of-phase) soliton family. This experiment was the first observation of bright staggered solitons in any physical system. Our scheme paved the way for reconfigurable focusing and defocusing photonic lattices where low-power (mW) discrete solitons can be thoroughly investigated.

6.2. Holographic solitons.

In this chapter, I was collaborating with Oren Cohen on the first prediction of a new type of soliton namely, holographic solitons. the relevant publication is:

O. Cohen, **T. Carmon**, M. Segev, and S. Odoulov, *Holographic solitons*, ,**Optics Letters**, **27**, 2031 (2002)

Optical spatial solitons are supported by many different nonlinear mechanisms, which generally can be classified into two generic types. The first type is the self-phase-modulation self-focusing mechanism through which an optical beam modifies the refractive index and induces an effective positive lens; i.e., the refractive index in

the center of the beam is larger than at the beam's margins. The refractive-index structure of the medium then resembles that of a graded-index waveguide. When the optical beam that has induced the waveguide is also a guided mode of the waveguide that it has induced, the beam's propagation becomes stationary [113]. The vast majority of the physical phenomena that support optical solitons belong to this self-focusing type, which has been implemented for observation of solitons in optical Kerr media [114,115], atom vapor [116,117], photorefractive [118,119] and photovoltaic [120] crystals, thermal nonlinearities [121], liquid crystals [122], and more. The second mechanism that can support spatial solitons arises from the nonlinear phase coupling that results from symmetric energy exchange between two or more mutually coherent beams. Each beam continuously loses some of its energy and regains the same amount, such that the net power in each beam is conserved. In this interaction, the field that constitutes the acquired energy (to each of the beams) is phase retarded relative to the primary field of each beam. Thus, as the acquired field is added to the primary field, it effectively slows the phase velocity of the beam. Hence, if the interaction occurs in such a way that the effect is more intense at the center of the beam (or for the lowest spatial frequencies of the beam), then it reduces or eliminates the broadening effects of diffraction. Currently, the only known soliton that is supported solely by such a mechanism is the quadratic soliton, which consists of multifrequency beams [123,124]. However, phase coupling between two mutually coherent beams can also be established through a grating in the refractive index that is induced (in real time) by interference between the beams. In this case the beams are coupled through Bragg ref lections. Each beam is Bragg reflected and coherently added into the other beam. This can lead to focusing of narrow beams when the reflected beams are $\pi/2$ phase retarded with respect to the primary beams and the index change that gives rise to the grating is an increasing function of the optical intensity. Inasmuch as the focusing effect results from the induced index grating, we termed the effect holographic focusing [125]. In this project we have proposed a soliton for which the focusing mechanism is solely holographic focusing, that is, a holographic soliton that does not rely on self-phase-modulation self-focusing but exists only by virtue of the induced periodic modulation of the refractive index.

6.3. Soliton in photorefractive CdZnTe:V

In this chapter, I was collaborating with Tal Schwartz on the observation of a soliton in photorefractive CdZnTe:V, exploiting the intensity-resonant behavior of the space-charge field. The relevant publication is:

T. Schwartz, Y. Ganor, **T. Carmon**, R. Uzdin, S. Schwartz, M. Segev and U. El-Hanany, *Photorefractive solitons and light-induced resonance control in CdZnTe,* **Optics Letters** 27, 1229 (2002).

In general, the formation time of solitons in most photorefractives is rather long, except when very high intensities are used [126]. This is because the photorefractive nonlinearity relies on charge separation, for which the response time is inversely proportional to the product of the mobility and the optical intensity, and the mobility in photorefractive oxides is low (~1 $cm^2/V\ s$). In principle, photorefractive semiconductors have high mobility and could offer formation 1000 times faster than other photorefractives. However, the electro-optic coefficient in these semiconductors is tiny, which implies that solitons that are as narrow as ~20 optical wavelengths necessitate very large applied fields, making soliton formation in them almost impossible [127]. But, in some of these materials (InP, CdZnTe) that have both holes and electrons as charge carriers, a unique resonance mechanism enhances the space-charge field by as much as ten times the applied electric field. This enhancement yields large enough self-focusing effects that can support narrow spatial solitons. The resonant enhancement of the space-charge field was first observed in photorefractive two-wave-mixing in such materials [128,129], and has led to the observation of solitons in photorefractive InP [130,131] and to their theoretical explanation [132] Resonant enhancement of the space-charge field occurs in materials with both types of charge carrier, both of which are excited from a common trap level: One is excited optically, and the other is excited by temperature. At steady state, when a focused beam illuminates a biased crystal of this kind, and the beam intensity is such that the excitation rates are comparable, the concentrations of both free carriers at the illuminated region decrease drastically [132]. The intuitive explanation for this is as follows: Under proper conditions, the ratio between the concentrations of electrons and holes is equal to their ratio in the absence of light and thus has a constant

(coordinate independent) value. The net excitation rate of the traps is the difference between the thermal (holes) and optical (electrons) excitation rates. At resonance, the net excitation rate goes to zero. At the same time, at steady state the excitation rate must be equal to the recombination rate, which, in turn, is proportional to the free-charge concentration. Hence, at resonance (when the excitation rates of holes and electrons are comparable) the free-charge concentration goes to zero. Consequently, the local electric field is highly enhanced because the current at steady state must remain constant throughout the crystal. For a given temperature, this enhancement occurs at a specific intensity (the resonance intensity), for which the thermal and optical excitation rates are comparable. It is a resonant enhancement, although it is an intensity resonance and not an atomic resonance. The enhanced electric field compensates for the smallness of the electro-optic coefficient and permits a sufficiently large change in the refractive index to support narrow solitons. Solitons based on the resonance enhancement in photorefractive semiconductors have thus far been demonstrated experimentally only in InP:Fe, for which the thermal excitation rate of electrons resonates with the optical excitation rate of holes in the close vicinity of a resonance intensity. In this project, we have used this mechanism to demonstrate narrow and fast solitons in photorefractives semiconductors. We demonstrated the formation of (1+1) Dimensional and (2+1) Dimensional solitons in CdZnTe:V, and showed optical control over the resonance intensity, facilitating formation times of 10 ms with very low optical power.

6.4. Collisions between (2+1)D rotating propeller solitons

In this chapter, I was collaborating with Claude Pigier on the theoretical study of Collisions between (2+1)D rotating propeller solitons. The relevant publication is:

C. Pigier, R. Uzdin, **T. Carmon**, M. Segev, A. Nepomnyashchy and Z. H. Musslimani, *Collisions between (2+1)D rotating propeller solitons,* **Optics Letters** 26, 1577 (2001).

Solitons are self-trapped wave packets that behave as particles: A single soliton propagates without loss of energy and momentum. This analogy holds also for collisions between solitons, which can be elastic or inelastic. Soliton collisions have been thoroughly investigated over the years [133]. In the (1+1) Dimensional integrable case, collisions are elastic: The number of solitons is conserved, each soliton conserves its power and linear momentum, and no energy is transferred from bound (soliton states to radiation states. But, in such a system, (2+1) Dimensional solitons are unstable, so soliton interactions are restricted to a single plane. In saturable nonlinearities collisions are much richer, because in such media (2+1) Dimensional solitons are stable. Thus, collisions can occur between (2+1) Dimensional solitons with trajectories in different planes. Also, in this system collisions at shallow angles are inelastic [133]. Thus far, various types of interaction, e.g., fusion, fission, and spiraling of solitons, have been identified in saturable nonlinearities [133]. The ability of saturable nonlinearities to support stable (2+1)Dimensional self-trapping can be exploited to launch (2+1)Dimensional composite (or multimode) solitons (See description of composite solitons in chapter 2). Here, we study numerically collisions between propeller solitons (on propeller solitons, see section 2.3) and show that they exhibit interesting features that result from exchanges of angular momentum between the field constituents of the interacting solitons.

The angular momentum gave rise to fascinating features, which arise from transmission of angular momentum from the input solitons to the fission products, which are solitons themselves. We have found that propeller solitons are robust, as they can survive collisions, much like their stationary cousins. In fact, in all the collisions between rotating propeller solitons for which the outcome is self-trapped beams, at least one of the emerging beams is a (stationary or rotating) dipole-type composite soliton.

6.5. Collisions between optical spatial solitons propagating in opposite directions

In this chapter, I was collaborating with Oren Cohen on the theoretical study of Collisions between optical spatial solitons propagating in opposite directions. The relevant publication is:

O. Cohen, R. Uzdin, **T. Carmon**, J. W. Fleischer, M. Segev and S. Odoulov, *Collisions between optical spatial solitons propagating in opposite directions,* **Physical Review Letters** 89, 133901 (2002).

We formulate the theory describing the evolution and interactions between optical spatial solitons that propagate in opposite directions. We have shown that coherent collisions between counterpropagating solitons (on collision, see previous section) gave rise to a new focusing mechanism resulting from the interference between the beams, and that interactions between such solitons are insensitive to the relative phase between the beams.

6.6. Spatial vector solitons consisting of counter-propagating fields

In this chapter, I was collaborating with Oren Cohen on the observation of a new type of vector soliton consisting of counter-propagating fields. the relevant publication is:

O. Cohen, S. Lan, **T. Carmon**, J. A. Giordmaine, and M. Segev, *Spatial vector solitons consisting of counter-propagating fields*, **Optics Letters**, **27**, 2013 (2002).

Thus far, all experimental observations of spatial vector solitons have displayed a multicomponent localized wave packet whose intensity induces a waveguide that is nonvarying in the propagation direction and whose field components

properly populate the guided modes of the induced waveguide (for more on composite solitons, see chapter 2 and section)

In this project we have demonstrated experimentally a spatial vector soliton formed by counterpropagating coherent optical fields. This was the first vector soliton in which the induced waveguide (potential) is periodic in the propagation direction.

6.7. Information transfer through cascaded collisions of vector solitons

In this chapter, I was collaborating with Charalambos Anastassiou on the experimental investigation of Information transfer through cascaded collisions of vector solitons. the relevant publication is:

C. Anastassiou, J. W. Fleischer**, T. Carmon**, M. Segev, and K. Steiglitz, *Information transfer through cascaded collisions of vector solitons*, Optics Letters 26, 1498 (2001).

Scalar solitons of an integrable system, such as those generated from the nonlinear Schrödinger equation (which describes propagation in Kerr media) remain unaffected during collisions with other solitons. In fact, collisions merely lead to a phase shift (a lateral displacement in the case of spatial solitons) [134]. This result is intuitive if one views a soliton as a guided mode of its own induced waveguide (induced potential) [135]. Thus, interactions between solitons can be viewed as interactions between guided modes of induced waveguides in close proximity [136]. For a Kerr soliton, the induced potential is reflectionless [135], so scalar Kerr solitons (solitons that have only one field component) colliding at a nonzero angle cannot exchange energy. This is why solitons suffer only a phase shift (displacement) upon collisions. This phase shift depends only on the soliton power and velocity, which are both conserved quantities. Therefore, when two scalar soliton collisions occur sequentially, the outcome of the first collision does not affect the second collision (except for the uniform shift). On the one hand, this property is a manifestation of the robustness of solitons, which is important for communication applications in which

temporal solitons are used as data bits carried along an optical fiber [137]. On the other hand, there are cases when it is desirable for solitons to interact more strongly, so that information may be transferred from one to another. For example, solitons have been suggested for information processing purposes, in which case they behave as particles in cellular automata [138]. For these purposes, scalar Kerr solitons offer very little [138]. One could, of course, employ collisions of solitons in nonintegrable systems, such as spatial solitons in saturable nonlinearities [136], in which case the number of solitons is not always conserved. But it is very difficult to define a state in such systems, because even the main building block, the presence of a soliton, is not always guaranteed in a fusion process in which two solitons merge into one. For this reason, and for reasons of zero radiation loss, it would be highly desirable to find a method to transfer information between solitons in an integrable system. As it turns out, vector solitons (solitons that consist of more than one field component) in the integrable ideal Manakov system can do just that; the field components that make up the vector solitons exchange energy (symmetrically) upon collision, and at the same time they retain all other conserved quantities of integrable systems [139,140]. (For temporal Manakov solitons, this energy exchange is commonly referred to as polarization switching.) This energy-exchange effect has been proposed as a mechanism for performing computation through interaction of vector solitons [141]. Experimentally, however, thus far to our knowledge only a single collision (a single energy-exchange process) has been demonstrated [142], and there is no experimental evidence of the ability to cascade information from one collision to the next.

Here we have studied experimentally the energy-exchange interactions between vector solitons. Such interactions are not possible with scalar solitons. We have shown how, in a sequence of two collisions, the outcome of the first collision directly affects the outcome of the second. One potential application of this arrangement is as a tunable, directional beam splitter, in which the signal is successively split off after each collision. Another application is coding of information bits that are carried by solitons and then transferred through collisions to numerous layers of interacting solitons. This cascaded interaction of vector solitons lays the experimental foundations for computation with solitons [141].

Appendix A: Solitons in photorefractive media.

The solitons presented here are universal to any saturable nonlinearity (e.g. two level system of dilute gas). However, since the experiments presented here were done in a photorefractive medium using the screening nonlinearity, it is important for the completeness of this thesis to give here a short description of the photorefractive effect.

Photorefractive materials have been studied for almost 35 years now. Typically, these are dielectric of semi-insolating single crystals that are non-centrosymmetric, that give rise to $\chi^{(2)}$ nonlinearity. Through the electrooptic effect, an electric field (E) modifies the index of refraction in the medium as $\Delta n = -n^3 r_{eff} E/3$, where r_{eff} is the relevant term from the electrooptic tensor determined by the directions of E and the polarization of the optical beam. Photorefractive materials always have some foreign atoms (dopants) hosted in the crystalline matrix, with energy levels deep inside the forbidden gap. These dopants are in the form of donors and acceptors thus they can contribute or trap free charges. Consider an optical beam incident upon such a crystal with optical photons that are not energetic enough to cause valence-to-conduction band excitation, but can excite charges (say, electrons) from the deep dopant levels. Once excited into the conduction band, the electrons are free to move. If the intensity of the optical beam is not uniform in space, these photo-excited electrons experience transport: they diffuse from higher concentrations to lower ones and drift if an external bias field is applied. Immobile ions are 'left' behind, stuck in their original position. Eventually, after some characteristic time, the electrons are re-trapped by ionized donors at locations that are different from their original donor ion sites. The resulting charge separation establishes an electric field within the medium, which varies is space. This internal space-charge field E_{sc}, gives rise to a change in the refractive index, via the electrooptic effect, that varies in space, In other words, non-uniform illumination incident upon a photorefractive medium results in a non-uniform change in the

refractive index. In this way, photorefractive materials are used to record volume holograms for applications such as optical data storage, realization of phase conjugate mirrors, etc. In the usual context, photorefractive nonlinear optics deals with a periodic interface grating whose envelope is (slowly) modulated by some pictorial information. A soliton is a different creature: it entails self-action of a single beam and has nothing to do with holography.

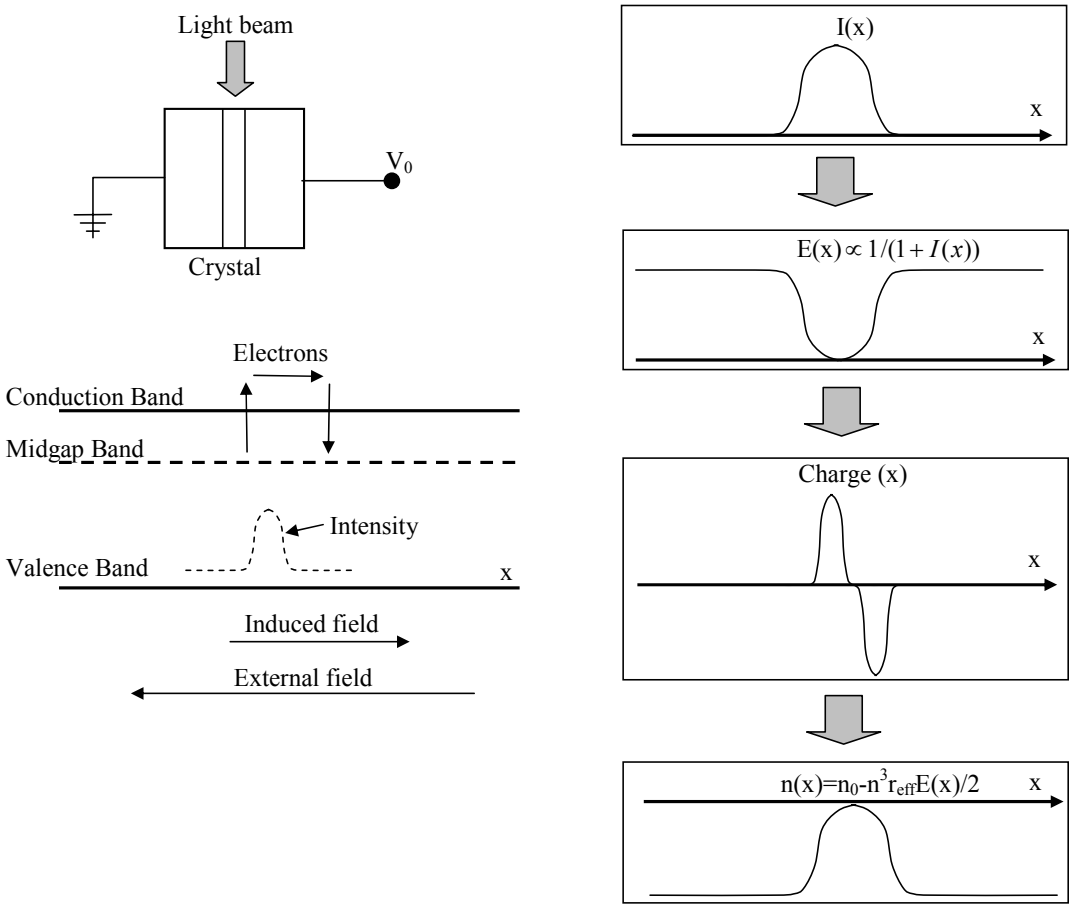


Figure 30: Formation of photorefractive spatial soliton.

One may view the formation of bright photorefractive screening solitons [72-74] is illustrated in Figure 32. An optical beam enters a photorefractive crystal which is biased by an external electric field. The light causes excitations of electrons from midgap states into the conduction band where they experience drift in one direction due to the external field. The electrons thus move and leave immobile positively charged donors behind. As the electrons drift into darker regions of the beam they start to fall back into the midgap levels, giving rise to a charge separation. The charge variation inside the crystal creates a space charge field that partially screens the

external field. The amount of screening depends upon the ratio between the local density of electrons due to the optical beam intensity and the background electron density (density in the dark regions). The total electric field modulates the refractive index via the electro-optic (Pockels') effect and creates an induced waveguide. The optical beam thus creates an electric space-charge field that induced an effective graded-index waveguide, which in a self-consistent manner is able to guide the beam itself. Here, we gave the physical principles for spatial solitons in photorefractive, for a complete mathematical description of these principles, the reader is referred to [72-74].

Appendix B: different aspects of angular momentum of light beams.

As much of this research thesis is about beams carrying angular momentum, I find it very important to distinguish between the different kinds of angular momentum light can carry.

In general, one can distinguish between two different kinds of angular momentum of light: (1) The *internal angular momentum* which is associated with the spin of the photon (either) $-\hbar$ or $+\hbar$. This type of angular momentum is determined by polarization state of the field. (2) *The orbital angular momentum* which can be associated with an $l\lambda$ phase jump (where l is an integer) of the phase upon making one closed loop in the transverse plane. One example of *orbital angular momentum* is the family of Gauss-Laguerre modes (also called optical vortices). However, in general, *orbital angular momentum* does not necessitate this phase jump and as long as

$$L\hat{z} = \int r \times p \; dxdy = \frac{1}{2}\int r \times (A\nabla_{\perp} A^{*} - C.C.)dxdy \quad (46)$$

is not zero, the beam contains orbital angular momentum. Here, $\boldsymbol{r}$ stands for the transverse coordinate, $\boldsymbol{p}$ stands for the linear momentum, and A stands for the slowly varying amplitude of the electric field.

A typical equation for light with *internal angular momentum* is the field equation of a right-hand circularly polarized beam:

***** Here, C.C. stands for the complex conjugate.

$$\vec{E}(x,y,z) = A(x,y)\begin{pmatrix}\cos(\omega t - kz)\hat{x}\\ \sin(\omega t - kz)\hat{y}\end{pmatrix} \qquad (47)$$
$$= A(x,y)\begin{pmatrix}i\hat{x}\\ \hat{y}\end{pmatrix}\exp[i(\omega t - kz)]$$

This equation[†††††] reflects the fact that the electric field direction rotates as the beam propagates (see Figure 31).

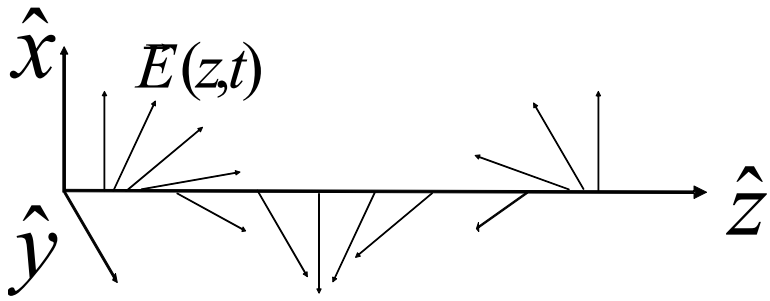


Figure 31: For circularly polarized light the direction of the field rotates along propagation.

It is easy to measure whether or not the beam has *internal angular momentum* with a circular polarizer.

For a linearly polarized beam with *orbital angular momentum* (and phase jump) a typical equation might look like:

$$\vec{E}(x,y,z) = A(x,y)\begin{pmatrix}1\hat{x}\\ 0\hat{y}\end{pmatrix}\cos(\omega t - kz + \phi l) \qquad (48)$$
$$= A(x,y)\begin{pmatrix}1\hat{x}\\ 0\hat{y}\end{pmatrix}\exp[i(\omega t - kz + \phi l)]$$

were ϕ is the azimuthal angle and l is an integer. This equation reflects the fact that the equal phase-plane of the electric field has a screw shape (see Figure 32)

[†††††] shown here in the traditional form and in the phasoric form, on the difference between the two presentation see for example [22]

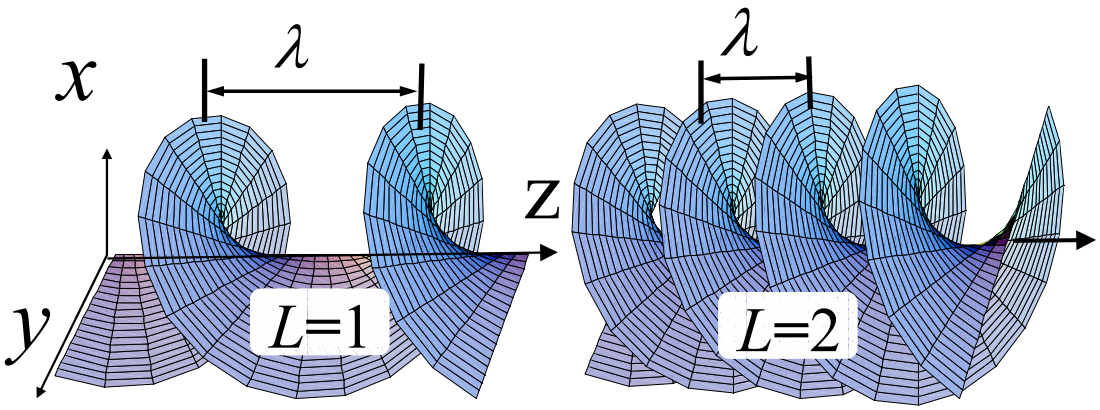


Figure 32: Equal-phase plane of beams with orbital angular momentum. If l is a positive integer, then the equal-phase plane is right-hand screw, and if l is a negative integer, then this plane describes a left-hand screw. The screw contain $|l|$ threads, the distance between the threads is λ and the screw pitch is $l\lambda$.

Measuring l is done by interfering the signal beam with a plane wave or with a spherical wave (see Figure 33)

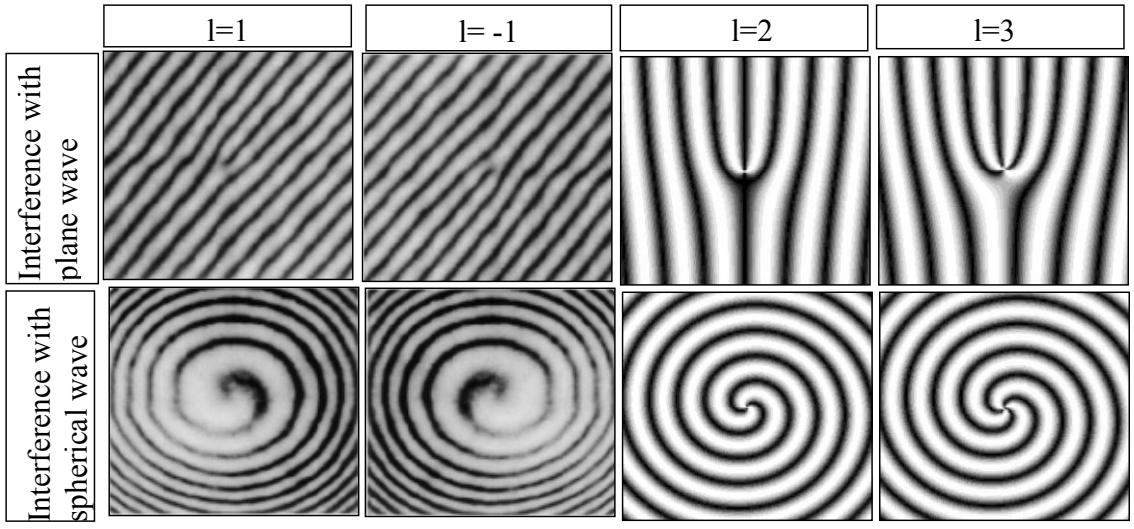


Figure 33: Interferograms of beams carrying orbital angular momentum. The spiral rotation direction denotes whether the angular momentum is right- or left-handed. The number of threads is equal to the absolute value of l. (Pictures on the left were taken by Chen [143])

The techniques for measuring l been much improved over the last few years, and in a recent paper, Leach shows that one can even measure l of a single photon [144].

A beam can also carry both internal and angular momentum. For example:

$$\vec{E}(x,y,z) = A(x,y)\begin{pmatrix}\cos(\omega t - kz + \phi l)\hat{x}\\ \sin(\omega t - kz + \phi l)\hat{y}\end{pmatrix} \quad (49)$$
$$= A(x,y)\begin{pmatrix}i\hat{x}\\ \hat{y}\end{pmatrix}\exp[i(\omega t - kz + \phi l)]$$

In this case, the internal and the orbital angular momenta can either add to or subtract from each other.

Conservation of total angular momentum stands for the total angular momentum only. It has been shown that a beam can change its *l* from positive to negative in free space [145]. In this case the total orbital angular momentum (equation (28)) does not change as the beam propagates. The question whether angular momentum can pass from the orbital to internal type in free space is still an open question.

Light emitted from a rotating source will have a Doppler shift, The spin and orbital angular momenta of the light beam acting in an additive way [146]. For a source rotating at Ω , the Doppler frequency shift for a photon is $\Omega(l+\sigma)$ where l is associate with the orbital angular momentum (an integer) and σ is associate with the spin angular momentum (-1 or 1).

It is worth noting that even for a linearly polarized Gaussian beam, half of the photons carry spin 1 and other half carry spin -1 (the photon spin cannot be 0), and therefore when we rotate this linearly polarized Gaussian source, half of the photons will be Doppler red-shifted and the other half will be Doppler blue-shifted. In the same manner, a linearly polarized TEM_{01} mode is a superposition of two counter-rotating vortices, and consequently a rotating TEM_{01} mode will be split into two counter-rotating vortices, one red-shifted and the other blue-shifted.

The geometric phase of *orbital beams* has also received considerable attention in the past few years. Geometric phase was first introduced by Berry in quantum systems [147], and its presence in image rotation was proposed by Segev

[148]. It was only very recently that Galvez measured the dependence of the orbital angular momentum on the geometric phase [149]. In this paper, Galvez uses the Guoy phase shift to manipulate a TEM_{01} mode into a vortex with cylindrical lenses.

Appendix C: List of publications.

Presented here is a list of papers and oral presentations.

1) **T. Carmon**, M. Soljacic and M. Segev, *Pattern formation in a cavity without resonant frequencies,* **Physical Review Letters, 89**, 183902 (2002).
2) **T. Carmon**, R. Uzdin, C. Pigier, Z. Musslimani, M. Segev, and A. Nepomnyashchy, *Rotating propeller solitons,* **Physical Review Letters** 87, 143901 (2001).
3) **T. Carmon**, C. Anastassiou, S. Lan, D. Kip, Z. Musslimani, M. Segev, and D. N. Christodoulides, *Observation of two-dimensional multimode solitons,* **Optics Letters** 25, 1113 (2000)
4) **T. Carmon**, R. Uzdin, C. Pigier, Z. Musslimani, M. Segev, and A. Nepomnyashchy, *Rotating propeller solitons*, Optics and Photonics News, Special Issue: Optics in 2001, vol. 12 (12), 613 (2001).
5) Hrvoje Buljan, Marin Soljac̆ic´, **Tal Carmon**, and Mordechai Segev. *Cavity pattern formation with incoherent light*, **Physical Review E, 68**, 0266XX ,(2003).
6) O. Cohen, R. Uzdin, **T. Carmon**, J. W. Fleischer, M. Segev and S. Odoulov, *Collisions between optical spatial solitons propagating in opposite directions,* **Physical Review Letters** 89, 133901 (2002).
7) J. W. Fleischer, **T. Carmon**, M. Segev, N. K. Efremidis, and D. N. Christodoulides, *Observation of discrete solitons in optically-induced real time waveguide arrays*, Physical Review Letters, 90, 023902 (2003).
8) J. W. Fleischer, N. K. Efremidis, **T. Carmon**, D. N. Christodoulides and M. Segev, *Solitons in optically-induced nonlinear photonic lattices,* **Optics and Photonics News**, ***Special Issue: Optics in 2002.*** **13**,49 (2002)
9) O. Cohen, **T. Carmon**, M. Segev, and S. Odoulov, *Holographic solitons*, to appear ,**Optics Letters**, **27**, 2031 (2002)
10) T. Schwartz, Y. Ganor, **T. Carmon**, R. Uzdin, S. Schwartz, M. Segev and U. El-Hanany, *Photorefractive solitons and light-induced resonance control in CdZnTe,* **Optics Letters** 27, 1229 (2002).

11) O. Cohen, S. Lan, **T. Carmon**, J. A. Giordmaine, and M. Segev, *Spatial vector solitons consisting of counter-propagating fields*, **Optics Letters**, **27**, 2013 (2002).
12) C. Pigier, R. Uzdin, **T. Carmon**, M. Segev, A. Nepomnyashchy and Z. H. Musslimani, *Collisions between (2+1)D rotating propeller solitons,* **Optics Letters** 26, 1577 (2001).
13) C. Anastassiou, J. W. Fleischer**, T. Carmon**, M. Segev, and K. Steiglitz, *Information transfer through cascaded collisions of vector solitons*, Optics Letters 26, 1498 (2001).

Oral Presentations:

1) Barak Freedman, **Tal Carmon**, Demetrios Christodoulides, Mordechai Segev, *Observation of optical spatial shock-waves,* (CLEO) 2003, Baltimore, Maryland, USA.
14) Oren Cohen, **Tal Carmon**, J Giordmaine, Song Lan, Mordechai Segev, *Spatial vector solitons consisting of counter-propagating mutually coherent fields,* (CLEO) 2003, Baltimore, Maryland, USA .
15) Hrvoje Buljan, **Tal Carmon**, Mordechai Segev, and Marin Soljacic, *Cavity pattern formation and cavity solitons with incoherent light,* (QELS) 2003, Baltimore, Maryland, USA
16) Tal Schwartz, Hrvoje Buljan, **Tal Carmon**, and Mordechai Segev, Experimental observation of modulational instability and spontaneous pattern formation with incoherent white light, (QELS) 2003, Baltimore, Maryland, USA
17) M. Segev, D. N. Christodoulides, J. W. Fliescher, N. Efrimidis, and **T. Carmon**, *.Discrete Solitons in Optically-induced Photonic Lattices*., Invited talk in Conference on Lasers and Electro-Optics / European Quantum Electronics Conference (CLEO®/Europe.EQEC) 2003, Munich, Germany.
18) J. W.Fleischer, **T. Carmon**, M. Segev, *.Discrete solitons in optically-induced photonic lattices*., Nonlinear Optics, (NLO) 2002, Maui, USA.
19) M. Segev, D. N.Christodoulides, M. Soljacic, Z. Chen, D. Kip,S. M. Sears, **T. Carmon**, *.Pattern formation and clustering of solitons in nonlinear weakly-correlated wave-systems*., Nonlinear Optics, (NLO) 2002, Maui, USA

20) J. W. Fleischer, **T. Carmon**, M. Segev, N. K. Efremidis, D. N. Christodoulides, *.Discrete solitons in optically-induced real-time waveguide arrays*., Nonlinear Guided Waves and Their Applications, (NLGW) 2002, Stresa, Italy.

21) **T. Carmon**, M. Segev, M. Soljačić, *.Pattern Formation in a Cavity Without Resonant Frequencies*., Nonlinear Guided Waves and Their Applications, (NLGW) 2002, Stresa, Italy.

22) T. Swartz, Y. Ganor, **T. Carmon**, R. Uzdin, S. Swartz, M. Segev and U. El-Hanany *.Photorefractive solitons and light induced resonance control in semiconductor CdZnTe*.. Quantum Electronics and Laser Science, (QELS), 2002, Long Beach, California, USA.

23) **T. Carmon**, M. Segev, M. Soljačić., H. Buljan, *.Cavity Pattern Formation with Incoherent Light*., Conference on lasers and Electro Optics, (CLEO), 2002, Long Beach, California, USA

24) J. W. Fleischer, **T. Carmon**, M. Segev, *.Discrete diffraction in optically-induced real-time waveguides in photorefractives*., Quantum Electronics and Laser Science, (QELS), 2002, Long Beach, California, USA.

25) C. Pigier, R. Uzdin, **T. Carmon**, M. Segev, A. Nepomnyashchy and Z. H. Musslimani, *Collisions between (2+1)D rotating propeller solitons,*Nonlinear Guided Waves and Their Applications, (NLGW) 2001, Florida, USA.

26) **T. Carmon**, R. Uzdin, C. Pigier, Z. H. Musslimani, M. Segev, A. Nepomnyashchy, *.Rotating propeller solitons*., Nonlinear Guided Waves and Their Applications, (NLGW) 2001, Florida, USA.

27) **T. Carmon**, R. Uzdin, C. Pigier, Z. H. Musslimani, M. Segev, A. Nepomnyashchy, *.Rotating propeller solitons*., Quantum Electronics and Laser Science, (QELS) 2001 Baltimore, Maryland, USA

28) C. Anastassiou, J. W. Fleischer, **T. Carmon**, M. Segev and K. Steiglitz. *"Cascading collisions between vectors solitons",* Nonlinear Guided Waves and Their Applications, (NLGW) 2001, Florida, USA. **T. Carmon**, R. Uzdin, C. Pigier, Z. H. Musslimani, M. Segev, A. Nepomnyashchy, *.Two dimensional dipole vector solitons*., NATO Advanced study institute on Soliton Driven Photonics 2000. Swinoujscie, Poland.

Observation of two-dimensional multimode solitons

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We present the first experimental observation of (2 + 1)-dimensional multimode (composite) solitons. A single-hump component and a double-hump (dipole-type) component are jointly self-trapped as a composite soliton in a biased photorefractive crystal. © 2000 Optical Society of America

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An optical beam propagating in a self-focusing nonlinear medium is self-trapped and forms a spatial soliton when it is guided in its own self-induced waveguide.[1,2] When the soliton constitutes a single optical field (a scalar soliton), the induced waveguide can guide a single mode only (e.g., in Kerr media) or multiple modes (in a saturable nonlinearity such as a photorefractive[3]). In general, however, a soliton can involve more than one optical field, in which case it is called a vector soliton. This happens when the field components jointly induce a waveguide and trap themselves in it by properly populating the guided (eigen)modes.[2] In the degenerate case, all the field components populate the fundamental mode, and the total intensity structure of the soliton has a single hump.[4,5] Vector solitons, however, can also form when the field components belong to different modes of their jointly induced waveguide, as predicted in the temporal[6] and spatial[7] domains and observed in photorefractives.[8] Such solitons are called composite or multimode solitons, and their intensity can exhibit a single-hump or a multihump structure.[8] Single-hump and double-hump (1 + 1)-dimensional [(1 + 1)D] multimode solitons were found to be stable for a large range of parameters.[8,9] Following the progress in the development of saturable nonlinearities that can support (2 + 1)-dimensional [(2 + 1)D] scalar solitons,[2] it is natural to seek (2 + 1)D composite solitons. Such multimode solitons were first predicted in cylindrically symmetric systems, in which at least one of the components carries a topological charge.[10] Later, a dipole vector soliton was proposed, in which one of the modes resembles a TEM_{01} Gaussian mode.[11] Thus far, however, all experimentally observed multimode solitons have been in a (1 + 1)D geometry.[8,12]

Here we present the first experimental observation of composite (2 + 1)D solitons. We generate single- and double-hump bimodal solitons in which the first component is a TEM_{00}-type mode and the second component is a dipole mode exhibiting a TEM_{01}-like structure.

If vector or multimode solitons are to form, the interference between the field components must not contribute to the nonlinear index change, Δn. Otherwise, the induced waveguide changes during propagation, and thus the field components self-trapped within this waveguide, are not stationary. One method of satisfying this condition is to have the field components polarized orthogonally to each other.[4,6] Another method is to have the components at two widely spaced optical frequencies, so that the interference term is asynchronous with either of the components.[13] A third method relates to nonlinearities that have a noninstantaneous temporal response with a time constant τ. In this technique the components are of the same wavelength and polarization but are mutually incoherent with respect to each other.[14] Thus the phase of the interference term varies (randomly) in time much faster than the nonlinearity can respond, and therefore the contribution of the interference to Δn averages out. Here, by employing the coherence-based method[14] and the photorefractive screening nonlinearity,[15] we generate composite multimode solitons.

We expand and collimate an Ar^+ laser beam and split it into ordinary, *o*, and extraordinary, *e*, polarized beams, using a polarizing beam splitter. The *o* and *e* beams are polarized perpendicular and parallel, respectively, to the *c* axis of the crystal. The *o* beam is used as background illumination covering the crystal uniformly, as is necessary for screening solitons.[16] We then split the *e* beam in two to create the first (TEM_{00}) and the second (dipole) components. The dipole component is created by inserting a thin piece of glass into half of the beam. The glass is rotated until it induces a $(2m + 1)\pi$ phase shift (m is an integer) between the two halves of the beam, thus forming a dipole (TEM_{01}-like) mode. We make the first component incoherent with the dipole by having the optical length difference between the beams (1 m) exceed the coherence length of the laser (10 cm). Since the phase between these modes varies much faster than the response time of the crystal ($\tau \approx 1$ s), no stationary interference pattern forms on the time scale of the crystal's response.[14,15] The two modes are combined (with a beam splitter) and focused

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to the input face of a 1.1-cm-long $Sr_{0.6}Ba_{0.4}Nb_2O_6$ crystal. The input and output faces of the crystal are imaged on a CCD camera. To view the individual components, we block one component and sample the other in a time interval (~1 ms) much shorter than τ so that Δn does not change within the observation window.

In the first experiment we launch a 19-μm FWHM fundamental component, u, and a 20-μm (peak-to-peak separation) dipole, d. The peak fundamental intensity (normalized to the background intensity) is $|u_o|^2 = 14$ and is equal to the peak dipole intensity, $|d_{\text{peak}}|^2$. The intensities of $|u|^2$ (fundamental), $|d|^2$ (dipole), and $|u|^2 + |d|^2$ (total intensity) at the input face of the crystal are shown in Figs. 1(a), 1(b), and 1(c), respectively. The normally diffracting outputs after 1.1-cm propagation (~3 diffraction lengths) of $|u|^2$, $|d|^2$, and $|u|^2 + |d|^2$ are shown in Figs. 1(d), 1(e), and 1(f), respectively. When 5500 V is applied (in the crystalline c-axis direction) between the electrodes separated by 1.5 cm, a composite soliton forms. The intensities at the output face of the crystal, $|u|^2$, $|d|^2$, and $|u|^2 + |d|^2$, are shown in Figs. 1(g), 1(h), and 1(i), respectively. This composite soliton is double humped and looks exactly like the input, apart from a slight rotation around its center of mass.

The two modes self-trap when they are launched together, forming a composite soliton. However, the stand-alone components do not trap on their own at the value of the nonlinearity that supports the composite soliton. The reason for this has to do with the parameter range that supports a composite soliton: the existence range. Scalar solitons are governed by the soliton existence curve: A soliton of a given peak intensity and width-to-wavelength ratio forms at a particular value of the maximum index change.[17] A two-mode soliton, however, has an existence range that, in addition, depends on the relative strength of the first to the second mode.[8] From the theory of (1 + 1)D multimode solitons,[8] we know that for a given intensity of the first mode there is a range of intensities of the second mode for which self-trapping occurs. Intuitively, if the second-mode constituent is too large, then the composite soliton turns into two separate solitons that, having a relative phase of π, repel each other. To show this experimentally, we launch the first and the second modes separately and apply the same nonlinearity that gave rise to the composite soliton. For the stand-alone first mode the nonlinearity is too high (because a composite soliton requires a higher nonlinearity than a scalar soliton of the same width[10]) and the output is distorted, as shown in Fig. 1(j). On the other hand, the stand-alone dipole forms two separate solitons that repel each other to an output separation of 75 μm (~4 times larger than the input), as shown in Fig. 1(k).

In the next set of experiments (Fig. 2) we reduce the intensity of the dipole component to $|d_{\text{peak}}|^2 = 10$ so that the composite soliton will be single humped, and we go deeper into saturation by increasing $|u_o|^2$ to 30. The input intensities of the fundamental mode, the dipole mode, and total input intensity are shown in Figs. 2(a), 2(b), and 2(c), respectively. The corresponding normally diffracting output beams are shown in Figs. 2(d), 2(e), and 2(f), respectively. A composite soliton forms when 5300 V is applied. The corresponding self-trapped fundamental, dipole, and total intensity are shown in Figs. 2(g), 2(h), and 2(i). From Fig. 2(i) we can see that this composite soliton is single humped. Again the stand-alone components do not self-trap if they are not simultaneously launched: The first mode becomes elliptical [Fig. 2(j)], whereas the separation between the peaks of the dipole increases to 55 μm [Fig. 2(k)] from the input 17-μm separation of Fig. 2(h).

In conclusion, we have experimentally demonstrated the first composite (2 + 1)D soliton. Self-trapping of multimode [(2 + 1)D] solitons opens up the possibility of distortionless image transmission through highly

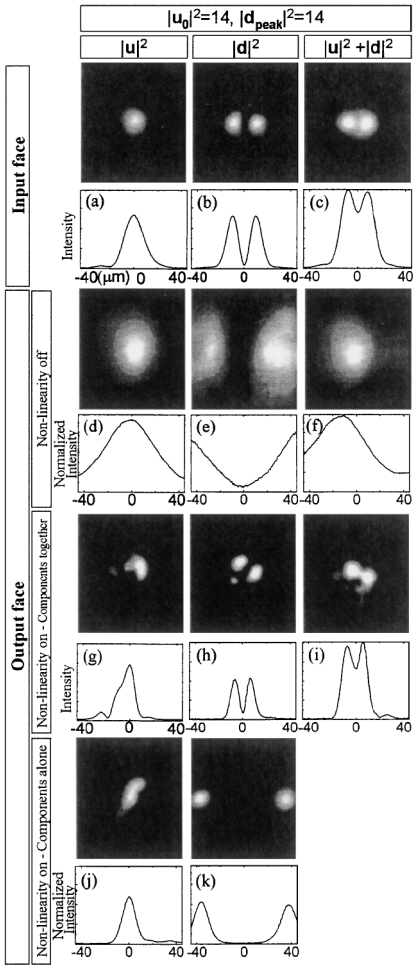


Fig. 1. Double-hump multimode [(2 + 1)D] soliton with $|u_o|^2 = |d_{\text{peak}}|^2 = 14$. (a)–(c) Input intensities: (a) $|u|^2$, (b) $|d|^2$, (c) $|u|^2 + |d|^2$. (d)–(f) Diffracted output intensities with nonlinearity off: (d) $|u|^2$, (e) $|d|^2$, (f) $|u|^2 + |d|^2$. (g)–(i) Composite soliton output intensities at 4230 V/cm: (g) $|u|^2$, (h) $|d|^2$, (i) total intensity of the double-hump soliton, $|u|^2 + |d|^2$. The individual stand-alone components at 4230 V/cm do not self-trap by themselves: (j) $|u|^2$ by itself is distorted because the nonlinearity its too high for a scalar soliton, and (k) the stand-alone dipole forms two separate solitons that repel each other. The intensity in each part of the figure is normalized to the peak intensity.

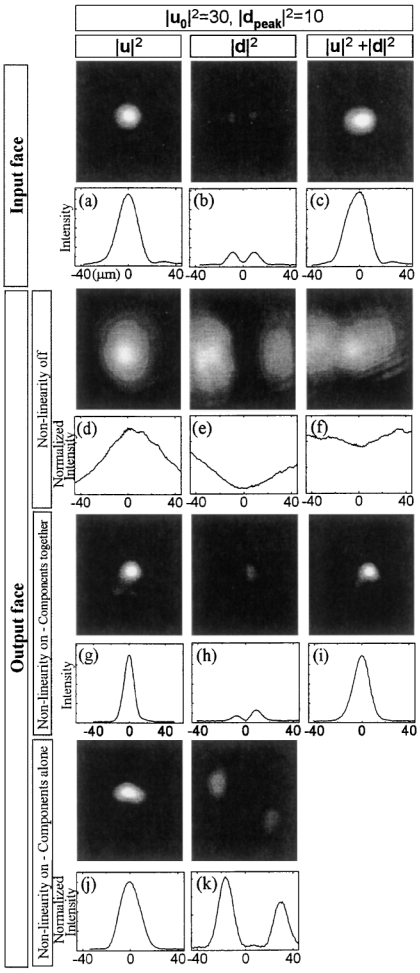


Fig. 2. Same as in Fig. 1 but with $|u_o|^2 = 30$ and $|d_{\mathrm{peak}}|^2 = 10$. This single-hump soliton forms for an applied field of 4060 V/cm.

nonlinear self-focusing media.[18] If the envelope of a highly multimode soliton is modulated to contain an image (superimposed on the intensity profile of the soliton), then this image can be transmitted through the self-focusing medium and remain unchanged through propagation. This method is in contradistinction to image transmission through a multimode fiber, for which the fiber modes are coherent with each other, yet they propagate at different velocities (intermodal dispersion) and thus destroy the image.

We are grateful to Yuri Kivshar for sending us the reprint of his theoretical paper[11] making the first prediction of dipole-type (2 + 1)D composite solitons. This work was supported by the Israeli Science Foundation, the U.S. Army Research Office, the U.S. Air Force Office of Scientific Research, and the Deutsche Forschungsgemeinschaft.

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Rotating Propeller Solitons

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We demonstrate experimentally and theoretically (both analytically and numerically) a new type of spatial soliton: a rotating "propeller" soliton. This is a composite soliton made of a *rotating* dipole component jointly trapped with a bell-shaped component. We observe as much as 239° of rotation over 13 mm of propagation (6.5 diffraction lengths).

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Self-trapping of an optical beam occurs when the beam induces, through a nonlinearity, a waveguide structure and, at the same time, is guided in its own induced waveguide [1]. Such beams are commonly referred to as spatial solitons [2]. The simplest (so-called scalar) soliton occurs when the soliton constitutes a single field, which populates the lowest mode of its own induced waveguide. A more complex soliton, a vector soliton, occurs when more than one field populates the lowest waveguide mode. These were first suggested by Manakov [3] in Kerr media, when the self- and the cross-phase modulations are equal. Manakov-like solitons were recently observed experimentally [4]. Moreover, solitons can also be composite: They can constitute optical fields that populate *different* modes of their jointly induced waveguide. In their simplest realization, composite solitons are made of a bell-shaped (bright) component populating the lowest mode and a dark component being the second mode [5,6]. A more interesting situation occurs when the field constituents of the composite soliton populate different bound modes of their jointly induced potential as suggested for temporal [7], as well as spatial [8], solitons. These multimode solitons, first observed in [9], offer many new features, e.g., shape transformation [10]. All of these studies refer to $(1 + 1)$D solitons which trap in a single transverse dimension.

Composite solitons in $(2 + 1)$D (which self-trap in both transverse dimensions) were proposed only recently: first in a vortex-type form [11], in which one component is bell shaped and the other is a ring carrying topological charge, and second in a dipole form, where the second component is a 2D dipole [12]. Experimental observations of $(2 + 1)$D dipole-type composite solitons followed soon thereafter [13]. The possibility of realizing $(2 + 1)$D composite solitons experimentally offers many new features, e.g., in their interaction [14] and stability [15] properties. Such solitons also directly relate to two-component Bose-Einstein condensates (BEC) [16], described by similar equations. In the BEC case, the components correspond to hyperfine states of the cooled atomic gas in a spherical trap, and resemble vortex-type composite solitons [11]. It is already obvious that the interactions between composite solitons, especially those that carry angular momentum, whether in the optical or atomic domains, offer unique properties.

Here we present a new type of composite soliton: a *rotating* dipole soliton. The intensity structure of the soliton constituents rotates throughout propagation. In particular, the dipole mode has a double-helix structure, and its equal-phase planes resemble propeller blades [Fig. 1(a)]. Hence, we name it a "*propeller soliton*." *This is the first soliton whose intensity structure (not only its phase) rotates during propagation.* The dimensionless equations for the slowly varying envelopes, Ψ_1 and Ψ_2, are

$$i(\partial\Psi_{1,2}/\partial\zeta) + \nabla_\perp^2\Psi_{1,2} + \Delta n(I)\Psi_{1,2} = 0, \quad (1)$$

where ζ is the propagation distance, $\nabla_\perp^2 = \partial^2/\partial\xi^2 + \partial^2/\partial\eta^2$, $\Delta n(I) = -1/(1 + I)$ is the refractive-index change, and $I = |\Psi_1|^2 + |\Psi_2|^2$ is the total intensity. This choice of Δn corresponds to propagation in a homogeneously broadened electronic two-level system, and is also a reasonable approximation of the photorefractive

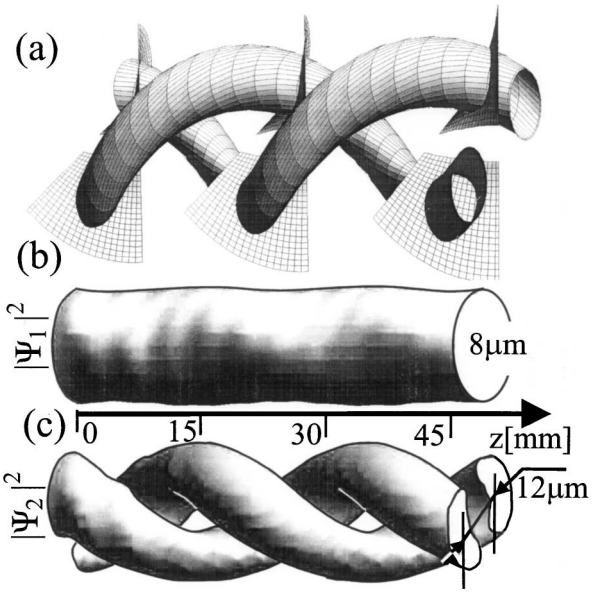


FIG. 1. (a) Illustration of the rotating-dipole mode of a propeller soliton. The propeller-blade surfaces represent the equal-phase surfaces. (b),(c) *Calculated* dynamics of the fundamental and dipole modes of a propeller soliton. Shown are the modal intensities for $|\Psi_1(\rho = 0)|^2 = 4$ and $|\Psi_2|^2_{\text{peak}} = 0.45$, in units of the saturation intensity. The slightly elliptic, 8 μm FWHM, fundamental mode (b) rotates in unison with a 12 μm peak-to-peak dipole (c).

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screening nonlinearity. The nonlinearity must be saturable, otherwise, e.g., in Kerr media, such 2D beams undergo catastrophic collapse. We emphasize that propeller-type composite solitons should exist in *any* saturable nonlinearity in a conservative (loss-free) system. Interference terms (e.g., $\Psi_1\Psi_2^*$) should not contribute to Δn, so the induced waveguide does not vary during propagation. There are several methods of eliminating the contribution of such terms to Δn [2,13]. We employ the method in which Ψ_1 and Ψ_2 are mutually incoherent, so the phase of $\Psi_1\Psi_2^*$ terms varies randomly in time much faster than the nonlinearity can respond, and their contribution to Δn averages out [17]. We seek self-trapped solutions, including those rotating about ζ, in the form

$$\Psi_{1,2}(\rho,\theta,\zeta) = \psi_{1,2}(\rho,\theta,\zeta)\exp(i\mu_{1,2}\zeta), \qquad (2)$$

where $\mu_{1,2}$ are the (real) propagation constants. The first, and simplest, solution is the radially symmetric, scalar (stationary in ζ) soliton, for which $\psi_1 = u_1(\rho)$ and $\psi_2 = 0$. A second case is the vortex-type composite soliton [11,14] for which $\psi_1 = u_1(\rho)$ and $\psi_2 = u_2(\rho)\exp(im\theta)$, where m (integer) is the topological charge. A third case is the dipole-type composite soliton [12] $\psi_1 = u_1(\rho,\theta)$ and $\psi_2 = u_2(\rho,\theta)$. For all of these three cases, the intensity structure is *stationary* in ζ. Here we are interested in a new case: *dipole solitons that rotate about* ζ (propeller-type composite solitons). Thus, neither ψ_1 nor ψ_2 have radial symmetry, and their intensity structure rotates while propagating.

Deriving the solutions for the rotating propeller soliton, we first present a *weakly nonlinear* analysis of the problem. It can be shown that this new class of solutions [in the form of Eq. (2)] appears through a *bifurcation* (in parameter space) from the family of radially symmetric scalar solutions. Near the bifurcation point, ψ_1 and ψ_2 can be expanded as $\psi_1(\rho,\theta,Z) = \psi_1^{(0)} + \varepsilon^2\psi_1^{(2)} + \ldots$ and $\psi_2(\rho,\theta,Z) = \varepsilon\psi_2^{(1)} + \varepsilon^3\psi_2^{(3)} + \ldots$, respectively, and the characteristic scale for the longitudinal coordinate is $Z = \varepsilon^2\zeta$. Here, $\psi_1^{(0)} = u_1(\rho)$ is radially symmetric, and $\psi_2^{(1)}(\rho,\theta,Z)$ satisfies the linear eigenvalue problem

$$\nabla_\perp^2\psi_2^{(1)} - \mu_2\psi_2^{(1)} + \Delta n(|u_1(\rho)|^2)\psi_2^{(1)} = 0 \qquad (3)$$

with the solution $\psi_2^{(1)} = u_2(\rho)[C_+(Z)\exp(i\theta) + C_-(Z)\exp(-i\theta)]$, where $C_\pm(Z)$ *cannot be determined in the framework of a linear theory*. The asymptotic weakly nonlinear theory, which takes into account the second order correction $\psi_1^{(2)}(\rho,\theta,Z)$, provides closed amplitude equations $idC_\pm/dZ = -C_\pm(a|C_\pm|^2 + b|C_\mp|^2)$, where a and b are constants calculated from the solvability condition in the third order in ε. Defining $C_\pm(Z) = R_\pm\exp(i\phi_\pm)$ yields $\partial\phi_\pm/\partial Z = (aR_\pm^2 + bR_\mp^2) \equiv \Omega_\pm$, where $R_\pm$ and $\Omega_\pm$ are real constants. Hence, $C_\pm(Z) = R_\pm\exp(i\Omega_\pm Z)$ and

$$\psi_2^{(1)} = u_2(\rho)[R_+\exp i(\Omega_+ Z + \theta) + R_-\exp i(\Omega_- Z - \theta)]. \qquad (4)$$

This family of solutions includes the vortex-type composite soliton [11,14], $\psi_2^{(1)} = u_2(\rho)[R_+\exp i(\Omega_+ Z + \theta)]$ (recovered for $R_- = 0$), whose phase rotates, but whose intensity structure, $|\psi_2^{(1)}|^2$, has radial symmetry and displays no rotation. Another member of this family is the nonrotating dipole-type composite soliton [12], $\psi_2^{(1)} = u_2(\rho) \times [R\exp i(\Omega Z)\cos(\theta)]$, recovered when $R_\pm = R/2$, $\Omega_\pm = \Omega$. This radially asymmetric dipole does not rotate, not even in its phase. Except for these particular cases, all other solutions described by Eq. (4) are new solutions having both their intensity structure and their phase rotating during propagation. These are the propeller-type solutions. The rotation rates of the intensity and the phase are different: The intensity, $|\psi_2^{(1)}|^2 = u_2^2(\rho)\{R_+^2 + R_-^2 + 2R_+R_- \times \cos[(\Omega_+ - \Omega_-)Z + 2\theta]\}$, rotates with an angular velocity $(\Omega_- - \Omega_+)/2$, whereas the phase of $\psi_2^{(1)}$ rotates with a *different* angular velocity. The rotation direction is determined by which is larger, R_+ or R_-. The Poynting vector forms a helix during propagation, because the energy flow is in a direction perpendicular to the equal-phase planes [Fig. 1(a)]. *Because the rotation rates of the intensity and the phase are different, there are no stationary solutions in any rotating frame.* One cannot simply generalize the stationary-dipole composite solitons found earlier [12] and obtain the propeller solitons by transforming the equations into a rotating frame. Propeller solitons are new creatures and can be considered as more general than the stationary vortex- and dipole-type composite solitons [11,12].

To find a propeller-type composite soliton solution, one can choose the parameters determining the rotation rates rather freely (one can increase $|\psi_2|^2$ as long as $|\Omega_\pm| \ll |\mu_2|$) and use a 1D relaxation code to find $u_1(\rho)$, $u_2(\rho)$, μ_1, and μ_2 from Eq. (1) by replacing $\Psi_{1,2}$ by $u_{1,2}(\rho) \times \exp(i\mu_{1,2}\zeta)$. This procedure holds as long as $|\psi_1|^2 \gg |\psi_2|^2$. When the modal intensities are comparable, a 2D relaxation code should be used to find $\Psi_{1,2}(\rho,\theta,\zeta)$. To confirm and demonstrate the propeller-type soliton, we investigate the propagation of the solution given by Eq. (4), by launching this solution into the nonlinear medium described by Eq. (1), and observe the propagation dynamics using a standard (2 + 1)D beam propagation code for two coupled fields [14]. The results are shown (in dimensional units [18]) in Figs. 1(b) and 1(c) and display the propagation of a solution with $|\Psi_1|^2_{\text{peak}}$ nine times larger than the peak intensity of the dipole mode $|\Psi_2|^2_{\text{peak}}$. In this example, $R_+ = 0.9$, $R_- = 0.15$, displaying an 8 μm FWHM almost circular bell-shaped Ψ_1, and a rotating dipole-type Ψ_2 with a 12 μm separation between the "poles" of the dipole. The two modes ($\Psi_{1,2}$) rotate together, in a self-trapped fashion, with a rate of $\sim 9^\circ$ per mm [18]. The propagation results [Figs. 1(b) and 1(c)] convincingly indicate that the solution of Eq. (4) is a good approximation, since both Ψ_1 and Ψ_2 exhibit stable self-trapped propagation and rotate in unison. In our simulations, we find that the *total* power of *each* field is conserved, and the linear and angular momenta of all fields *together* are conserved.

We emphasize that, although the solution of Eq. (4) is valid only within the weakly nonlinear regime, composite propeller solitons do exist in a larger range of parameters.

In fact, our simulation [Figs. 1(b) and 1(c)] shows that Ψ_1 has a slightly elliptic shape thus extending beyond the $|\Psi_1|^2 \gg |\Psi_2|^2$ regime.

Experimentally, we generate the propeller solitons in the photorefractive media, employing the screening nonlinearity [19], and the mutual incoherence method to observe multicomponent solitons [17]. We expand and collimate a 488 nm laser beam and split it into Ψ_1 and Ψ_2. The Ψ_2 beam is first passed through a helical phase mask, which multiplies the beam by $\exp(i\theta)$, providing the angular momentum necessary for rotation. Then, to shape it as a dipole, the plane of the mask is reproduced (with a $4f$ system) and a metal wire is introduced so that it crosses the beam from side-to-side through its center. The components are combined (with a beam splitter) after ensuring that the path difference between them exceeds the coherence length of the laser [9,13]. The combined structure is imaged onto the input face of the crystal, so that the input face is a demagnified image plane of the vortex crossed by the wire. This structure is propagating along the a axis of an $Sr_{0.6}Ba_{0.4}Nb_2O_6$ (SBN:60) crystal, in the standard configuration [20]. The crystal is also illuminated uniformly with a background beam (to tune the degree of saturation of the nonlinearity), which is polarized orthogonally to the Ψ_1, Ψ_2 beams. The background beam is made spatially incoherent (with a rotating diffuser), and remains uniform even when large fields are applied. The intensity distributions at the input and output faces are imaged on a CCD camera. The modes can be inspected individually by using the method described in [9,13].

We measure the rotation rate and propagation dynamics of the dipole mode at various distances, by launching the soliton in a series of samples of 1, 6, and 13 mm lengths, all of which being SBN:60 crystals of the same properties. *These samples have been verified to support solitons under the same parameters.* Figures 2(a)–2(d) show photographs of $|\Psi_2|^2$ for $m = -1$ at the crystal input and after propagating 1, 6, and 13 mm, respectively, when jointly trapped with $|\Psi_1|^2$ (not shown here), together forming a soliton. The dipole rotation rate vs propagation distance is shown in the graph (Fig. 2), displaying a rotation rate of 20° per mm, and a total rotation (for the 13-mm-long sample) of up to 239°. The discrepancy between this rate and the calculated rate (of Fig. 1) results from the fact that the calculation is conducted in the weakly nonlinear regime using a 1D relaxation code to find the wave functions. The approximation leads to rather slow rotation rates. Experimentally, we are motivated to display as much rotation effects as possible, even though at present we cannot simulate them. To properly simulate our experimental results, a two-dimensional relaxation code is necessary. We also expect the anisotropy of the photorefractive screening nonlinearity [21] to modify the rotation rate from that in a fully isotropic nonlinearity (as assumed in our model). To a large extent, the very fact that the propeller composite soliton survives (experimentally), in spite of the anisotropy, highlights its stability and robustness.

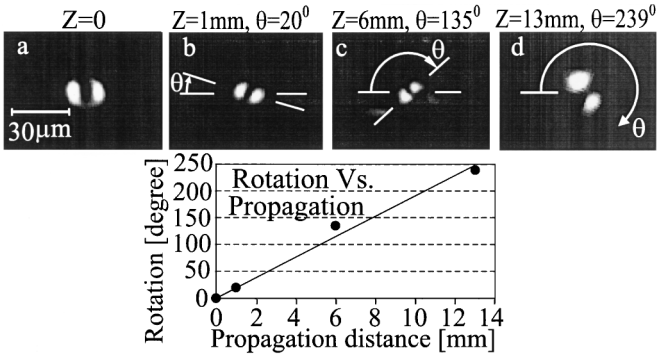


FIG. 2. Experimentally observed propagation of the dipole mode of a propeller soliton with $|\Psi_1|^2_{\max} = 10$, $|\Psi_2|^2_{\max} = 1$, and $m = -1$. (a) Input, with 13.5 μm separation between the dipole peaks. (b) Trapped $|\Psi_2|^2$ output after 1 mm propagation, displaying 20° rotation. (c) Trapped $|\Psi_2|^2$ output after 6 mm propagation, displaying 135° rotation. (d) Trapped $|\Psi_2|^2$ output after 13 mm propagation, displaying 239° rotation. The graph displays the rotation angle vs distance, extracted from (a) to (d), with a linear fit; the measurement errors are ± 0.05 mm for the propagation length and $\pm 3°$ for the rotation angle.

In order to prove that the angular momentum of the dipole is the reason for its rotation, we compare the rotation of identical composite solitons with opposite topological charges. The input fields [Figs. 3(a) and 3(f)], Ψ_1 and Ψ_2, are the same, except that the topological charge is reversed from $m = -1$ to $m = 1$. After 6 mm propagation, the former rotates 135° clockwise [Fig. 3(b)], whereas the latter rotates 142° counterclockwise [Fig. 3(c)].

We emphasize that the propeller soliton consists of two modes: The rotating dipole mode cannot self-trap all by it-

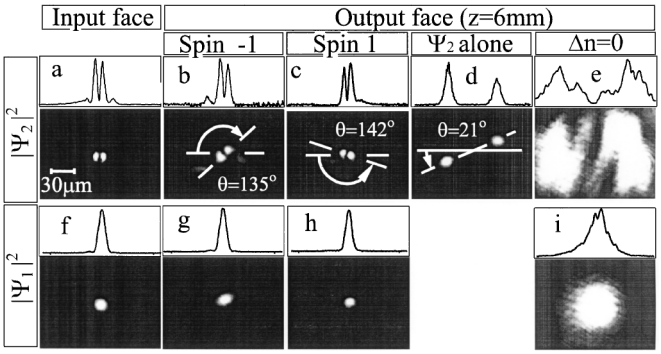


FIG. 3. Experimentally observed propagation of a propeller soliton with $|\Psi_1|^2_{\max} = 10$, $|\Psi_2|^2_{\max} = 1$ for 6 mm propagation (three diffraction lengths). (a),(f) Input intensities of the modes. The separation between the dipole peaks in (a) is 11 μm, and the FWHM of the fundamental mode in (f) is 11 μm. (b),(g) Self-trapped output with 3300 V/cm applied. The dipole mode rotates by 135° and its peak-to-peak separation is 13 μm. The fundamental mode is slightly elliptic. (c),(h) Same as (b),(g) with opposite spin $m = 1$. Inverting the topological charge leaves all dimensions and rotation angles similar to those of (b)-(g), but leads to rotation in the opposite direction. (d) Output dipole mode at 3300 V/cm when the fundamental mode is blocked. The dipole evolves into two solitons separated by a distance of 114 μm and overall rotation of 21°. (e),(i) Output intensities of the dipole mode and the fundamental mode when the nonlinearity is set to zero. Separation between peaks in (e) is 113 μm, and the FWHM of the fundamental mode in (i) is 48 μm.

self when the fundamental component is blocked. Instead, when we turn off the fundamental mode Ψ_1 and examine the propagating stand-alone dipole mode Ψ_2, the input dipole evolves into a pair of repelling (2 + 1)D solitons [Fig. 3(d)] with an output separation about 10 times larger than the distance between the poles of the composite propeller soliton [Fig. 3(c)]. Note that the input Ψ_2 has angular momentum, and thus the output Ψ_2 is rotated, but the rotation angle is small. This is because the angular momentum is conserved (as we indeed find in our numerical simulations). Thus, the rotation rate of the "repelling poles" of Fig. 3(d) decreases with the peak-to-peak separation distance squared (the "skater on the ice effect").

Finally, we examine the propagation of both modes (Ψ_1 and Ψ_2) in the absence of nonlinearity, that is, during linear diffraction. When we turn the nonlinearity off (by turning off the bias voltage), the beams diffract to the structures shown in Figs. 3(e) and 3(i) and the modes are no longer trapped. The beams expand considerably, and the separation between the dipole peaks increases by more than tenfold. This shows that the 6 mm propagation distance in our experiments is $\approx$3 diffraction lengths.

In conclusion, we have demonstrated propeller-type composite solitons: bimodal solitons that contain a rotating dipole. This idea can be extended to include rotating quadrupoles, hexapoles, etc. What makes propeller-type solitons so interesting is the fact that they display two different rotations: the rotation of their intensity structure (the double helix) and the rotation of their phase. To a large extent, propeller-type composite solitons resemble rotating molecules. This adds another feature to the analogy between solitons and particles, and the extension to multisoliton systems, including collisions, seems promising. Finally, we point out the relevance to other systems that support two-component solitons, and specifically to two-component BECs [16]. For BECs, self-trapping occurs because of the balance between kinetic energy and repulsive atom-atom collisions. Two-component BECs form by trapping atoms at different electronic states [22]. Since dark solitons in BECs have already been observed experimentally [23], we envision that multicomponent solitons, in particular those that carry angular momentum, can be constructed with atoms. The phenomenon, however, of composite solitons carrying angular momentum is universal.

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[18] The transformation of the dimensionless Eq. (1) into dimensional units is through $\Delta n(I) = -\Delta n_0/(1 + I)$. (ξ, η) coordinates are scaled by $\frac{\lambda}{2\pi}(\frac{n}{2\Delta n_0})^{1/2}$, and ζ is scaled by $(\frac{\lambda}{2\pi}\frac{n}{\Delta n_0})$. Typically, as in our experiments, $\Delta n_0/n$ is of the order of 2×10^{-4} for 10 μm FWHM solitons at $\lambda =$ 0.5 μm. Note, that Eq. (1) is not integrable, so some power could escape from the soliton to radiation modes. This may lead to nonstationary propagation. In our simulations, power escape is small so there is no observable change in the soliton shape up to distances in excess of 30 diffraction lengths.
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Pattern Formation in a Cavity Longer than the Coherence Length of the Light in It

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We study, theoretically and experimentally, the evolution of patterns in a passive nonlinear cavity that is longer than the coherence length of the light circulating in it. The patterns exhibit spatial line narrowing as the feedback is increased, resembling the line narrowing in lasers.

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Pattern formation in optical cavities has been the subject of continuing interest since lasers were discovered [1–4]. Patterns naturally start from noise, from which some frequencies experience higher gain, and, through the feedback in the cavity, they stabilize and form a pattern. Such patterns appear not only in laser cavities, but also in passive cavities employing optical nonlinearities [2,4]. Pattern formation in cavities possesses several common features that make it distinct from the patterns arising [via modulation instability (MI)] during propagation in the absence of feedback [2]. First, the instability leading to pattern formation in a cavity is "global" (absolute), whereas MI without feedback arises due to a "convective" instability. Second, cavity pattern formation (including a passive cavity) has a definite threshold. In all passive cavities studied thus far patterns form only above the pattern-formation threshold. Third, a cavity gives rise to a set of resonant frequencies, and the detuning between the frequency of the light and the nearest resonant frequency critically affects the pattern. These features are universal and also appear in the temporal domain [5]. However, all resonators studied previously in the context of pattern formation are fully coherent; i.e., the beam is monochromatic, spatially coherent, and coherent with the feedback beams circulating in the resonators for many cycles. That is, the cavity round-trip time is much shorter than the coherence time of the light.

Here we explore pattern formation in a different kind of cavity: a passive ring cavity for which the coherence length of the light is much shorter than the cavity length; that is, the coherence time is much shorter than the time it takes the light to go once around the cavity. In this low-finesse cavity, the feedback beams from different cycles are mutually incoherent with one another (the relative phase among them is random). The nonlinear medium in our cavity responds much slower than the characteristic phase fluctuation time between beams from different cycles. Thus, the resonant frequencies of our cavity, as well as the relative phase between beams from different cycles, do not play any role in the pattern-formation process. The beams from different cycles interact with one another through a nonlinear index change, Δn, that is a function of the sum of their intensities but does not depend on interference cross terms. This interaction resembles cross-phase modulation [6], yet it leads to different phenomena. We find that the patterns emerging in our cavity exhibit spatial line narrowing as the feedback and/or the nonlinearity are increased, closely resembling the narrowing of the frequency linewidth in lasers [7].

Consider the system in Fig. 1(a). A plane wave enters, passes through a nonlinear medium, and circulates in the cavity. The cavity length greatly exceeds the coherence length of the laser, so interference terms between beams from different cycles fluctuate at characteristic times equal to the coherence time of the laser, t_c. The nonlinearity has a response time $\tau \gg t_c$; thus Δn is a function of the intensity I averaged over τ. The input and output faces of the nonlinear crystal are $z = 0$ and $z = L$. A fraction, ϵ^2, of I exiting the crystal is "recycled" in the cavity.

We model our cavity when the feedback is small and the pattern is of a low modulation depth (visibility), so the entire dynamics (in temporal steady state) results from the interaction between the input beam Ψ, and the feedback beam Φ that went one cycle around the cavity. Our Δn is a function of $I = \langle|\Psi + \Phi|^2\rangle = \langle|\Psi|^2\rangle + \langle|\Phi|^2\rangle$, where the time average $\langle\,\rangle$ is taken over τ. Since in MI studies, any saturable nonlinearity initially acts as a Kerr nonlinearity for the perturbation (with the nonlinear coefficient evaluated at the intensity of the input beam) [8], we take $\Delta n \propto I$ (Kerr self-focusing). The

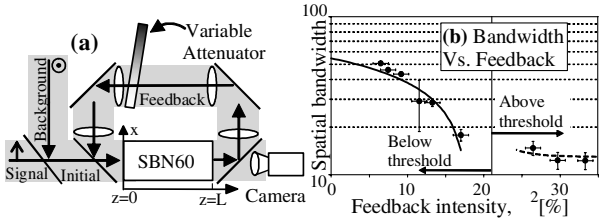


FIG. 1. (a) The incoherent resonator. (b) Bandwidth of the spatial frequency power spectrum (FWHM) as a function of feedback intensity. The experimental results are represented by crosses. The theoretical results below threshold (solid line) are based on Eq. (4) and the experimental parameters. The dashed curve above threshold is a guide to the eye.

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dimensionless equations in one transverse dimension are

$$i\frac{\partial\Psi}{\partial z}+\frac{1}{2}\frac{\partial^2\Psi}{\partial x^2}+I\Psi=0; \qquad i\frac{\partial\Phi}{\partial z}+\frac{1}{2}\frac{\partial^2\Phi}{\partial x^2}+I\Phi=0. \tag{1}$$

We seek solutions of the form $\Psi(x, z, t) = \psi(x, z)\exp(i\mu z)$, $\Phi(x, z, t) = \phi(x, z)\exp(i\nu z)\exp[i\alpha(t)]$, where μ, ν are real. $\alpha(t)$ is a stochastic phase that varies much faster than τ. Taking a fraction $\epsilon(\ll 1)$ of $\Psi(x, z = L)$ from the crystal output and imaging it as feedback to the input $z = 0$ implies $\epsilon^2|\psi(x, z = L)|^2 = |\phi(x, z = 0)|^2$. Ψ starts as a plane wave ψ_0 and develops a periodic pattern due to MI. When the modulation depth of this pattern is low, ψ and ϕ each can be written as a sum of a plane wave and perturbation: $\Psi(x, z) = [\psi_0 + \psi_1(x, z)]\exp(i\mu z)$ and $\Phi(x, z) = [\phi_0 + \phi_1(x, z)]\exp(i\nu z)$, where $|\psi_1| \ll |\psi_0|$, $|\phi_1| \ll |\phi_0|$. Substituting Ψ, Φ in Eq. (1) yields $\mu = \nu = \psi_0^2$. The boundary condition gives $\phi_0 = \epsilon\psi_0$. The linearized equations for ψ_1, ϕ_1 are

$$i\frac{\partial\psi_1}{\partial z}+\frac{1}{2}\frac{\partial^2\psi_1}{\partial x^2}+\psi_0^2(\psi_1+\psi_1^*)+\phi_0(\phi_1^*+\phi_1)\psi_0=0, \tag{2}$$

$$i\frac{\partial\phi_1}{\partial z}+\frac{1}{2}\frac{\partial^2\phi_1}{\partial x^2}=0. \tag{3}$$

Here, $|\psi_0| \gg |\phi_0|$, thus ϕ is propagating mostly linearly. Without loss of generality, we seek $\phi_1(x, z) = \tilde{\phi}_a\exp[if]\cos[qx]\exp[-iq^2z/2]$, $\psi_1(x, z) = \tilde{\psi}_a\exp[if]\cos[qx]\exp[-iq^2z/2] + \psi_b\exp[i(qx + \Omega z)] + \psi_c\exp[-i(qx + \Omega^* z)]$, where $\tilde{}$ denotes the (real) modal amplitude and f is real [9]. Substitution of ψ_1 in Eq. (3) yields $\tilde{\psi}_a = -\epsilon\tilde{\phi}_a$ and $\Omega = \pm q\sqrt{(q^2/2) - (\psi_0^2)}$. For a proper Ω this pattern (with $2\pi/q$ periodicity) grows exponentially. We impose $\epsilon\Psi(x, z = L) = \Phi(x, z = 0)\exp[i\alpha(t)]$. Because $\Delta n \propto \langle I\rangle$, $\alpha(t)$ has no influence on the evolution of ψ. Using the boundary condition, the dependence of the spatial spectrum, $\hat{\psi}_1$, on q, is

$$\hat{\psi}_1(q, z = L) \propto \frac{\exp(|q|L\sqrt{\psi_0^2 - q^2/4})}{[1 - \epsilon^2\exp(|q|L\sqrt{\psi_0^2 - q^2/4})]}. \tag{4}$$

For $\epsilon = 0$ Eq. (4) converges to the known case of MI without feedback [10]. As ϵ^2 is increased, Eq. (4) yields band narrowing of the spatial spectrum, and there is a particular q at which Eq. (4) diverges, and this q dominates. This is apparent in Fig. 1(b) which shows the calculated width of $|\hat{\psi}_1|^2$ as a function of ϵ^2 (solid curve). This bandwidth narrowing means that only a single, well-defined, spatial frequency (the one with the highest growth rate) is visible at the output. Increasing ϵ^2 further causes $\hat{\psi}_1$ to diverge. This behavior is characteristic of feedback systems with gain, such as lasers [7]. The divergence indicates the presence of an oscillation threshold [11].

The intuition behind our system is subtle. First, the propagation of ϕ is mostly linear. This is manifested by Eqs. (1): ϕ and ψ cannot exchange energy because they are incoherent with one another (the temporal fluctuations in their fields are uncorrelated). ϕ_1 could grow if it could drain energy from ϕ_0. But the feedback is weak: $|\phi_0| \ll |\psi_0|$, so the MI in ϕ is much smaller than MI in ψ. Thus, MI in ϕ can be ignored. The only way ϕ_1 can evolve is due to Δn produced by ψ_1. But this happens only for large z's, where it does not have much influence on ψ, since ψ dominates its own propagation there. Nevertheless, the propagation of ϕ, albeit mostly linear, establishes a grating of period $2\pi/q$ at $z = 0$, providing a preferential "noise" for ψ. Hence, the MI in ψ does not evolve from random noise (as for $\epsilon = 0$), but instead is channeled by a particular periodic Δn induced by ϕ_1 at $z = 0$. This is embedded in the last term in Eq. (2). The contribution of ϕ_1, albeit small, is crucial. Another way to think about this is to consider the periodic Δn induced by ϕ_1 as a grating, and ψ scatters from it, changing its transverse momentum by q. Thus the q component in ψ is much bigger than it would have been if the seed grating was absent. Third, through the self-consistent process in the cavity, the q that has the largest gain is the one that grows the fastest. From this picture it is obvious that the MI growth rate is the same as for $\epsilon = 0$, but the noise gets a preferential periodicity $2\pi/q$, and this q is the one that grows fastest in MI. The larger the feedback, the stronger the seed "grating" in the input, and the stronger the amplification of that preferential grating with respect to all other gratings arising from noise. Finally, when the amount of feedback is identical to the noise amplitude in the q of highest gain, this specific q oscillates, just as in lasers. This oscillation occurs when the nonlinear gain is equal to the loss in a single cycle, as the denominator of Eq. (4) implies. When the expression in Eq. (4) diverges, the system oscillates and exhibits bandwidth narrowing [Fig. 1(b)].

In our experiment [Fig. 1(a)], we use a 488 nm Ar^+ laser beam with a 10 cm coherence length and circulate it in a 80 cm long ring cavity. The transmission spectrum of our cavity never drops more than 55% (at threshold) allowing all the bandwidth of our laser to circulate in it [12]. The nonlinear medium is an $Sr_{0.6}Ba_{0.4}Nb_2O_6$ crystal, employing the screening nonlinearity [13]. We split the laser beam into an extraordinarily polarized "signal" beam and an ordinarily polarized "background" beam [not shown in Fig. 1(a)].(The background beam is necessary for the photorefractive screening nonlinearity [13].) The ratio between the intensities of the signal and background beams is 2.1, and the applied field is 340 V/cm. The background beam is made to be highly spatially incoherent (to eliminate possible MI on it [8]), by passing it through a diffuser rotating much faster than τ. The signal and background beams are combined by a second beam splitter, expanded to a width greater than the crystal width, launched into the 5.5 mm long

crystal and propagate along its a axis. We reflect a fraction of both signal and background beams exiting the crystal, and "recycle" them in the cavity (keeping the ratio between their intensities fixed when the feedback is varied). In the feedback loop we use four lenses forming two sequential $4f$ systems, so that the intensity at the crystal output face is imaged 1:1 onto the input face (without mirror inversion). The feedback is controlled by a variable attenuator. All apertures are large enough to pass all the spatial frequencies in the system.

Starting with zero feedback, we observe emergence of low-visibility stripes (from MI), which do not have a well-defined periodicity [14]. Typical results are shown in the upper row of Fig. 2 (left) along with the spatial (Fourier) power spectrum (right). When we unblock the feedback and increase its intensity, ϵ^2, we observe obvious linewidth narrowing. We take data with ϵ^2 ranging from 0% to 33%, while keeping all other experimental parameters constant, especially the value of the nonlinearity (determined by the applied field, and the ratio between signal and background beams [13]). The linewidth narrowing is obvious (Fig. 2), as the stripes become sharper and regular with increasing feedback. The spatial power spectrum goes from multiple low-intensity peaks (at $\epsilon = 0$) to two isolated narrow peaks at high feedback. We compare theory and experiments by plotting the measured spatial bandwidth of the spectrum as a function of ϵ^2 [Fig. 1(b)]. The measured power spectrum is taken from the right column of Fig. 2. We took many more data points of different ϵ^2 values and from other regions of the crystal output, confirming that the behavior is always consistent with Figs. 1 and 2. The threshold behavior is clear in Fig. 2: the visibility of the stripes and the energy in the first spatial harmonic exhibit a clear jump at the threshold value. Comparing theory and experiments [Fig. 1(b)] shows much similarity [15]. At 6% feedback, a specific spatial frequency starts to dominate, and one can determine its bandwidth and plot it [Fig. 1(b)]. Increasing ϵ^2 further makes the bandwidth narrower with an increasing slope. The increasing negative slope in Fig. 1(b) between 0% and 15% feedback is a direct quantitative measure of the line narrowing of the stripes. When we increase ϵ^2 beyond 15%, we observe a clear inflection point in the graph, at which the second derivative of the bandwidth changes sign. This is the oscillation threshold. Increasing the feedback further results in more narrowing, but with a decreasing slope. The theoretical curve in Fig. 1(b) is calculated from Eq. (4) with the actual experimental parameters, adjusting only one parameter, ϕ_0, which cannot be measured independently. We measure the total intensity I and from it evaluate ψ_0 to be 53% of $\sqrt{I}$ via the best fit to our experiments.

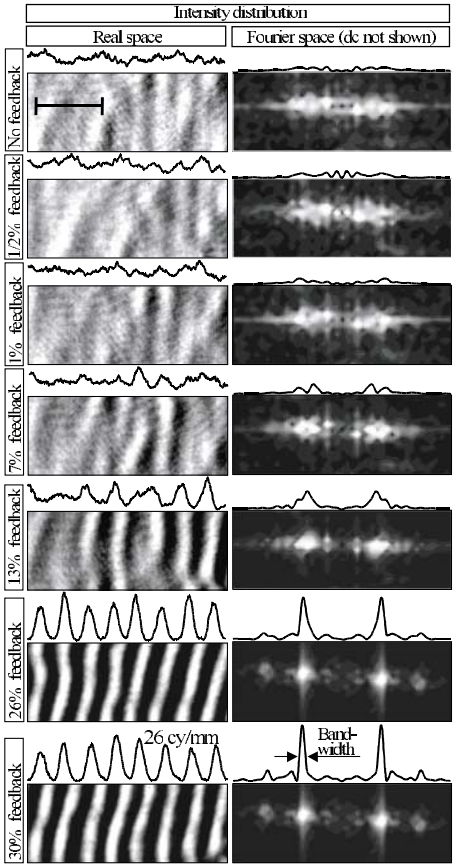


FIG. 2. Experimental results, bandwidth narrowing while increasing feedback. Shown are photographs of the intensity distribution in a specific region at the crystal output (left), along with the calculated spatial power spectrum (right). All measurements are taken without moving the crystal and without changing the nonlinearity. Bandwidth narrowing is obvious: the stripes become sharper and regular with increasing feedback, and the spatial power spectrum goes from multiple peaks (at no feedback) to two narrow peaks at high feedback.

Figure 1(b) proves that our theory truly describes the behavior of our system from zero feedback to the threshold. The threshold value, 14%, is beyond the $\epsilon \ll 1$ limit. In spite of that, our experiments agree well with the theory up to the threshold. Above threshold, we have no theory, but experimentally, we observe line narrowing down to the narrowest linewidth possible in our system. The characteristic behavior almost one-to-one resembles linewidth narrowing in lasers [7]. This is characteristic of systems undergoing a phase transition. We believe that other phase transition features occur in our system, with different characteristics than those of coherent cavities. For example, critical slowing down near the threshold should have different features from those in coherent cavities [16] and self-oscillators [17]. We emphasize that in our system the visibility of the pattern is always enhanced in the cavity as compared to MI without feedback under the same parameters. This is true even *below* threshold, in contrast to passive coherent cavities [2,4], where below threshold patterns are suppressed. Clearly, pattern formation in our cavity has different features than in a coherent cavity.

Let us discuss the differences between pattern formation in our cavity and in a single-mirror nonlinear

feedback system [4,18,19]. Most single-mirror systems rely on a fully coherent feedback, for which pattern formation is dominated by reflection gratings forming via interference between the input and feedback beams [19]. The difference between these cases and ours is obvious, as our cavity interference of beams from different cycles does not lead to an index grating. Some single-mirror systems do show patterns driven by incoherent feedback [18], which is seemingly similar to our cavity. Nevertheless, there are several major differences. For the single mirror, the spatial frequency of the pattern is fully determined by Talbot self-imaging (upon reflection from the mirror), which critically depends on the (small) spacing between the nonlinear sample and the mirror. This small spacing limits the range of spatial frequencies that can evolve. In our cavity, the cavity length is unimportant (as long as it is $\gg ct_c$), and there are no hard restrictions on the spatial frequencies, as we have a double $4f$ imaging system in the feedback. The spatial frequency of the pattern in our cavity depends only on Δn, the length of the nonlinear medium, and the Q factor of the cavity. Therefore, our cavity reveals fully the natural frequencies of the pattern arising from the MI, unaffected by bounday conditions. This is why we can study line narrowing of the natural spatial frequencies of the pattern. The single-mirror system is restricted by external boundary conditions, limiting the spatial frequencies that can evolve, whereas our cavity lets the spatial frequency of the highest nonlinear gain win.

In conclusion, we studied the evolution of patterns in a nonlinear cavity which is longer than the coherence length of the light circulating in it. The patterns exhibit spatial line narrowing as feedback is increased, resembling linewidth narrowing in lasers. This effect is characteristic of phase transitions, yet unlike other previously studied optical cavities, our cavity is incoherent. Our cavity relates to classical phase transitions, because its behavior does not depend on the phase, whereas pattern formation in coherent cavities is inherently phase dependent, thus related to nonclassical phase transitions. The distinctions between coherent and incoherent cavities and the parallels with lasers and phase transitions offer many exciting ideas. The ability to study pattern formation in systems of varying correlation has implications beyond optics; e.g., we envision partially incoherent cavities containing cooled atoms. We note the relation to patterns in granular materials (sand) [20], which displays the formation of stripes in a phase-independent system. In optics, however, the next idea is to study pattern formation in a fully incoherent cavity, that is, a cavity in which the circulating beam is spatially and temporally incoherent.

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Cavity pattern formation with incoherent light

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We study the propagation dynamics of an incoherent light beam circulating in a passive cavity containing noninstantaneous nonlinear media. It is shown that patterns form in this cavity in spite of spatial incoherence of the light. We show that the pattern formation process is always associated with two consecutive thresholds. The first (instability) threshold is unaffected by the cavity boundary conditions, whereas the second threshold is induced by the feedback through the interplay of nonlinear gain and cavity loss.

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I. INTRODUCTION

Nonlinear optical systems with feedback and associated phenomena, such as pattern formation [1] and cavity solitons [2,3], have been continuously drawing attention for several reasons. On the fundamental side, the understanding of nonlinear optical phenomena contributes immensely to the understanding of nonlinear dynamical systems in general, and has a direct impact on other fields [4]. Equivalent nonlinear phenomena appear in various areas of physics, chemical, and biological systems [4]. From the applications standpoint, these systems can be engineered to perform as useful devices for switching, storing, and manipulating information [5]. The phenomenon of pattern formation refers to the fact that in an extended nonlinear medium, above a suitable threshold, any uniform intensity distribution of light becomes unstable, and splits into space (time) correlated domains [1]. In nonlinear optical cavities, patterns can assume a variety of forms: stripes, hexagons, rolls etc. [6]. Nonlinear optical cavities can also give rise to cavity solitons [2,3,5]. However, all previous studies of nonlinear optical cavities have considered only spatially coherent light [7].

Here we present the study of pattern formation in a nonlinear optical cavity with spatially incoherent light. The system is a passive ring cavity of length L_c, containing a nonlinear medium (crystal) of length $L \ll L_c$. The intensity structure from the output face of the crystal is attenuated by a factor ϵ and imaged to the input face of the crystal by using conventional optics. The light entering the cavity is partially spatially incoherent yet quasimonochromatic, with temporal coherence length l_{coh} much shorter than the cavity length: $L \ll l_{coh} \ll L_c$. The finesse of the cavity is low, of order one or less, which ensures that the temporal coherence length of the light is not increased by any filtering process in the cavity. Experimentally, this requirement can be achieved simply by making the length of the cavity large enough, since this reduces the separation between the resonant frequencies thereby decreasing finesse. The nonlinear medium has a noninstantaneous response; its response time is much longer than (i) the characteristic time of phase fluctuations across the beam and (ii) the average time of phase fluctuations between the beams from different cycles. The medium responds only to the time-averaged intensity [8,9]. This is the key physical mechanism responsible for the pattern formation and cavity solitons in this *incoherent cavity*. Examples for noninstantaneous nonlinear media are photorefractive crystals [8], liquid crystals [9], polymers, etc.

The main goal of this manuscript is to analyze the early stage of pattern formation process in the incoherent cavity. By using the stability analysis of a uniform intensity beam in the cavity, it is shown that the pattern formation process is always associated with two consecutive thresholds which are determined by the degree of spatial coherence, the strength of the nonlinearity, and the cavity feedback parameter. At the first threshold the beam becomes unstable, as self-focusing overcomes diffusive tendency of spatially incoherent light. The second threshold occurs when the nonlinear gain overcomes the loss in a single pass. The first (instability) threshold is independent of the cavity boundary conditions, which is in contrast to the coherent cavities (e.g., see Ref. [10]), whereas the second threshold is an inherent cavity feature. As an interesting feature of our system, we point out that if the nonlinearity is at low (high) saturation, an increase in feedback leads to forward (backward) crossing over of the two thresholds, i.e., to switching the pattern on (off).

II. PROPAGATION EQUATIONS AND BOUNDARY CONDITIONS

For the analysis of the incoherent cavity, the quantities arising from coherence, such as the resonant frequencies (modes) of the cavity [11], are unimportant. Hence, the theoretical description cannot resort to the commonly used mean-field theory [11]. Instead, a new approach with new parameters (e.g., the degree of spatial coherence) has to be adopted.

We begin by deriving equations governing the dynamics. We assume that the light circulating through the cavity is linearly polarized. The slowly varying amplitude of the electric field cycling through the cavity for the jth time is described by a complex amplitude $\psi_j(x,z,t)$, where x denotes the spatial coordinate, z is the propagation axis, and t denotes time. The spatial coherence properties and intensity of the field cycling for the jth time through the cavity are described by the mutual coherence function $B_j(x_1,x_2,z) = \langle \psi_j(x_2,z,t)^* \psi_j(x_1,z,t) \rangle$ [12], where brackets $\langle \cdots \rangle$ de-

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note the time average taken over the response time of the medium. Since the finesse of the cavity is low, and the temporal coherence length of the light l_{coh} is much smaller than the length of the cavity L, the phases of the fields that are circulated through the cavity a different number of times are mutually uncorrelated, that is, $\langle \psi_j^* \psi_l \rangle = 0$ for $j \neq l$.

Under the paraxial approximation [13], the propagation dynamics of the mutual coherence functions B_j within the nonlinear medium is given by an infinite set of coupled partial differential equations

$$\frac{\partial B_j}{\partial z} - \frac{i}{k}\frac{\partial^2 B_j}{\partial r\,\partial \rho} = \frac{ik}{n_0}\{\delta n(I_+) - \delta n(I_-)\}B_j(r,\rho,z), \quad (1)$$

where $j = 1,2,\ldots$. In Eq. (1), the new set of spatial coordinates is defined by $r = (x_1 + x_2)/2$ and $\rho = x_1 - x_2$, $I_\pm$ denotes the time-averaged intensity $I_\pm = \Sigma_l B_l(r \pm \rho/2,0,z)$, the nonlinear response of the material is $n^2(I) \simeq n_0^2 + 2n_0\delta n(I)$, and k is the wave number of the carrier (in the medium). In the feedback loop, the light is imaged from the $z = L$ face of the crystal to the incident plane $z = 0$. Hence, besides the propagation equations (1), the mutual coherence functions B_j are also subject to the boundary conditions

$$B_{j+1}(r,\rho,z=0) = \epsilon B_j(r,\rho,z=L), \quad (2)$$

where ϵ denotes the cavity feedback parameter. From Eq. (1), we see that diffraction is accounted for in the nonlinear part of the cavity [the term $ik^{-1}\,\partial^2 B_j/\partial r\,\partial\rho$]. The boundary conditions consider only losses since the conventional optical system in the feedback loop images light from the output to the input face of the crystal.

III. LINEAR STABILITY ANALYSIS OF A UNIFORM INTENSITY BEAM

To study the stability of the uniform intensity beam, we express the mutual coherence functions as

$$B_j(r,\rho,z) = B_j^{(0)}(\rho) + B_j^{(1)}(r,\rho,z), \quad (3)$$

where $B_j^{(1)}(r,\rho,z)$ denotes small perturbations upon the uniform intensity component $B_j^{(0)}(\rho)$. Boundary conditions (2) must be satisfied by the uniform intensity components $B_{j+1}^{(0)}(\rho) = \epsilon B_j^{(0)}(\rho)$ and by the small perturbations $B_{j+1}^{(1)}(r,\rho,0) = \epsilon B_j^{(1)}(r,\rho,L)$. From this, we express the uniform component of the jth cycle in terms of the uniform component of the incident beam, $B_j^{(0)}(\rho) = \epsilon^{j-1} B_1^{(0)}(\rho)$, and find the total uniform intensity of the light in the cavity, $I_{total} = \Sigma_l B_l^{(0)}(0) = I^{(0)}/(1-\epsilon)$; the intensity of the incident beam is denoted by $I^{(0)} = B_1^{(0)}(0)$. As long as perturbations are small enough, i.e., $|B_j^{(1)}(r,\rho,z)| \ll |B_j^{(0)}(\rho)|$, Eq. (1) can be linearized:

$$\frac{\partial B_j^{(1)}}{\partial z} - \frac{i}{k}\frac{\partial^2 B_j^{(1)}}{\partial r \partial \rho} = \frac{ik\kappa}{n_0} B_j^{(0)}(\rho) \sum_l \left\{ B_l^{(1)}\left(r + \frac{\rho}{2},0,z\right) - B_l^{(1)}\left(r - \frac{\rho}{2},0,z\right)\right\}, \quad (4)$$

where $\kappa = \partial \delta n(I)/\partial I|_{I = I_{total}}$. Equations (4) are linearized, however, they are still coupled. To uncouple Eq. (4), we seek their solution through the superposition of modes

$$B_j^{(1)}(r,\rho,z) = \epsilon^{j-1} M_1(r,\rho,z) + \sum_{l \geq 2} h_{jl} M_l(r,\rho,z), \quad (5)$$

where $h_{jl} = \delta_{jl-1} - \delta_{jl}$, and δ_{jl} denotes the Kronecker delta. From Eqs. (4) and (5), we find that the modes evolve according to

$$\frac{\partial M_j}{\partial z} - \frac{i}{k}\frac{\partial^2 M_j}{\partial r \partial \rho} = \delta_{1j}\frac{ik\kappa B_1^{(0)}(\rho)}{n_0(1-\epsilon)}\left\{ M_1\left(r + \frac{\rho}{2},0,z\right) - M_1\left(r - \frac{\rho}{2},0,z\right)\right\}. \quad (6)$$

Equation (6) for the first mode M_1 is equivalent to the equation describing modulation instability (MI) in a spatially incoherent single-pass system [14], which implies that the first mode M_1 can experience nonlinear gain and destabilize the beam. In contradistinction, since the right-hand side of Eq. (6) is zero for $j \geq 2$ [due to the term $\delta_{1j} = 0$], all other modes M_j, $j \geq 2$, satisfy equation describing the evolution of small perturbations in a *linear* medium; consequently, they do not experience any nonlinear gain.

Equation (6) can be solved with Fourier analysis. In order to satisfy condition $B_j(r,\rho,z) = B_j(r,-\rho,z)^*$, we express the modes as $M_j(r,\rho,z) = P_j(r,\rho,z) + P_j(r,-\rho,z)^*$, where

$$P_j(r,\rho,z) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} d\alpha\, dK\, e^{g_j(\alpha)z}\hat{L}_j^\alpha(K) A_j(\alpha) e^{-iK\rho} e^{i\alpha r}. \quad (7)$$

Here, α denotes the spatial wave number, $g_j(\alpha)$ the growth rates, $A_j(\alpha)$ the amplitudes, and $\hat{L}_j^\alpha(K)$ the spatial coherence properties of the perturbations corresponding to the jth mode M_j. The functions $\hat{L}_j^\alpha(K)$ are normalized so that $\int dK \hat{L}_j^\alpha(K) = 1$. From Eqs. (6) and (7) for $j = 1$, and with the use of $\int dK \hat{L}_1^\alpha(K) = 1$, we obtain an implicit integral equation for the growth rate of the first mode $g_1(\alpha)$:

$$\frac{k\kappa}{n_0(1-\epsilon)}\int_{-\infty}^{\infty}\frac{h(K,\alpha)}{ig_1 + \dfrac{\alpha K}{k}} dK = -1, \quad (8)$$

where $h(K,\alpha) = \hat{B}_1^{(0)}(K + \alpha/2) - \hat{B}_1^{(0)}(K - \alpha/2)$, and $\hat{B}_j(K)$ denotes the Fourier transform of $B_j(\rho)$. The growth rate of the first mode $g_1(\alpha)$ can assume real values greater than zero. The growth rate of the nongrowing eigenmodes is purely imaginary: $g_j(\alpha) = iK\alpha/k$, for $j \geq 2$.

The boundary conditions $B_{j+1}^{(1)}(r,\rho,0) = \epsilon B_j^{(1)}(r,\rho,L)$ can be written as

$$\epsilon^j A_1(\alpha)\hat{L}_1^\alpha(K)[e^{g_1 L} - 1] = \sum_{l=2}^{\infty} a_{jl} A_l(\alpha)\hat{L}_l^\alpha(K), \quad (9)$$

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where $j=1,2,\ldots$, $a_{jj}=\epsilon e^{g_2 L}$, $a_{jj+1}=-(\epsilon e^{g_2 L}+1)$, $a_{jj+2}=1$, and $a_{jl}=0$ otherwise. From Eq. (9), it follows that the coherence properties $\hat{L}_j(K)$ and amplitudes $A_j(\alpha)$ of the non-growing modes ($j\geqslant 2$) can be expressed in terms of the coherence properties $\hat{L}_1(K)$ and amplitude $A_1(\alpha)$ of the first mode. From Eqs. (6) and (7) for $j=1$, and by using $\int dK \hat{L}_1^{\alpha}(K)=1$, it follows that $\hat{L}_1^{\alpha}(K)$ is determined [via $h(K,\alpha)$] by the coherence of the source:

$$\hat{L}_1^{\alpha}(K)=-\frac{k\kappa}{n_0(1-\epsilon)}\frac{h(K,\alpha)}{ig_1+\dfrac{\alpha K}{k}}. \tag{10}$$

The perturbations upon the incident beam can be written as $B_1^{(1)}(r,0,0)=\int_{-\infty}^{\infty}A_I(\alpha)e^{i\alpha r}d\alpha+\text{c.c.}$, and via boundary conditions as $B_1^{(1)}(r,0,0)=\Sigma_{j=1}^{\infty}[B_j^{(1)}(r,0,0)-\epsilon B_j^{(1)}(r,0,L)]$. From these two identities and Eqs. (5) and (7), we obtain

$$A_1(\alpha)=(1-\epsilon)\frac{A_I(\alpha)}{1-\epsilon e^{g_1(\alpha)L}}. \tag{11}$$

From the analysis above, we conclude that the coherence properties of the source $B_1^{(0)}(\rho)$, and the perturbations $A_I(\alpha)$ upon the incident beam, determine the coherence properties and the perturbations corresponding to the fields that circulate in the cavity. Although the linear stability analysis performed in Eqs. (3)–(11) resembles the linear stability analysis from Ref. [14], we emphasize that there is a significant difference. Namely, together with linearized propagation equations (4), here the solution has to obey the cavity boundary conditions [Eq. (2)], which results in a specific cavity threshold (to be explained below) that accompanies pattern formation in a cavity. Such a threshold does not have a counterpart in a single-pass system such as the one in Ref. [14]. We also emphasize that Eqs. (1)–(11) can be straightforwardly generalized to include $(2+1)D$ systems.

IV. THE INSTABILITY THRESHOLD

The linear stability analysis of the uniform intensity beam above is applicable for any type of input beam correlation statistics and nonlinearity. From now on we assume that the correlation statistics is given by Lorenzian in Fourier space, $\hat{B}_1^{(0)}(K)\sim(K^2+K_0^2)^{-1}$, and that the nonlinearity is $\delta n(I)=\gamma I/(1+I/I_S)$. In the limit $I_S\to\infty$, the nonlinearity is of Kerr type, and saturable otherwise. The nonlinear gain $g_1(\alpha)$ follows from Eq. (8)

$$|g_1(\alpha)|=-\theta|\alpha|+|\alpha|\sqrt{\frac{\kappa}{n_0}\frac{I^{(0)}}{1-\epsilon}-\left(\frac{\alpha}{2k}\right)^2}, \tag{12}$$

where $\theta=K_0/k$ denotes the degree of spatial coherence. From Eq. (10), we find the functional dependence of the maximally destabilizing perturbation α_{max} (pattern wave vector in the low visibility regime) on the degree of coherence θ and other parameters:

$$\frac{\alpha_{max}}{k}=\left(\frac{2\kappa I^{(0)}}{n_0(1-\epsilon)}-\frac{\theta^2}{2}-\sqrt{\frac{2\kappa I^{(0)}}{n_0(1-\epsilon)}\theta^2+\frac{\theta^4}{4}}\right)^{1/2}. \tag{13}$$

Furthermore, from Eq. (12), it follows that there exists a well-defined threshold at which the beam becomes modulationally unstable; if

$$\kappa I^{(0)}/(1-\epsilon)>\theta^2 n_0, \tag{14}$$

the beam is unstable.

We point out that the incoherent beam incident on the cavity is unstable if and only if the same beam with intensity enhanced by the factor $(1-\epsilon)^{-1}$ is unstable in a single-pass system [14]. The factor $(1-\epsilon)^{-1}$ simply reflects the increase of total intensity in the cavity due to feedback, $I_{total}=I^{(0)}/(1-\epsilon)$. Thus, below and at the instability threshold, one cannot distinguish from the cavity system with incident intensity $I^{(0)}$, and the single-pass system with intensity $I^{(0)}/(1-\epsilon)$. Physically, this equivalence is a consequence of the mutual incoherence between the fields of different cycles. The perturbations and the uniform parts of beams from different cycles do not interfere, but simply add up. This means that apart from the trivial enhancement of total intensity, the cavity boundary conditions do not affect the position of the MI threshold. This is also confirmed by the fact that the position of the instability threshold, given by inequality (14), does not depend on the length of the sample L, which is embedded in boundary conditions (2). This result is in a sharp contrast to coherent cavities [10], where the instability process is influenced by the boundary conditions through the interference of fields from difference cycles.

V. THE CAVITY THRESHOLD

In contrast to the instability threshold, the boundary conditions, through the interplay of nonlinear gain and cavity loss, induce a transition from the low to high visibility pattern. To see this, we calculate the modulation depth of the intensity pattern. From Eqs. (5), (7), and (11), it follows that the intensity pattern at $z=L$ is

$$\sum_{i=1}^{\infty}B_i^{(1)}(r,0,L)=\int\frac{e^{g_1(\alpha)L}}{1-\epsilon e^{g_1(\alpha)L}}A_I(\alpha)e^{i\alpha r}d\alpha+\text{c.c.} \tag{15}$$

From Eq. (15), we can estimate the modulation depth (visibility) $m=(I_{max}-I_{min})/(I_{max}+I_{min})$ of the pattern.

First, we calculate integral (15) numerically. We assume that the noise upon the incident beam does not have any preferential spatial frequency α, i.e., $|A_I(\alpha)|$ is independent of α, while the phase of $A_I(\alpha)$ is random. For the illustrations of the pattern visibility, we use the following parameters. The length of the nonlinear medium is $L=5$ mm, the wavelength of the carrier wave is $\lambda=488$ nm (in vacuum), and the linear part of the refractive index is $n_0=2.3$. The dependences of the modulation depth on the feedback ϵ and the strength of the nonlinearity $\Delta n=\gamma I^{(0)}$ are displayed in Figs. 1 and 2, respectively; solid lines represent the values of

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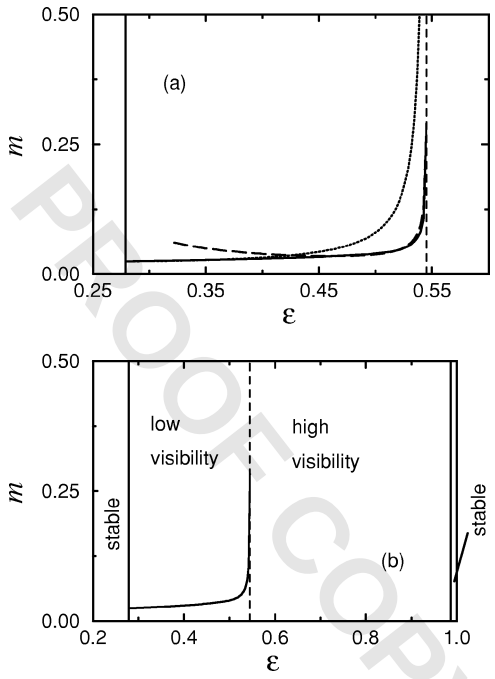


FIG. 1. The modulation depth m of the pattern vs the feedback parameter ϵ. Vertical solid (dashed) lines show the position of the first (second) threshold. Solid curves show m calculated numerically from Eq. (15). (a) Dotted curve represents $m_{\alpha_{max}}$ [see Eq. (16)], and dashed curve represents modulation depth calculated from expression (18). (b) The region of the stable output, and of a low and high visibility pattern. For ϵ close to 1, the nonlinearity is highly saturated, and the pattern is switched off. The parameters used are $\Delta n = 10^{-4}$, $\theta = 0.0068$, and $I^{(0)}/I_S = 0.1$.

m calculated numerically from integral (15). Vertical lines represent the positions of the thresholds as is explained in figure captions. From Figs. 1 and 2, we see that just after the instability threshold, the modulation depth grows until it comes close to the second threshold where the modulation depth experiences a sudden jump. This second threshold is the cavity threshold which appears when the denominator of the integrand (15) becomes zero.

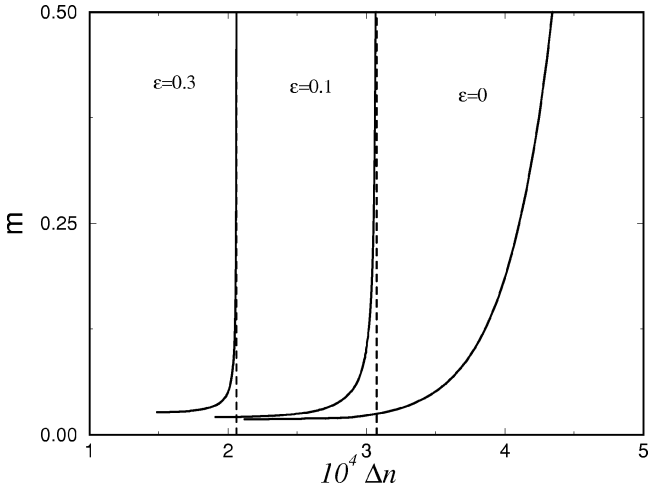


FIG. 2. The modulation depth vs $\Delta n = \gamma I^{(0)}$ for different values of ϵ; $\theta = 0.0096$, and $I_S = \infty$. m is shown from the instability threshold up to the cavity threshold (indicated by vertical dashed lines).

Before explaining the cavity threshold, let us gain more insight into the modulation depth of the pattern by analytical formulas. From Eq. (15), it follows that the maximally destabilizing perturbation α_{max} has the largest contribution to integral (15). The modulation depth of the maximally destabilizing perturbation α_{max} (not the whole pattern) is

$$m_{\alpha_{max}} \simeq C_1(1-\epsilon)\frac{e^{g_1(\alpha_{max})L}}{1-\epsilon e^{g_1(\alpha_{max})L}}, \tag{16}$$

where parameter C_1 corresponds to the strength of the small intensity perturbations upon the incident beam. The dotted curve in Fig. 1 displays the behavior of $m_{\alpha_{max}}$. We see that $m_{\alpha_{max}}$ describes well the trend of the modulation depth behavior in between the two thresholds, however, it diverges faster than the modulation depth calculated numerically.

A more accurate description can be achieved by approximately integrating expression (15). The growth rate $g(\alpha)$ can be Taylor expanded around the maximally destabilizing perturbation α_{max}, $g(\alpha) \simeq g(\alpha_{max}) + \frac{1}{2}g''(\alpha_{max})(\alpha-\alpha_{max})^2$, and the integrand (15) is approximately

$$\frac{e^{g_1(\alpha)L}}{1-\epsilon e^{g_1(\alpha)L}} \simeq \frac{1}{\epsilon}\sum_{j=1}^{\infty} \epsilon^j e^{g(\alpha_{max})Lj} e^{-j/2|g''(\alpha_{max})|L(\alpha-\alpha_{max})^2}. \tag{17}$$

Approximate integration over α gives the following expression for the modulation depth:

$$m = C\frac{1-\epsilon}{\epsilon}\frac{\mathrm{Li}_{1/2}(\epsilon e^{g(\alpha_{max})L})}{\sqrt{|g''(\alpha_{max})|L}}, \tag{18}$$

where $Li_k(x) = \Sigma_{j=1}^{\infty} x^j/j^k$ denotes the polylogarithm function. We see that Eq. (18) for the modulation depth contains the term $1/\sqrt{|g''(\alpha_{max})|L}$ which expresses the spatial wave number selectivity. Namely, if the absolute value of the derivative $|g''(\alpha_{max})|$ is larger, integrand (17) will be more peaked around α_{max}, and only a small number of spatial frequencies close to α_{max} will contribute to the pattern. The functional dependence of the modulation depth from Eq. (18) on the feedback is displayed by a dashed curve in Fig. 1. Evidently, the functional form (18) does not increase monotonically from the first to the second threshold but has a minimum. This is a consequence of the spatial wave number selectivity term $1/\sqrt{|g''(\alpha_{max})|L}$. Thus, the functional form (18) can be used to describe the modulation depth only from that minimum up to the second threshold, i.e., when $|g''(\alpha_{max})|L$ becomes sufficiently large. From Fig. 1, we observe that the functional form (18) gives a good description for the behavior of the modulation depth below the second threshold. In Fig. 1, the parameter C is chosen such that the minimum of the curve from expression (18) intersects the numerically calculated curve for the modulation depth.

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Now we explain the cavity threshold in simple words. In the cavity, the output pattern, with preferred periodicity determined by α_{max}, is imaged (with some loss) to the input plane, thus affecting the output pattern. From the denominator in Eq. (15), it follows that the modulation depth of the intensity pattern is determined by the relation between the nonlinear gain and cavity loss. If the nonlinear gain is smaller than the cavity loss, the modulation depth at the input is small enough, so that the pattern at the input is regarded as noise with preferential periodicity determined by α_{max}. However, if the nonlinear gain is larger than the cavity loss, the intensity structure at the input is more than just preferential noise; this structure guides the light from the input beam into its shape. This significantly differs from the single-pass MI, where the modulation depth at the input is always small, and the pattern builds from small noise [14]. From Figs. 1 and 2, we see that the feedback of the maximally destabilizing perturbation induces a rapid increase of the modulation depth, which is referred to as the cavity threshold; it occurs approximately when the nonlinear gain becomes equal to the cavity loss, i.e., when $\epsilon \exp g_1(\alpha_{max})L = 1$. The transition will be sharper for larger values of the feedback parameter ϵ (see Fig. 2). This threshold is analogous to that in many feedback systems with gain (e.g., see Ref. [15]). The behavior of the intensity pattern in between the two thresholds corresponds to noisy precursors in one-dimensional patterns observed in Ref. [16].

We note that the features of the intensity structure above the cavity threshold cannot be determined from the stability analysis of Sec. III. However, this analysis shows that formation of any such structure is preceded by the two consecutive thresholds, and it predicts the positions of these thresholds in parameter space spanned by Δn, θ, and ϵ (see Fig. 3). From inequality (14), it follows that the increase of coherence and/or the strength of the nonlinearity always leads to pattern formation. The dependence of the stability on the feedback ϵ depends on the saturation of the medium. If the medium is Kerr, or in the regime of low saturation $I^{(0)}(1-\epsilon)^{-1} I_S^{-1} \ll 1$, the increase of feedback acts destabilizing. However, if the medium is in the regime of high saturation $I^{(0)}(1-\epsilon)^{-1} I_S^{-1} \gg 1$, then the increase of feedback leads to the stabilization of the beam. Thus, as an interesting feature of the cavity system, we show that in the limit of low (high) saturation, the increase of feedback leads to switching the pattern on (off) [see Figs. 1(b) and 3].

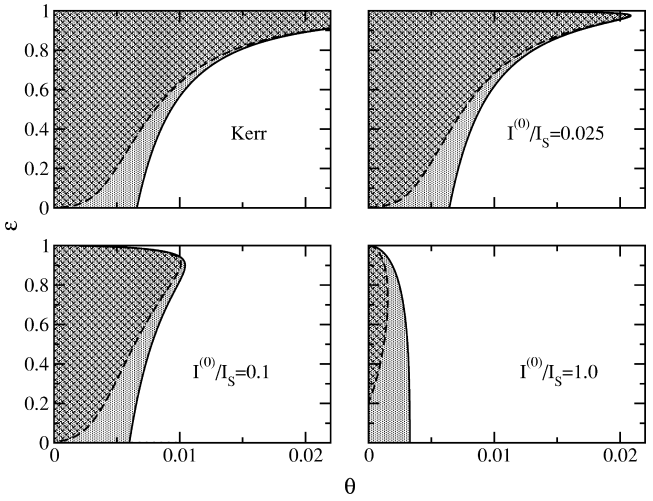


FIG. 3. The positions of the modulation instability threshold (solid line) and the cavity threshold (dashed line) in the (θ, ϵ) plane depending on the saturation intensity I_S; $\Delta n = 10^{-4}$. White (cross-hatched) region denotes stable output (pattern, respectively).

VI. CONCLUSION

In conclusion, we have analyzed the early stage of pattern formation process in a nonlinear optical cavity with incoherent light. The nonlinear medium within the cavity has a noninstantaneous response, that is, it is unable to follow fast random fluctuations of incoherent light. We have demonstrated that two consecutive thresholds always accompany the process of pattern formation. In contrast to the coherent cavity cases [10], the instability threshold is unaffected by the incoherent cavity boundary conditions. The second (cavity) threshold is determined by the interplay of nonlinear gain and cavity loss. It appears when the nonlinear gain in single pass overcomes the cavity loss. For future work on incoherent optical cavities, such as the one described here, we envision the study of incoherent cavity solitons, whose features are yet to be determined.

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Jensen *et al.*, Phys. Rev. Lett. **81**, 1614 (1998).

[7] In fact, almost all previous studies on pattern formation in optical cavities considered cases where the light was *both spatially and temporally coherent*. There are only two exceptions. The first relates to some of the single-mirror feedback systems—those with the distance between the mirror and nonlinear medium much larger than l_{coh} [R. MacDonald *et al.*, Opt. Commun. **89**, 289 (1992)]; The second is a recent study [T. Carmon *et al.*, Phys. Rev. Lett. **89**, 183902 (2002)] with a passive nonlinear ring cavity with $l_{coh} \ll L_c$. We emphasize that the light in both of these studies was spatially coherent, in contrast to our current study with spatially incoherent light.

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Addendum

Prior to my research, composite solitons were demonstrated experimentally only in a (1+1)D configuration. At the beginning of my doctoral studies, I made the first experimental observation of a *dipole composite soliton* in (2+1)D geometry [68]. This opened the way for my observation of the first bright soliton carrying angular momentum - the *propeller soliton* [63], for which the energy structure rotates during propagation. During all this time, there was, and there still a desire to observe vortex soliton of finite digestion. While other groups in the world were focusing their efforts on investigating the breakup of a vortex (due to medium anisotropy), I came up with the solution of using isotropic thermal nonlinearity to make a *bright vortex soliton* possible. I am currently instructing a student to find a numerical solution for a thermal vortex by using self-consistency methods. I hope that in the near future, a whole world of vortices and rotating multi-poles will be opened.

In parallel, in this research, I presented the first experimental investigation of *pattern formation in a temporally incoherent cavity* [91] and afterwards the first experimental investigation of pattern formation in *a temporally and spatially incoherent cavity* [94,95]. Recently I have taken a part in an experiment in which we experimentally observed (for the first time) *pattern formation with incoherent white light* [32] from a regular arc light bulb. I hope that, in the future, this research on incoherent modulation instability will let us answer the important question: Are pulses of light, coming from distant galaxies, pulsed because they traveled the whole distance as pulses, or, is because a continuous source evolved into pulses via modulation instability.

While writing this thesis, I have tried to describe my own contributions. Yet, I also felt that while there are many texts on solitons in various systems, there is no text containing a "periodic table" of solitons, allowing one to get a broader perspective and to compare solitons from different fields of physics. In the same manner, I had found it important to discuss the uncertainty origins of diffraction and to make comparison

between the different kinds of angular momenta. I will give the next generation of students all the tools needed to continue this research.

At the end, I would like to note that I enjoyed doing the experiments presented here and I hope the reader also enjoyed (and got inspiration from) reading this thesis.

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בסוף המחקר מוצגת הצעת ניסוי לסוליטון מערבולת בהיר. מערבולת אופטית בהירה היא טבעת אור המסתובבת תוך כדי התקדמותה ובמרכזה (בעין הסערה) נקודה חשוכה. עד עתה לא הודגם ניסיונית סוליטון מערבולת בהיר בשום מערכת אופטית. ההצעה מתבססת על שימוש בתווך המשנה את מקדם השבירה שלו בהשפעת חם ומאמצים מכאניים.

תקציר

אלומות אור נוטות בדרך כלל להתרחב תוך כדי התקדמות בתווך לינארי (עקיפה). אולם, אם אלומת האור מתקדמת בתווך לא לינארי ממקד (בו מהירות האור עומדת ביחס הפוך לעוצמתו) יכולה אלומת האור ליצור שינוי במקדם השבירה של התווך ששקול למבנה של סיב אופטי, ובה בשעה להוליך את עצמה ב"סיב" שיצרה תוך שמירת רוחבה וצורתה. המיקוד במקרה זה מתקזז עם ההתרחבות. אלומות שכאלה השומרות על רוחבן וצורתן תוך כדי התקדמות נקראות סוליטונים אופטיים מרחביים. אלומת האור היוצרת את הסיב יכולה להיות מורכבת ממספר אופנים, במקרה זה נקרא לסוליטון: "סוליטון וקטורי" כאשר האופנים המרכיבים את הסוליטון שונים זה מזה בצורתם הסוליטון נקרא "סוליטון מורכב".

בתחילת המחקר יצרנו לראשונה סוליטון מרחבי וקטורי בשלושה מימדים (שניים להתרחבות ואחד להתקדמות) המורכב משני אופנים: הראשון בצורת פעמון והשני בצורת כיפה כפולה (שני כתמי אור). להתפלגות העצמה הכללית הייתה צורת פעמון או צורת כיפה כפולה. המעבר משני מימדים לשלושה מימדים איפשר יצירת סוליטון מורכב (ווקטורי) המסתובב תוך כדי התקדמות, כאשר אופן אחד שלו בצורת פעמון אליפטי מסתובב, ואופן שני בצורת שני כתמי אור (דיפול) מסתובבים. שני האופנים של הסוליטון מסתובבים יחדיו. סוליטון מורכב זה היה הסוליטון הבהיר הראשון הנושא תנע זוויתי תוך כדי התקדמות, שהודגם ניסיונית אי פעם. בהמשך חקרנו את תוצרי ההתנגשות בין סוליטונים מסתובבים כאלה הנושאים תנע זוויתי.

מערך סוליטונים מרחביים יכול להתפתח תוך כדי התקדמות (בתווך לא לנארי) מגל אור מישורי, כלומר מהתפלגות עוצמה אחידה במרחב. ההתפתחות זו של מערך סוליטונים מגל מישורי קשורה לתופעה הנקראת "התפתחות ספונטאנית של צורות". צורות שכאלה יכולות להתפתח גם מאור לא קוהרנטי אחיד בעוצמתו.

בחלקו השני של מחקר זה הדגמנו התפתחות צורות במהוד אופטי לא קוהרנטי. תחילה חקרנו מהוד בו האור אינו קוהרנטי זמנית, ולאחר מכן חקרנו התפתחות ספונטאנית של צורות במהוד בו האור איננו קוהרטי זמנית ומרחבית גם יחד .במהוד בו סבב אור לא קוהרנטי זמנית גילינו הופעת צורות והצרת סרט של הספקטרום המרחבי של הצורות עם הגדלת המשוב במהוד. במהוד השני, בו האור לא קוהרטי זמנית ומרחבית גם יחד, גילינו שקיימים שני ערכי סף להתפתחות צורות. הסף הראשון מוגדר כאשר המיקוד העצמי גובר על העקיפה הלא קוהרנטית והסף השני קורה כאשר ההגבר בתווך הלא לנארי גדול מההפסדים במהוד. ניסוי זה הינו הניסוי הראשון בו הודגמה התפתחות ספונטאנית של צורות במהוד אופטי לא קוהרנטי כלשהו.

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המחקר נעשה בהנחיית פרופסור מרדכי שגב בפקולטה לפיסיקה.

אני מודה לטכניון ולמשרד המדע על התמיכה הנדיבה בהשתלמותי.

אופטיקה לא לנארית וסוליטונים

חיבור על מחקר לשם מילוי חלקי של הדרישות
לקבלת תואר דוקטור לפילוסופיה

טל כרמון

הוגש לסנט הטכניון – מכון טכנולוגי לישראל

חשון תשס"ד חיפה נובמבר 2003

אופטיקה לא לינארית וסוליטונים

טל כרמון